CHANGE OF AXES 4

Art. 31. Transformation of Co-ordinates.

The co-ordinates of a point or the equation of a curve are alway given with reference to a fixed origin and a set of axes of co-ordinates. The above co-ordinates of the equation of the curve changes when the origin is changed or the direction of axes changed or both. The process of changing the co-ordinate of a point or the equation of a curve is called **transformation of co-ordinates.** Now we have to investigate the mode of the change of the co-ordinates or the equation of the curve according to the transfer from one set to another set of co-ordinate axes.

Art. 32. Change of origin (Translation of axes)

To find the change in the co-ordinates of a point when the origin is shifted to another point O (α, β) where the direction of axes remains unalterad.

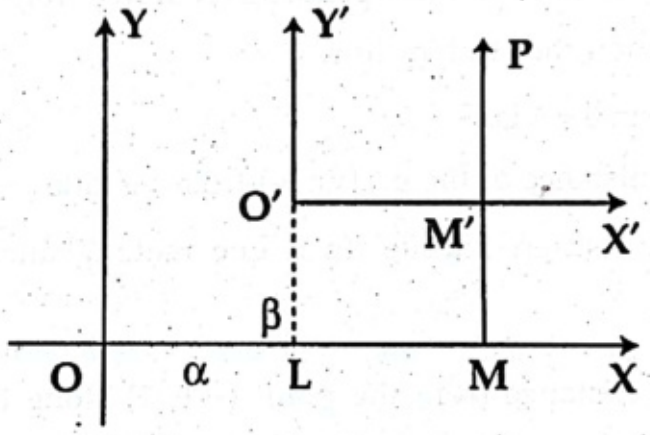


Fig. 10

Let us take a new pair of axes O′ X′ and O′Y′ parallel to the old pair OX and OY; O′ being the new origin whose co-ordinate are (α, β) referred to O′X′ and O′Y′.

let (x', y') be the co-ordinates referred to the axes O′X′ and O′Y′ or a point P, whose co-ordinate referred to the old axes are (x, y).

It is required to transform the co-ordinates (x, y) in terms or (x', y').

From O′ and P draw O′L and PM perpendiculars to OX. Let PM meet O′X′ in M′.

Then $OL = \alpha, LO' = \beta, OM = x, MP = y$

Also $O'M' = x'$ and $M'P = y'$

Therefore, $OM = OL + LM = OL + O'M'$

$\therefore x = \alpha + x'$ ... ... (i)

Similarly, $MP = MM' + M'P = LO' + M'P$

$\therefore y = \beta + y'$ ... ... (ii)

The transformed co-ordinates are

$$\text{and } \left.\begin{matrix} x' = x - \alpha \\ y' = y - \beta \end{matrix}\right\} \quad \ldots \quad \ldots \quad \ldots \quad (2)$$

Rule : *In order to shift the origin to (α, β) the transformation is obtained by replacing x by $x + a$ and y by $y + \beta$. If from the transformed equation we want to get the old equations then replace x by $x - \alpha$ and y by $y - \beta$. This is known as shifting the origin back.*

Art. 33. Rotation of axes (origin fixed)

To find the change in the co-ordinates of a point when the direction of axes is turned through an angle θ where as the origin of co-ordinates remains the same.

Let OX and OY be the old axes and OX′ and OY′ set the new axes. O is the common origin for the two sets of axes. Let the angle X′ OX through which the axes have rotated be represented by θ.

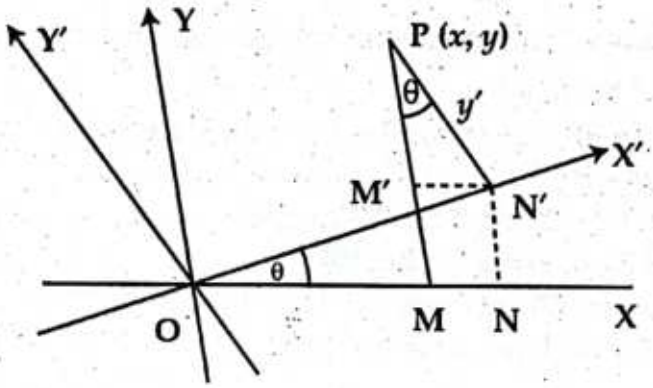


Fig. 11

Let P be any point in the plane and let its co-ordinates referred to the old axes be (x, y), and referred to the new axes be (x', y').

Let us try to determine x and y in terms of x', y' and θ, Draw PM perpendicular to OX, PN′ perpendicular to OX′, and N′N perpendicular to OX, and N′M′ parallel to OX.

Then $x = OM = ON - MN = ON - M'N'$

$$= ON' \cos\theta - PN' \sin\theta = x' \cos\theta - y' \sin\theta$$

$y = MP = MM' + M'P = NN' + M'P$

$$= ON' \sin\theta + PN' \cos\theta = x' \sin\theta + y' \cos\theta$$

Hence the formula for the rotation of the axes through an angle θ are.

$$\text{and } \left.\begin{matrix} x = x' \cos\theta - y' \sin\theta \\ y = x' \sin\theta + y' \cos\theta \end{matrix}\right\} \quad \ldots \quad \ldots \quad \ldots \quad (3)$$

Art. 34. Change of origin with the change of the direction of axes.

It is really the combination of Art. 32 and Art. 33. The best method is to apply Art. 32 first and then Art. 33. Of course the two transformations may also be made simultaneously.

Let us suppose that the system of axes be rectangular. The origin is shifted to the point (α, β) and then the axes are rotated through an angle θ. If the co-ordinates of any point be (x, y) in the old system, and (x', y') in the new system, from Art. 32, and Art. 33.

we have.

$$\text{and } \left.\begin{matrix} x = \alpha + x' \cos\theta - y' \sin\theta \\ y = \beta + x' \sin\theta + y' \cos\theta \end{matrix}\right\} \quad \ldots \quad \ldots \quad \ldots \quad (4)$$

Art. 35. Invariants. [R.U. 1983]

If by the rotation of the rectangular co-ordinate axes about the origin, the expression $ax^2 + 2hxy + by^2$ *changes to*

$a'x'^2 + 2h'x'y' + b'y'^2$, $a + b = a' + b'$ and $ab - h^2 = a'b' - h'^2$

Let (x, y) be the co-ordinates of a point P referred to a set of rectangular axes. If the axes are roated through an angle θ about the origin, let the co-ordinates of the same point P be (x', y') referred to the new system of rectangular axes. Then we have by

(3) Art. 33. $x = x' \cos\theta - y' \sin\theta$ and $y = x' \sin\theta + y' \cos\theta$.

By using the above transformations the expression

$ax^2 + 2hxy + by^2$ becomes

$a(x' \cos\theta - y' \sin\theta)^2 + 2h\,(x'\cos\theta - y' \sin\theta)\,(x' \sin\theta + y' \cos\theta) + b\,(x' \sin\theta + y' \cos\theta)^2$

$= a'x'^2 + 2h'x'y' + b'y'^2$

where $a' = a\cos^2\theta + 2h \sin\theta \cos\theta + b \sin^2\theta$ ... ... (1)

$b' = a \sin^2\theta - 2h \sin\theta \cos\theta + b \cos^2\theta$ ... ... (2)

$h' = h\,(\cos^2\theta - \sin^2\theta) - (a - b) \sin\theta \cos\theta$ ... ... (3)

From (1) and (2) we have on simplification

$a' + b' = a + b$

Again $2a' = 2a \cos^2\theta + 4h \sin\theta \text{ cis } \theta + 2b \sin^2\theta$

$= a\,(1 + \cos 2\theta) + 2h \sin 2\theta + b(1 - \cos 2\theta)$

$= a + b + 2h \sin 2\theta + (a - b) \cos 2\theta$ ... ... (4)

Similarly. $2b' = a + b - 2h \sin 2\theta - (a - b) \cos 2\theta$ ... ... (5)

From (4), (5) and (3) we have

$4(a'b' - h'^2) = (a + b)^2 - \{2h \sin 2\theta + (a - b)\cos 2\theta\}^2$

$- \{2h \cos 2\theta - (a - b) \sin 2\theta\}^2$

$= (a + b) - 4h^2 - (a - b)^2 = 4ab - 4h^2$

or, $a'b' - h'^2 = ab - h^2$

The two quantities $a + b$ and $ab - h^2$ for the expression $ax^2 + 2hxy + by^2$ are called **invariants** of transformation from one system of rectangular axes (Without change of the origin), because their values remain unchanged by the transformation.

Cor. Removal of the-xy-term from the expression $ax^2 + 2hxy + by^2$

If we rotate the rectangular axes through an angle θ about the origin, we have transformations by Eq. 3, Art. 33.

$x = x' \cos\theta - y' \sin\theta$, $y = x' \sin\theta + y' \cos\theta$

Putting the values of x and y in $ax^2 + 2hxy + by^2$

we have, $a'x'^2 + 2h'x'y' + b'y'^2$

where the values of a'. b' and h' are available from (1), (2) and (3) of Art 35.

The co-efficient of $x'y'$ term in the new expression is $2h'$ where $h' = h(\cos^2\theta - \sin^2\theta) - (a-b)\sin\theta\cos\theta$. In order to remove the $x'y'$ term, the co-efficient h' is zero i.e.

$h(\cos^2\theta - \sin^2\theta) - (a-b)\sin\theta\cos\theta = 0$

$\therefore h\cos 2\theta = \frac{1}{2}(a-b)\sin 2\theta$ or, $\tan 2\theta = \frac{2h}{a-b}$

$\therefore \theta = \frac{1}{2}\tan^{-1}\frac{2h}{a-b}$ ... ... (6)

Hence if the axes are rotated through an angle $\theta = \frac{1}{2}\tan^{-1}\frac{2h}{a-b}$ then xy term in the expression $ax^2 + 2hxy + by^2$ vanishes.

Ex. 1. Determine the equation of the curve $2x^2 + 3y^2 - 8x + 6y - 7 = 0$ when then origin is transferred to the point $(2, -1)$.

when then origin is transferred to the point $(2, -1)$ Put $x = x' + 2$, $y = y' - 1$ in the above equation. Then $2(x'+2)^2 + 3(y'-1)^2 - 8(x'+2) + 6(y'-1) - 7 = 0$

or, $2x'^2 + 3y'^2 = 18$

Now removing suffixes the equation referred to new axes is $2x^2 + 3y^2 = 18$

Ex. 2. Determine the equation of the parabola

$x^2 - 2xy + y^2 + 2x - 4y + 3 = 0$ ... ... (i)

after rotating of axes through 45°

When the axes have been rotated through an angle 45°; then

$x = x'\cos 45° - y'\sin 45° = (x' - y')/\sqrt{2}$

$y = x'\sin 45° + y'\cos 45° = (x' + y')/\sqrt{2}$

Put them in (1), then $\left(\frac{x'-y'}{\sqrt{2}}\right)^2 - 2\left(\frac{x'-y'}{\sqrt{2}}\right)\frac{x'+y'}{\sqrt{2}} + \left(\frac{x'+y'}{\sqrt{2}}\right)^2 + 2\left(\frac{x'-y'}{\sqrt{2}}\right) - 4\left(\frac{x'+y'}{\sqrt{2}}\right) + 3 = 0$

On simplification the equation reduces to

$2y'^2 - \sqrt{2}x' - 3\sqrt{2}y' + 3 = 0$

Now dropping the suffixes, the equation is

$2y^2 - \sqrt{2}x - 3\sqrt{2}y + 3 = 0$ Ans.

Ex. 3. Remove the first degree terms in

$3x^2 + 4y^2 - 12x + 4y + 13 = 0$

Complete the square,

or, $3(x^2 - 4x) + 4\left(y^2 + y + \frac{1}{4}\right) = 0$

or, $3(x-2)^2 + 4\left(y + \frac{1}{2}\right)^2 = 0$ ... ... (ii)

Put $x-2=x'$ and $y+\frac{1}{2}=y'$ in (ii)

$\therefore\ 3x'^2+4y'^2=0$

This is satisfied only by $x'=0, y'=0$, which is the new origin.

By shifting origin at $\left(2, \frac{-1}{2}\right)$, the first degree terms can be removed.

Ex. 4. By transforming to parallel axes through a properly chosen point (h, k), prove that the equation.

$12x^2-10xy+2y^2+11x-5y+2=0$

can be reduced to one containing only the terms of the 2nd degree.

Transforming to parallel axes through (h, k) we have

$12(x'+h)^2-10(x'+h)(y'+k)+2(y'+k)^2+11(x'+h)-5(y'+k)+2=0$

or, $12x'^2-10x'y'+2y'^2+(24h-10k+11)x'+(-10h+4k-5)y'+(12h^2-10hk+2k^2+11h-5k+2)=0$ ... ... (i)

Now equate the co-efficients of x', y' to zero then

$24h-10k+11=0$ also $-10h+4k-5=0$

and $12h^2-10hk+2k^2+11h-5k+2=0$

Solve the first two equations $h=-\frac{3}{2}$, $k=-\frac{5}{2}$ and it is clear that these values of h, k satisfy the equation (ii). Hence the equation. (i) becomes $12x'^2-10x'y'+2y'^2=0$

5. The clirection cosine of axes remaining the same, choose a new origin such that the new co-ordinates of the pair of points whose old co-ordinate are (5, –13) and (–3,11) may be of the forms (h, k) and (–h, –k). [N.U.H-1997]

Solution : Let the co-ordinate of new origin be (α, β).

Given that the old co-ordinates of the two points are (5, –13) and (–3, 11) whose new co-ordinates are (h, k) and (–h, –k).

So, $(5-\alpha, -13-\beta)=(h, k)$ and

$(-3-\alpha, 11-\beta)=(-h, k)$

Equating corresponding Components,

$5-\alpha=h,\quad -13-\beta=-k$

$-3-\alpha=-h\quad 11-\beta=k$

$\therefore 2-2\alpha=0\quad -2-2\beta=0$

$\therefore \alpha=1\quad \therefore \beta=-1$

Thus, The co-ordinate of new origin $(1, -1)$

6. Transfer the equation $11x^2 - 4xy + 14y^2 - 58x - 44y + 126 = 0$ to new axes of X and Y whose equation are $x - 2y + 1 = 0$ and $2x + y - 8 = 0$ respectively. [N.U.H-2000; 2008]

Solution : Given that

$11x^2 - 4xy + 14y^2 - 58x - 44y + 126 = 0$ ... ... (1)

Let $x - 2y + 1 = 0$ and $2x + y - 8 = 0$ be The new axes of X and Y.

New co-ordinate of (x, y) be (x_1, y_1).

So, $x_1 = \frac{2x + y - 8}{\sqrt{2^2 + 1^2}} = \frac{2x + y - 8}{\sqrt{5}}$.

$\Rightarrow 2x + y - 8 = \sqrt{5}x\text{-}1$ ... ... (2)

$y_1 = \frac{x - 2y + 1}{\sqrt{1^2 + 2^2}} = \frac{x - 2y + 1}{\sqrt{5}}$ and

$\Rightarrow x - 2y + 1 = \sqrt{5}y_1$ ... ... (3)

From (2) and (3)

$4x + 2y - 16 = 2\sqrt{5}\,x_1$

$x - 2y + 1 = \sqrt{5}y_1$

(+) $5x - 15 = \sqrt{5}y_1 + 2\sqrt{5}x_1$

$5x = 15 + \sqrt{5}y_1 + 2\sqrt{5}x_1$

$x = \frac{2\sqrt{5}x_1 + \sqrt{5}y_1 + 15}{5}$

$= \frac{2x_1 + y_1}{\sqrt{5}} + 3$

From (2) ; $y = -2x + \sqrt{5}x_1 + 8$

$= \frac{-4x_1 - 2y_1}{\sqrt{5}} - 6 + \sqrt{5}x_1 + 8$

$= \frac{-4x_1 - 2y_1 + 5x_1}{\sqrt{5}} + 2$

$= \frac{x_1 - 2y_1}{\sqrt{5}} + 2$

$\therefore \quad x = h + 3$

Where $h = \frac{2x_1 + y_2}{\sqrt{5}} + 3$

$y = k + 2\,k = \frac{x_1 - 2y_1}{\sqrt{5}} + 2$

Putting the values of x and y in (1)

we get

$11(h+3)^2 - 4(h+3)(k+2) + 14(k+2)^2 - 58(h+3) - 44(k+2) + 126 = 0$

$\Rightarrow 11(h^2 + 6h + 9) - 4(hk + 2h + 3k + 6) + 14(k^2 + 4k + 4) - 58h - 174 - 44k - 88 + 126 = 0$

$\Rightarrow 11h^2 - 4hk + 14k^2 - 5 = 0$

$$\Rightarrow 11\left(\frac{2x_1 + y_1}{\sqrt{5}}\right)^2 - 4\left(\frac{2x_1 + y_1}{\sqrt{5}}\right)\left(\frac{x_1 - 2y_1}{\sqrt{5}}\right) + 14\left(\frac{x_1 - 2y_1}{\sqrt{5}}\right)^2 - 5 = 0$$

$\Rightarrow 11(4x_1^2 + 4x_1y_1 + y_1^2) - 4(2x_1^2 - 4x_1y_1 - 2y_1^2) + 14(x_1^2 - 4x_1y_1 + 4y_1^2) - 25 = 0$

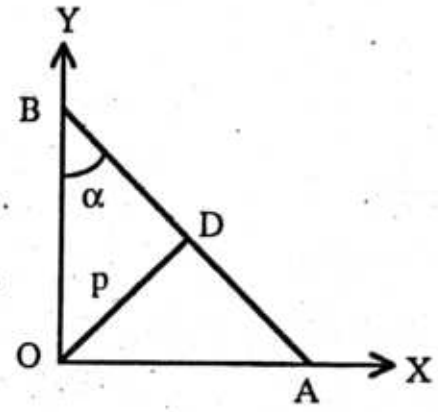
$= 50x_1^2 - 75y_1^2 = 25$

$= 2x_1^2 + 3y_1^2 = 1$ which is required equation.

7. Reduce the equation of a straight line in the polar form $r\cos(\theta - \alpha) = p$ [N.U.H. 2001]

Solution :

Let the line AB makes an angle α with Y-axis and OD= p. Form the figure, equation of AB is $\frac{x}{OA} + \frac{y}{OB} = 1$ ... ... (1)



Sinec OD perpendicular to AB and OD = P

From the Δ OBD ; $\sin\alpha = \frac{OD}{OB} = \frac{P}{OB}$

$\therefore OB = \frac{P}{\sin\alpha}$

Again, from the Δ AOD,

$\sin(90° - \alpha) = \frac{OD}{OA} = \frac{P}{OA}$

$\therefore OA = \frac{P}{\cos\alpha}$

Putting the values of OA and OB in (1) we get $\frac{x}{\frac{P}{\cos\alpha}} + \frac{y}{\frac{P}{\sin\alpha}} = 1$

$\Rightarrow x\cos\alpha + y\sin\alpha = p$ ... ... (2)

Relation between cartesian and polar Co-ordinates we can write,

$x = r\cos\theta, \quad y = r\sin\theta$

Equation (2) becomes

$r\cos\theta\cos\alpha + r\sin\theta\sin\alpha = p$

$\Rightarrow r\cos(\theta - \alpha) = p$ which is required polar equation.

8. The equation $3x^2 + 2xy + 3y^2 - 18x - 22y + 50 = 0$ is transformed to $4x^2 + 2y^2 = 1$ when refered to rectangular axes through the point (2, 3). Find the inclination of the later axes to the former. [N.U.H. 2004]

Solution : Given that

$3x^2 + 2xy + 3y^2 - 18x - 22y + 50 = 0$ ... ... (1)

Transfer the origin to a point (2, 3). Then putting $x = x_1 + 2$, $y = y_1 + 3$ in (1), we get (1) $\Rightarrow 3(x_1 + 2)^2 + 2(x_1 + 2)(y_1 + 3) + 3(y_1 + 3)^2 - 18(x_1 + 2) - 22(y_1 + 3) + 50 = 0$

$\Rightarrow 3(x_1^2 + 4x_1 + 4) + 2(x_1y_1 + 3x_1 + 2y_1 + 6) + 3(y_1^2 + 6y_1 + 9) - 18x_1 - 36 - 22y_1 - 66 + 50 = 0$

$= 3x_1^2 + 2x_1y_1 + 3y_1^2 = 1$

Removing suffixes, we get

$3x^2 + 2xy + 3y^2 = 1$ ... ... (2)

Let the axes rotated be an angle θ, Then we know, $\theta = \frac{1}{2}\tan^{-1}\frac{2h}{a-b}$; where $a = 3, b = 3, h = 1$

$\Rightarrow \theta = \frac{1}{2}\tan^{-1}\frac{2.1}{3-3}$

$\Rightarrow \theta = \frac{1}{2}\tan^{-1}\infty$

$\Rightarrow \theta = \frac{1}{2}.\frac{\pi}{2} = \pi/4 = 45°$

We know that the axes rotated be an angle θ, then

$x = x_1\cos\theta - y_1\sin\theta$, $y = x_1\sin\theta + y_1\cos\theta$

$x = x_1\cos45° - y_1\sin45°$ $y = x_1\sin45° + y_1\cos45°$

$x = \frac{x_1 - y_1}{\sqrt{2}}$ $x = \frac{x_1 + y_1}{\sqrt{2}}$

Putting the values of x and y in (2) we get

$$(2) \Rightarrow 3\left(\frac{x_1 - y_1}{\sqrt{2}}\right)^2 + 2\left(\frac{x_1 - y_1}{\sqrt{2}}\right)\left(\frac{x_1 + y_1}{\sqrt{2}}\right) + 3\left(\frac{x1 + y1}{\sqrt{2}}\right)^2 = 1$$

$\Rightarrow 3(x_1 - y_1)^2 + 2(x_1^2 - y_1^2) + 3(x_1 + y_1)^2 = 2$

$\Rightarrow 3(x_1^2 - 2x_1y_1 + y_1^2) + 2x_1^2 - 2y_1^2 + 3(x_1^2 + 2x_1y_1 + y_1^2) = 2$

$\Rightarrow 8x_1^2 + 4y_1^2 = 2$

$\Rightarrow 4x_1^2 + 2y_1^2 = 1$

Remoing suffixes,

$\therefore 4x^2 + 2y^2 = 1$

Which is the required transformed equation

9. If the direction of axes is turned through an angle 30° and the origin remains unchanged then find the transformation equation of $x^2 + 2\sqrt{3}xy - y^2 - 2a^2 = 0$. Idntify and sketch it. [N.U.H. 2009; 1999]

Solution : given that,

$x^2 + 2\sqrt{3}xy - y^2 - 2a^2 = 0$ ... ... (1)

Since the axes rotated be an angle 30° and origin be unchanged.

$$\text{So, } x = x_1\cos 30° - y_1\sin 30° = \frac{\sqrt{3}x_1 - y_1}{\sqrt{2}}$$

$$y = x_1\sin 30° + y_1\cos 30° = \frac{x_1 + \sqrt{3}y_1}{\sqrt{2}}$$

Putting this valus in (1) we get

$$(1) = \left(\frac{\sqrt{3}x_1 - y_1}{\sqrt{2}}\right)^2 + 2\sqrt{3}\left(\frac{\sqrt{3}x_1 - y_1}{\sqrt{2}}\right)\left(\frac{x_1 + \sqrt{3}y_1}{2}\right) - \left(\frac{x_1 + \sqrt{3}y_1}{2}\right)^2 - 2a^2 = 0$$

$$\Rightarrow \frac{1}{4}(3x_1^2 - 2\sqrt{3}x_1y_1 + y_1^2) + \frac{2\sqrt{3}}{4}(\sqrt{3}x_1^2 + 3x_1y_1 - x_1y_1 + \sqrt{3}y_1^2) - \frac{1}{4}(x_1^2 + 2\sqrt{3}x_1y_1 + 3y_1^2) - 2a^2 = 0$$

$$\Rightarrow 3x_1^2 - 2\sqrt{3}x_1y_1 + y_1^2 + 6x_1^2 + 4\sqrt{3}x_1y_1 - 6y_1^2 - x_1^2 - 2\sqrt{3}x_1y_1 - 3y_1^2 - 8a^2 = 0$$

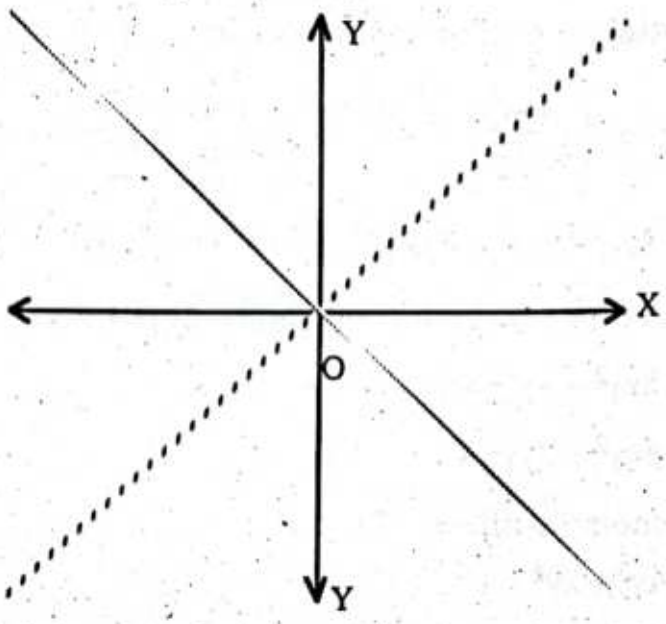
$$\Rightarrow 8x_1^2 - 8y_1^2 - 8a^2 = 0$$

$$= x_1^2 - y_1^2 = a^2$$

Removing suffixes, we get

$x^2 - y^2 = a^2$ which is the equation of rectangular hyperbola.

2nd part : we have. $x^2 - y^2 = a^2$. It is rectangular hyperbola whose asymptotes are $y = \pm x$. ie Asymptotes makes an angles with x – axis are ±45°. Its figure below :



10. Transform the equation $17x^2 + 18xy - 7y^2 - 16x - 32y - 18 = 0$ to one in which there is no term involuing x, y and xy both sets of axes being rectangular. [N.U.H. 2003]

Solution : Given equation

$17x^2 + 18xy - 7y^2 - 16x - 32y - 18 = 0$

... ... (1)

Let transfer the origin to a point (4, k)

Then $x = x_1 + h$, $y = y_1 + k$

Putting these values in (1) we get, $17(x_1 + h)^2 + 18(x_1 + h)(y_1 + k) - 7(y_1 + k)^2 - 16(x_1 + 4h - 32(y_1 + k) - 18 = 0$

$\Rightarrow 17(x_1^2 + 2x_1h + h^2) + 18(x_1y_1 + x_1k + y_1h + hk) - 7(y_1^2 + 2y_1k + k^2) - 16x_1 - 16h - 32y_1 - 32k - 18 = 0$

$\Rightarrow 17x_1^2 + 18x_1y_1 - 74y_1^2 + (34h + 18k - 16)x_1 + (18h - 14x - 32)y_1 + 17h^2 + 18hk - 7k^2 - 16h - 32k - 18 = 0$ ... ... (2)

Since there is no terms of x, y. So equating the coefficient of x_1, y_1 from both sides.

$\therefore 34h + 18k - 16 = 0$

$18h - 14k - 32 = 0$

By the rule of cross multiplication,

$\therefore \frac{h}{-576 - 224} = \frac{k}{-288 + 1088} = \frac{1}{476 - 324}$

$= \frac{h}{-800} = \frac{k}{800} = \frac{1}{-800}$

$\therefore h = 1, \quad k = -1$

Putting the values of h and k in (2) we get $17x_1^2 + 18x_1y_1 - 7y_1^2 + 0 + 0 + 17 + 18(1)(-1) - 7 - 16 - 32(-1) - 18 = 0$

$17x_1^2 + 18x_1y_1 - 7y_1^2 - 10 = 0$

Removing Suffixes,

$\therefore 17x^2 + 18xy - 7y^2 - 10 = 0$ ... ... (3)

If we rotated the axes through an angle θ

Then $\theta = \frac{1}{2}\tan^{-1}\frac{2h}{a - b}$

$\therefore 2\theta = \tan^{-1}\frac{2.9}{17 + 7}$

$\Rightarrow \tan 2\theta = \frac{18}{24} = \frac{3}{4}$

$\therefore \sin 2\theta = \frac{3}{5}$

$\therefore 2\sin\theta\cos\theta = \frac{3}{5}$ $\quad$ $\cos 2\theta = \frac{4}{5}$

$\therefore \sin\theta = \frac{3}{10\cos\theta}$ $\quad$ $2\cos^2\theta - 1 = \frac{4}{5}$

$= \frac{3\sqrt{10}}{10.3} = \frac{1}{\sqrt{10}}$ $\quad$ $\cos^2\theta = \frac{9}{10}$ $\therefore \cos\theta = \frac{3}{\sqrt{10}}$

We know,

$x = x_1\cos\theta - y_1\sin\theta$

$= \frac{3x_1 - y_1}{\sqrt{10}}$

$y = x_1\sin\theta + y_1\cos\theta$

$$= \frac{x_1 + 3y_1}{\sqrt{10}}$$

Putting the values of x, y in (3) we get

$$(3) \Rightarrow 17\left(\frac{3x_1 - y_1}{\sqrt{10}}\right)^2 + 18\left(\frac{3x_1 - y_1}{\sqrt{10}}\right)\left(\frac{x_1 + 3y_1}{\sqrt{10}}\right)$$

$$-7\left(\frac{x_1 + 3y_1}{\sqrt{10}}\right) - 10 = 0$$

$$\Rightarrow 17(9x_1^2 - 6x_1y_1 + y_1^2) + 18(3x_1^2 + 9x_1y_1 - 3y_1^2)$$

$$-7(x_1^2 \Rightarrow 6x_1y_1 + 9y_1^2) - 100 = 0$$

$$\Rightarrow (153 + 54 - 7)\, x_1^2 = (-102 + 144 - 42)\, x_1y_1 + (17 - 54 - 63)\, y_1^2 \Rightarrow 100$$

$$\Rightarrow 200x_1^2 - 100y_1^2 = 100$$

$$\Rightarrow 2x_1^2 - y_1^2 = 1$$

Removing suffixes, we get

$$2x^2 - y^2 = 1$$

Which is the required transformation.

PAIR OF STRAIGHT LINES 5

Art. 36. Homogeneous Quadratic Equations.

Let us consider the equation $ax^2 + 2hxy + by^2 = 0$ ... ... (1)

On multiplying it by a, it may written in the form

$a^2x^2 + 2ahxy + aby^2 = 0$, if $a \neq 0$

or, $a^2x^2 + 2ahxy + h^2y^2 - (h^2 - ab)\, y^2 = 0$

or, $\{ax + hy)^2 - \{y\sqrt{(h^2 - ab)}\}^2 = 0$

or, $(ax + hy) + \sqrt{(h^2 - ab)}\, y\}\ \{ax + hy - (h^2 - ab))y\} = 0$

The equation (1) therefore, represents the two straight lines whose equations are

$ax + hy + y\sqrt{(h^2 - ab)} = 0$ ... ... (2)

and $ax + hy - y\sqrt{(h^2 - ab)} = 0$ ... ... (3)

each of which passes through the origin. The two lines (2), (3) form the locus of the equation (1) for (1) is satisfied by the co-ordinates of all points which satisfy (2) and (3).

These two straight lines are real and different if $h^2 > ab$, real and coincident if $h^2 = ab$, and imaginary if $h^2 < ab$.

In the case when $h^2 = ab$, the straight lines though themselves imaginary intersect in a real point. For the origin lies on the locus given by (1) since the equation (1) is always satisfied by the values $x = 0$ and $y = 0$.

Alternative method

If $b \neq 0$ dividing both sides of the equation (1) by $x^2\, b$ we have

$$\left(\frac{y}{x}\right)^2 + 2\frac{h}{b}\frac{y}{x} + \frac{a}{b} = 0 \quad \ldots \quad \ldots \quad \ldots \quad (4)$$

Let m_1, m_2, be the roots of this quadratic equation in $\frac{x}{y}$.

Sum of the roots = $m_2 + m_2 = -2\,h/b$

and product of the roots = $m_1m_2 = a/b$ ... ... (5)

The Eq. (4) must be equivalent to

$$(\frac{y}{x} - m_1)\ (\frac{y}{x} - m_2) = 0 \quad \ldots \quad \ldots \quad \ldots \quad (6)$$

The two lines represented by (4) *i.e.* (1) are given

$\frac{y}{x} - m_1 = 0$, and $\frac{y}{x} - m_2 = 0$,

i.e. $y - m_1 x = 0$ and $y - m_2 x = 0$ which pass through the origin.

Thus the homogeneous quadratic equation $ax^2 + 2hxy + by^2 = 0$ *always represents a pair of straight lines, real or imaginary, through the origin.*

Art. 37. Prove that a homogeneous equation of the *n* th degree represents *n* straight lines, real or, imaginary, which all pass through the origin.

Consider the homogeneous equation

$$y^n + a_1 x y^{n-1} + a_2 x^2 y^{n-2} + a_3 x^3 y^{n-3} + \ldots + a_r x^r y^{n-r} + a_n x^n = 0 \quad \ldots \quad \ldots \quad \ldots \quad (1)$$

This can be written as (dividing by x^n)

$$(\frac{y}{x})^n + a_1 (\frac{y}{x})^{n-1} + a_2 (\frac{y}{x})^{n-2} + a_3 (\frac{y}{x})^{n-3} + \ldots + a_r (\frac{y}{x})^{n-r} + \ldots + a_n = 0 \quad \ldots \quad (2)$$

Since this is an equation of the n th degree in y/x it must have n roots. Let the roots of this equation be $m_1, m_2, m_3, \ldots m_r, \ldots m_n$.

Then the given equation (1) or (2) must be the same as

$$(\frac{y}{x} - m_1)(\frac{y}{x} - m_2) \ldots (\frac{y}{x} - m_r) \ldots (\frac{y}{x} - m_n) = 0$$

or, $(y - m_1 x)(y - m_2 x) \ldots (y - m_r x) \ldots (y - m_n x) = 0$

This equation is satisfied by the points which satisfy the separate equations.

$y - m_1 x = 0,\ y - m_2 x = 0, \ldots\ y - m_r x = 0, \ \ldots\ y - m_n x = 0$

which all pass through the origin. Conversely, the co-ordinates of all the points which satisfy these n equation satisfy equations (1) also. Hence the theorem.

Note : 1. An equation of the type

$$a_0 y^n + a_1 y^{n-1} x + a_2 y^{n-2} x^2 \ldots + a_r y^{n-r} x^r \ \ldots \ldots + a_n x^n = 0$$

in which the sum of the powers of x and y in every term is the same (say n) is called a Homogeneous equation (of degree n)

Note. 2. Since the given equation contains real co-efficients, the roots, if imaginary, must occur in pair.

Note. 3. The relation between the roots and the co-efficients can be determined by the symmetric functions of the roots (see author's Higher Algebra. Art. 53 and 54)

Note. 4. If two or more of the roots are equal the corresponding straight lines are coincident.

Art. 38. Angle between the lines represented by the equation. [N. U. H. 1998]

$$ax^2 + 2hxy + by^2 = 0 \quad \ldots \quad \ldots \quad \ldots \quad (1)$$

The axes are assumed to be rectangular. Let the lines represented by (1) be $y - m_1 x = 0$ and $y - m_2 x = 0$

So that (1) and $(y - m_1 x)(y - m_2 x)$ are the same.

$\therefore m_1 + m_2 = -2\frac{h}{b}$ and $m_1 m_2 = \frac{a}{b}$ by Art 36 ... ... (2)

If θ be the angle between the straight lines

$y - m_1 x = 0$ and $y - m_2 x = 0$, then by Art. 25.

$$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2} = \frac{\sqrt{[(m_1 + m_2)^2 - 4m_1 m_2]}}{1 + m_1 m_2}$$

$$= \sqrt{[(-2h/b)^2 - 4a/b]}\,(1 + a/b) = 2\sqrt{(h^2 - ab)} / (a + b) \quad \ldots \quad (2)$$

$$\therefore \tan\theta = \frac{2\sqrt{(h^2 - ab)}}{a + b} \quad \ldots \quad \ldots \quad \ldots \quad (3)$$

where θ the angle between the lines represented by the equation $ax^2 + 2hxy + by^2 = 0$

Cor. 1. Condition of Perpendicularity.

If the straight lines are perpendicular to each other then $\theta = 90°$ hence $\tan\theta = \infty$ then from (3)

$a + b = 0$

i. e. if the sum of the co-efficients of x^2 and y^2 is zero, then the straight lines will be perpendicular to each other.

Cor. 2. Condition of parallelism or Coincidence.

If the lines of $ax^2 + 2hxy + by^2 = 0$ be coincident, $\theta = 0$

From (3) $\tan\theta = 0$

i, e. $h^2 = ab$

which is the required condition for coincidence.

Here lines cannot be parallel as both of the lines pass through the origin, these are coincident straight lines.

Cor. 3. Two lines represented by $ax^2 + 2hxy + by^2 = 0$ will be real if $h^2 > ab$

Cor. 4. Two lines of above equation will be imaginary if $h^2 < ab$.

Art. 39. The bisectors of the angles between the lines represented by $ax^2 + 2hxy + by^2 = 0$

[N. U. H. 2008, 1998]

Let the lines $y - m_1 x = 0$ and $y - m_2 x = 0$ represented by

$ax^2 + 2hxy + by^2 = 0$ ... ... (1) have slopes m_1 and m_2 and make angles θ_1 and θ_2 with OX. Let one of the bisectors make an angle θ with OX. from the diagram we have

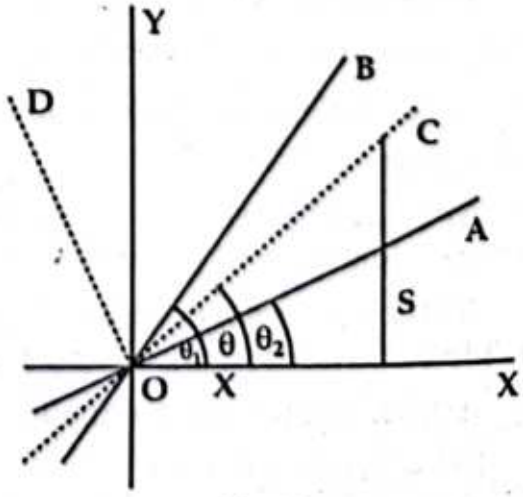


Fig. 12

$$\theta = \theta_1 + \frac{1}{2}(\theta_2 - \theta) \text{ or, } \frac{1}{2}\pi + \theta_1 + \frac{1}{2}(\theta_2 - \theta_1) = \frac{1}{2}(\theta_1 + \theta_2)$$

$$\text{or, } \frac{1}{2}\pi + \frac{1}{2}(\theta_1 + \theta_2)$$

$$\therefore 2\theta = \theta_1 + \theta_2$$

$$\text{or, } \pi + (\theta_1 + \theta_2) \quad \ldots \quad \ldots \quad \ldots \quad (2)$$

In either case

$$\tan 2\theta = \tan(\theta_1 + \theta_2) = \frac{\tan\theta_1 + \tan\theta_2}{1 - \tan\theta_1 \tan\theta_2} = \frac{m_1 + m_2}{1 - m_1 m_2}$$

or, $$\frac{2\tan\theta}{1 - \tan^2\theta} = \frac{m_1 + m_2}{1 - m_1 m_2} \quad \ldots \quad \ldots \quad \ldots \quad (3)$$

From Eq. (1), we have

$$b(y - m_1 x)(y - m_2 x) = by^2 + 2hxy + ax^2$$

$$\therefore \; m_1 + m_2 = -2\frac{a}{b}, \; m_1 m_2 = \frac{a}{b} \quad \ldots \quad \ldots \quad \ldots \quad (4)$$

Also if C (x, y) be any point on one of the bisectors, (see fig)

$$\tan\theta = y/x \quad \ldots \quad \ldots \quad \ldots \quad (5)$$

From (3) by (4) and (5), we have

$$\frac{2xy}{x^2 - y^2} = \frac{2h}{a - b}$$

i. e. the required equation of the bisectors is

$$\frac{x^2 - y^2}{a - b} = \frac{xy}{h} \quad \ldots \quad \ldots \quad \ldots \quad (6)$$

Note. 1. In (2), first value θ is the angle made by the internal bisector OC with OX and the second value of θ is the angle made by the external bisector OD with OX.

Alternative Method :

Let the given equation $ax^2 + 2hxy + by^2 = 0$ represent the line

$y - m_1 x = 0$ and $y - m_2 x = 0$,

$$\therefore \; m_1 + m_2 = -2\frac{h}{b}, \; m_1 m_2 = \frac{a}{b} \quad \ldots \quad \ldots \quad \ldots \quad (1)$$

The equations of the required bisectors are

$$\frac{y - m_1 x}{\sqrt{(1 + m_1^2)}} = \pm\frac{y - m_2 x}{\sqrt{(1 + m_2^2)}} \quad \ldots \quad \ldots \quad \ldots \quad (2)$$

Squaring, we have

$$(y - m_1 x)^2 (1 + m_2^2) = (y - m_2 x)^2 (1 + m_1^2)$$

or, $$(m_1^2 - m_2^2)(x^2 - y^2) = 2xy\,\{(m_1 - m_2) - m_1 m_2 (m_1 - m_2)\}$$

or, $$(m_1 + m_2)(m_1 - m_2)(x^2 - y^2) = 2xy\,(m_1 - m_2)(1 - m_1 m_2)$$

or, $$(m_1 + m_2)(x^2 - y^2) = 2xy\,(1 - m_1 m_2)$$

or, $$-2\frac{h}{b}(x^2 - y^2) = 2xy\frac{(b - a)}{b}$$

or, $$h(x^2 - y^2) = xy(a - b)$$

or, $$\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$$

GENERAL EQUATION OF SECOND DEGREE

Art. 40. Find the condition that the general equation of the second degree

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ ... ... (1)

may represent a pair of straight lines. [N. U. H. 1997, 2002, 2007, 2009]

If we transfer the origin to a point (α, β), the point of intersection of two straight lines and keep the direction of the axes unaltered, the eq. (1) reduces to

$a(x+\alpha)^2 + 2h(x+\alpha)(y+\beta) + b(y+\beta)^2 + 2g(x+\alpha) + 2f(y+\beta) + c = 0$ by Art. 32 (Rule)

or, $ax^2 + 2hxy + by^2 + 2(a\alpha + h\beta + g)x + 2(h\alpha + b\beta + f)y + a\alpha^2 + 2h\alpha\beta + b\beta^2 + 2g\alpha + 2f\beta + c = 0$

... ... (2)

The equation (2) may represent a pair of straight lines, if it is reduced to a homogeneous equation in x and y. This is possible if the co-effcients of x and y and the constant terms are saperately zero. *i. e.*

$a\alpha + h\beta + g = 0$ ... ... (3)

$h\alpha + b\beta + f = 0$ ... ... (4)

and $a\alpha^2 + 2h\alpha\beta + b\beta^2 + 2g\alpha + 2f\beta + c = 0$ ... ... (5)

The relation (5) may be written as

$\alpha(a\alpha + h\beta + g) + \beta(h\alpha + b\beta + f) + g\alpha + f\beta + c = 0$ ... ... (6)

By (3) and (4), the relation (6) becomes $g\alpha + f\beta + c = 0$ ... ... (7)

Now if we eliminate α and β from (3), (4) and (7), the required condition is

$$\Delta \equiv \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0 \quad \ldots \quad \ldots \quad \ldots \quad (8)$$

or, $\Delta \equiv abc + 2fgh - af^2 - bg^2 - ch^2 = 0$ ... ... (9)

Alternative method — By Calculus

Suppose $F(x, y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c$

$\equiv (lx + my + n)(l'x + m'y + n')$ ... ... (1)

This relation being true for all values of x and y, it must admit of partial differentiation w. r. to each of the variables x and y. So we have, by taking partial differentaiation of (1)

$\frac{\delta F}{\delta x} = 2(ax + hy + g) = l'(lx + my + n) + l(l'x + m'y + n')$ ... ... (2)

and $\frac{\delta F}{\delta y} = 2(hx + by + f) = m'(lx + my + n) + m(l'x + m'y + n')$ ... ... (3)

Let (α, β) be the point of intersection of the lines represented by the general equation (1)

$\therefore l\alpha + m\beta + n = 0, \quad l'\alpha + m'\beta + n' = 0$

Putting α, β respectively for x ; y in (2) and (3) we have

$a\alpha + h\beta + n = 0$ ... ... (4)

$h\alpha + b\beta + f = 0$ ... ... (5)

Again puttin (α, β) in the left hand side of (1) we have

$a\alpha^2 + 2h\alpha\beta + b\beta^2 + 2g\alpha + 2f\beta + c = 0$

or, $\alpha(a\alpha + h\beta + g) + \beta(h\alpha + b\beta + f) + (g\alpha + f\beta + c) = 0$

By (4) and (5)

$g\alpha + f\beta + c = 0$ ... ... (6)

Eliminating α, β from (4), (5) and (6) we have

$$\Delta \equiv \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

or, $\Delta \equiv abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

which is the required condition. It is both necessary and sufficient.

Cor. 1. Solving (4) and (5), we have

$$\alpha = \frac{hf - bg}{ab - h^2} = \frac{G}{C} \text{ and } \beta = \frac{gh - af}{ab - h^2} = \frac{F}{C}$$

where G, F, C are the co-factors of g, f, c, in

Note : **1.** From the above discussions we see that if $F(x, y) = 0$ be the equation of two lines, the point of intersection (α, β) would be obtained by solving the equations.

$$\frac{\delta F(\alpha, \beta)}{\delta \alpha} = 0, \frac{\delta F(\alpha, \beta)}{\delta \beta} = 0$$

Cor. 2. If $a = 0$ and $b \neq 0$, we solve it for y and proceed as before. We thus find the condition $2f\ gh - bg^2 - ch^2 = 0$

Cor. 3. If $a = 0$, $b = 0$ and $h \neq 0$, then the general equations become $2hxy + 2gx + 2fy + c = 0$

or, $xy + (\frac{g}{h})x + (\frac{c}{h})y + \frac{c}{2h} = 0$ $\therefore$ Dividing by $2h$

or, $(x + \frac{f}{h})\ (y + \frac{g}{h}) = (\frac{fg}{h^2}) - \frac{c}{2h}$

or, $(hx + f)(hy + g) = fg - \frac{1}{2}ch$

and the condition that it may represent a pair of lines is

$fg - \frac{1}{2}ch = 0$ or, $2fg - ch = 0$

Note : **2** There is another method of finding the condition that the general second degree equation may represent a pair of straight lines. This process will be demonstrated by a worked out example. See worked sum No. 1.

Art. 41. (A) Angle between the lines given by

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

Let $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = (lx + my + n)\ (l'x + m'y + n')$ ... (1)

$ll' = a, mm' = b, nn' = c$

$lm' + l'm = 2h, mn' + m'n = 2f, nl' + n'l = 2g$ ... ... (2)

$lx + my + n = 0$ and $l'x + m'y + n' = 0$

or, $y = -\frac{l}{m}x - \frac{n}{m}$ and $y = -\frac{l'}{m'}x - \frac{n'}{m'}$

If θ be the angle between the lines then

$$\tan\theta = \frac{-l/m - (-l'/m')}{1 + (-\frac{l}{m})(-\frac{l'}{m'})}$$

$$= \frac{l'm - lm'}{l'l + mm'}$$

$$\frac{\sqrt{(lm' + l'm)^2 - 4\,ll'mm'}}{ll' + mm'} = \frac{\sqrt{2h^2 - ab}}{a + b} \quad \ldots \quad \ldots \quad \text{by (2)}$$

On account of relation (2), we have $(lx + my)(l'x + m'y) = ll'x^2 + 2(lm' + l'm)xy + mm'y^2 = ax^2 + 2hxy + by^2$

hence it follows that the lines represented by

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

are parallel to the lines given by $ax^2 + 2hxy + by^2 = 0$

(B) Condition for perpendicularity of lines [N. U. H. 2007]

The lines will be perpendicular of $\theta = 90°$ *i. e.* $\tan\theta = \infty$

From (3), we have $ll' + mm' = 0$. *i. e.* if $a + b = 0$

This also follows from (4). Hence the given equation represents a pair of perpendicular lines, if $\Delta = 0$ and $a + b = 0$

(C) Condition for parallelism of the lines.

The lines in (4) will be parallel if $\frac{l}{l'} = \frac{m}{m'}$ or, $lm' = l'm$

or, $lm' - l'm = 0$

or, $(lm' + l'm)^2 - 4ll'mm' = 0$ or $h^2 - ab = 0$ by (4).

This also follows from (4). Hence the given equation represents a pair of parallel lines is $\Delta = 0$ and $h^2 = ab$

(D) Condition for coincidence of the lines. [N. U. H. 2007]

The line in (1) will be concident if $\frac{l}{l'} = \frac{m}{m'} = \frac{n}{n'}$

i. e. $lm' - l'm = 0$, $mn' - m'n = 0$, $n'l - nl' = 0$

or, $(lm' + l'm)^2 - 4ll'mm' = 0$, $(mn' + m'n)^2 - 4mm'nn' = 0$, $(nl' + n'l)^2 - 4nn'll' = 0$

or, $h^2 - ab = 0$, $f^2 - bc = 0$, $g^2 - ac = 0$ by (2)

In such a case, the given equation becomes a perfect square. Hence the given equation will represent a pair of coincident lines if

$\Delta = 0, f^2 - bc = 0, g^2 - ac = 0, h^2 - ab = 0$

which are equavalent to $gh = af$, $hf = bg$, $fg = ch$

(E) Bisectors of the angles between the lines.

Let (α, β) be the points of intersection of the two lines represented by the equation.

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

If we transfer the origin to (α, β) without changing the direction of axes, the first degree terms and the term independent of x and y will vanish. Then the transformed equation will be a homogeneous equation and will be a form $ax^2 + 2hxy + by^2 = 0$

The equations of the bisectors of the angle between the lines with (α, β) as origin by

$$\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$$

If we go back to our old axes, the above equations, of the bisectors are given by

$$\frac{(x-\alpha)^2 - (y-\beta)^2}{(a-b)} = \frac{(x-\alpha)(y-\beta)}{h}$$

which is the required (See Art. 39 Rule) equation of the bisectors of the angles between the given lines.

It should be noted that with reference to the new origin the co-ordinates of the old origin are $(-\alpha, -\beta)$

Art. 42. Lines Joining the origin to the intersection of a curve and a line.

Let the second degree equation representing the curve be

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \quad \ldots \quad \ldots \quad \ldots \quad (1)$$

and the first degree equation representing the line be

$$lx + my = n \ldots \ldots \text{ or } \frac{lx}{n} + \frac{my}{n} = 1 \quad \ldots \quad \ldots \quad \ldots \quad (2)$$

Let the straight line meet the curve (1) at P and Q. We are to find the equation of the pair of lines OP and OQ.

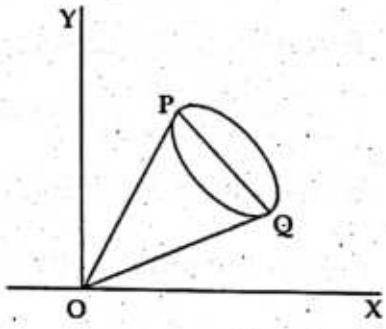


Fig. 13

Now making (1) homogeneous with the help of (3) we get

$$ax^2 + 2hxy + by^2 + 2(gx + fy)\left(\frac{lx + my}{n}\right) + c\left(\frac{lx + my}{n}\right)^2 = 0$$

Which, on simplification, reduces to an equation of the form

$$Ax^2 + 2Hxy + By^2 = 0 \quad \ldots \quad \ldots \quad \ldots \quad (4)$$

This is a Homogeneous equation of 2nd degree. It represents a pair of straight lines through the origin. Moreover the co-ordinates of P and Q satisfy (4), because the co-ordinates of P and Q satisfy (1) and (3). Therefore, (4) must represent OP and OQ and is our required equation.

Ex. 1. Prove that the equation $x^2 + 6xy + 9y^2 + 4x + 12y - 5 = 0$ represents a pair of parallel lines. [R. U. 1997; D. U. 1986]

$$x^2 + 6xy + 9y^2 + 4x + 12y - 5 = 0$$

or, $x^2 + x(6y + 4) + (9y^2 + 12y - 5) = 0$ quadratic in x

$$x = \frac{-(6y+4) \pm \sqrt{(6y+4)^2 - 4 \cdot 1 \cdot (9y^2 + 12y - 5)}}{2}$$

$$= (-6y - 4 \pm 6)/2 = -3y - 2 \pm 3$$

$$x + 3y + 5 = 0 \text{ and } x + 3y - 1 = 0$$

Hence the given equation represents two straight line $x + 3y + 5 = 0$ and $x + 3y - 1 = 0$. As these two lines differ only in constant terms, so these form a pair of parallel straight lines. See Art. 26 (1).

Ex 2. Show that the angle between one of the lines given by $ax^2 + 2hxy + by^2 = 0$ and one of the lines.

$ax^2 + 2xy + by^2 + \lambda(x^2 + y^2) = 0$ is equal to the angles between the other two lines of the system.

The equation of the straight lines bisecting the angles between

$ax^2 + 2hxy + by^2 = 0$ is $\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$

Again, the equation of the straight lines bisecting the angles between $ax^2 + 2hxy + by^2 + \lambda(x^2 + y^2) = 0$

or, $(a + \lambda)x^2 + 2hxy + (a + \lambda)y^2 = 0$ is

$\frac{x^2 - y^2}{(a + \lambda) - (b + \lambda)} = \frac{xy}{h}$ or, $\frac{(x^2 - y^2)}{a - b} = \frac{xy}{h}$

which is the same as the equation of bisectors for the first pair. Hence the result.

Ex. 3. If the pair of straight lines $x^2 - 2axy - y^2 = 0$ and $x^2 - 2bxy - y^2 = 0$ be such that each pair bisects the angle between the other pair, prove that $ab = -1$.

[D. U. 1953, 57, 61, R. U. 1964, 79]

Equation to the bisector of first pair is

$\frac{x^2 - y^2}{1 - (-1)} = \frac{xy}{-a}$ or, $x^2 - y^2 = \frac{2xy}{-a}$ ... ... (1)

By the given condition second pair $x^2 - y^2 = 2bxy$ ... ... (2)

bisects the angle between the first pair. Hence comparing we have $\frac{1}{-a} = b$

or, $ab = -1$ Proved.

Ex. 4 Prove that two of the lines represented by the equation $ax^4 + bx^3y + cx^2y^2 + dxy^3 + ey^4 = 0$ will be at right angles if

$(b + d)(ad + be) + (a - e)^2(a + e + c) = 0$

Let $ax^4 + bx^3y + cx^2y^2 + dxy^3 + ey^4 = (x^2 + kxy - y^2)(ax^2 + k'xy - ey^2) = 0$

$= ax^4 + (k' + ak)x^3y + (-ek - k')xy^3 + (kk' - a - e)x^2y^2 + ey^4$

Now compare the co-efficients of x^4, x^3y, xy^3, x^2y^2 and y^4 from both sides, then

$k' + ak = b$, ... (1)

$-ek - k' = d$, ... (2)

$-a - e + kk' = c$ ... ... (3)

Find the value of k and k' from 1st and 3rd equations.

$k = (b + d)/(a - e)$, $k' = -(ad + bc)/(a - e)$

Therefore from (3), we have $kk' = a + e + c$

$$\frac{b+d}{a-e} \times \frac{ad+be}{a-e} = a+e+c$$

or, $(b+d)(ad+be) + (a-e)^2(a+e+c)$ proved.

Ex. 5. Prove the equation $y^3 - x^3 + 3xy(y-x) = 0$. represents three straight lines equally inclined to one another. [C. U. (Hons) 1987]

Changing the equation $y^3 - x^3 + 3xy(y-x) = 0$

in polar co-ordinates ; $x = r\cos\theta$, $y = r\sin\theta$, we have $r^3\sin^3\theta - r^3\cos^3\theta + 3r^3\sin\theta.\cos\theta(\sin\theta - \cos\theta) = 0$

or, $\sin^3\theta - \cos^3\theta + 3\sin\theta\cos\theta(\sin\theta - \cos\theta) = 0$

or, $(\sin\theta - \cos\theta)\{\sin^2\theta + \cos^2\theta + 4\sin\theta\cos\theta\} = 0$

or, $(\tan\theta - 1)(\tan^2\theta + 4\tan\theta + 1) = 0$

Either $(\tan\theta - 1 = 0$ *i. e.* $\theta = 45°$. say $\theta_1 = 45°$ ... ... (1)

Again $\tan^2\theta + 4\tan\theta + 1 = 0$

or, $\tan\theta = \{-4 \pm \sqrt{(16-4)}\}/2 = -2 \pm \sqrt{3}$

Let $\tan\theta_2 = -2 + (\sqrt{3}) = \tan 165°$

$\therefore \theta_2 = 165°$

$\tan\theta_3 = -2 - \sqrt{3} = -\tan 75° = \tan(-75°) = \tan(360° - 75°) = \tan 285°$

$\therefore \theta_3 = 285°$

Now $\theta_2 - \theta_1 = 165° - 45° = 120°$

$\theta_3 - \theta_2 = 285° - 165° = 120°$

$\theta_3 - \theta_1 = 285° - 45° = 360° - 75° - 45° = -120°$

Hence the straight lines are inclined to each other at an angle of 120°. here RO is produced to OR'.

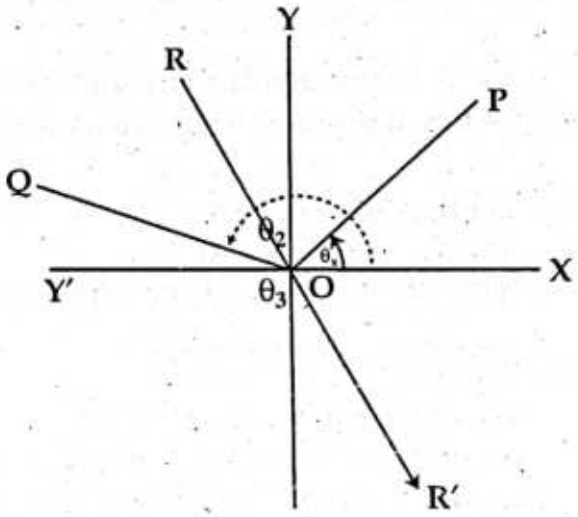


Fig : 14

Ex. 6. Show that the equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents two parallel lines if $a : h = h : b = g : f$. Also show that the distance between them is $2\sqrt{\left\{\frac{(g^2-ac)}{a(a+b)}\right\}}$

[N. U. H. 2006]

We know the condition that the general equation represents two straight lines is $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$ and it will represent two parallel straight lines if in addition, $ab - h^2 = 0$

Now $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

or, $h^2c + 2fgh - af^2 - bg^2 - ch^2 = 0$, Since $ab = h^2$

or, $bg^2 - 2fgh + af^2 = 0$

or, $bg^2 - 2fg\sqrt{(ab)} + af^2 = 0 \quad (\therefore h^2 = ab)$

or, $(\sqrt{bg} - \sqrt{af})^2 = 0$ or, $\sqrt{bg} = \sqrt{af}$

or, $\frac{g}{f} = \sqrt{(\frac{a}{b})} = \sqrt{(\frac{ab}{b^2})} = h/b \quad \therefore h^2 = ab.$ or, $\frac{a}{h} = \frac{h}{b}$

$\therefore \frac{a}{h} = \frac{h}{b} = \frac{g}{f}$ which are the equired conditions

Let AB and CD be the lines $lx + my + n = 0$, $lx + my + n_1 = 0$ respectively. Take any point P (x_1, y_1) on AB. Draw PM perpendicular to CD.

$l^2 = a, m^2 = bn_1 = c, l(n + n_1) = 2g.$

$\therefore lx_1 + my_1 + n_1 = 0 \quad \ldots \quad \ldots \quad \ldots \quad (1)$

$\therefore$ PM $= \frac{lx_1 + my_1 + n_1}{\sqrt{(l^2 + m^2)}} = \frac{n_1 - n}{\sqrt{(l^2 + m^2)}}$ from (1)

$= \frac{\sqrt{\{(n + n_1)^2 - 4nn_1\}}}{\sqrt{(l^2 + m^2)}} = \frac{\sqrt{(4g^2/l^2 - 4c)}}{\sqrt{(a + b)}} = \frac{\sqrt{(4g^2/a - 4c)}}{\sqrt{(a + b)}}$

$= \frac{2\sqrt{\{(g^2 - ac)}}{\sqrt{\{a(a + b)}} = 2\sqrt{\left(\frac{g^2 - ac}{a(a + b)}\right)}$

Ex. 7. Prove that the straight lines represented by the equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ will be equidistant from the origin, if $f^4 - g^4 = (bf^2 - ag^2)$

[D. U. 1956, C. U. (Hons) 1977, C. U. 1981]

Let the lines represented by the given equation be $lx + my + n = 0$

and $l'x + m'y + n' = 0$

Since they are equadistant from the origin.

$\frac{n}{\sqrt{(l^2 + m^2)}} = \frac{n'}{\sqrt{(l'^2 + m'^2)}}$

or, $n^2(l'^2 + m'^2) = n'^2(l^2 + m^2)$

or, $n^2l'^2 - n'^2l^2 = n'^2m^2 - n^2m'^2$

or, $(nl' + n'l)(nl' - n'l) = (nm' + n'm)(n'm - nm')$

or, $(nl' + n'l)^2 \{(nl' + n'l)^2 - 4nn'll'\} = (nm' + n'm)\}^2$

$\{(n'm + nm')^2 - 4nn'mm'\} \quad \ldots \quad \ldots \quad \ldots \quad (2)$

Let $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = (lx + my + n)(l'x + m'y + n')$ comparing the co-efficients we have

$ll' = a, mm' = b, nn' = c, mn' + m'n = 2f, nl' + n'l = 2g, lm' + l'm = 2h$

Putting the values in (2) we have

$g^2(g^2 - ac) = f^2(f^2 - bc)$ or, $f^4 - g^4 = c(bf^2 - ag^2)$

Ex. 8. Show that the lines joining the origin to the intersection of $7x^2 + 8x^2 - 7x^2 + 6x - 12y = 0$ and $2x + y - 1 = 0$ are at right angles.

Making the first equation homogeneus with the help of

$2x + y = 1$, we have

$7x^2 + 8xy - 7x^2 + (6x - 12y)(2x + y) = 0$ or, $19x^2 - 10xy - 19y^2 = 0$ which is the equation of the required straight lines.

Since the sum of the co-efficients of x^2 and y^2 is equal to zero, the straight lines are at right angles.

Ex. 9. The axes being rectangular, find the equation to the pair of sraight lines meeting at the origin which are perpendicular to the pair given by the equation $ax^2 + 2hxy + by^2 = 0$

If the lines represented by $ax^2 + 2hxy + by^2 = 0$ be $y - m_1x = 0$ and $y - m_2x = 0$, then we have $m_1 + m_2 = -2\frac{h}{b}$ and $m_1m_2 = \frac{a}{b}$ ... ... (1)

Now the equations of the lines perpendicular to the above lines and passing through the origin are $m_1y + x = 0$ and $m_2y + x = 0$

Their joint equation is

$(m_1y + x)(m_2y + x) = 0$, or $m_1m_2y^2 + (m_1 + m_2)xy + x^2 = 0$

or, $(\frac{a}{b})y^2 - (\frac{2h}{b})xy + x^2 = 0$ by (1)

or, $bx^2 - 2hxy + ay^2 = 0$ Ans.

Ex. 10. The line joining the origin to the point P (2,3) is the diameter of a circle which cuts the lines $5x^2 - 12xy + 3y^2 = 0$ at Q and R. Find the combined equation of the lines PQ, PR.

P is a point (2, 3) and O is the origin. The circle on OP as diameter cuts at Q and R, the lines OQ & OR and whose combined equations are $5x^2 - 12xy + 3y^2 = 0$. PQ and PR are perpendicular to the chord OQ, OR respectively.

Let $5x^2 - 12xy + 3y^2$

$= 3(y - m_1x)(y - m_2x)$

$\therefore m_1 + m_2 = \frac{12}{3}$, $m_1m_2 = \frac{5}{3}$

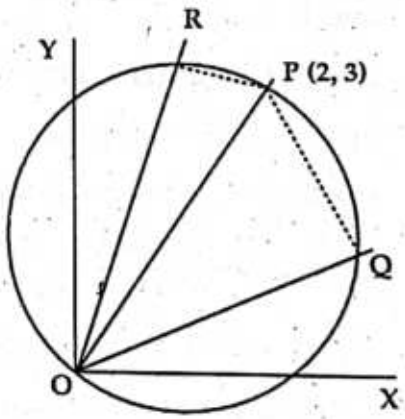


Fig. 15

The combined equation of the perpendiculars through O to OQ, OR is

$(m_1y + x)(m_2y + x) = 0$

or, $m_1m_2y^2 + (m_1 + m_2)xy + x^2 = 0$

by (1) ... ... (2)

or, $5y^2 + 12xy + 4x^2 = 0$

Now the combined equation of the lines PR, PQ parallel to the lines represented by (2) is obtained by changing the origin O to

P (2, 3) or by putting $x - 2$ for x and $y - 3$ for y in (2)

Therefore, $5(y - 3)^2 + 12(x - 2)(y - 3) + 3(x - 2)^2 = 0$

or, $3x^2 + 12xy + 5y^2 - 48x - 54y + 129 = 0$

Ex. 11. Find the area of the triangle formed by the lines. [N.U.H. 2005]

$ax^2 + 2hxy + by^2 = 0$ and $lx + my + n = 0$

[N.U.H. R. U. Hons. 1978; D. U. '60; C. U. Hons. 1991]

Let the lines represented by the given pair be $y = m_1x$ and $y = m_2x$, so that $m_1 + m_2 = -2\frac{h}{b}$ and $m_1m_2 = \frac{a}{b}$ ... ... (1)

These lines will meet the, third line at points.

$$\left(\frac{-n}{1+nm_1}, \frac{-nm_1}{1+n_1m_2}\right) \text{and} \left(\frac{-n}{1+mm_2} \frac{-nm_2}{1+mm_2}\right)$$

which will be two vertices of the triangle and the third vertex is clearly (0, 0). Hence the area of the triangle,

$$\Delta = \frac{1}{2}\begin{vmatrix} 0 & 0 & 1 \\ \frac{-n}{l+mm_1} & \frac{-nm_1}{1+mm_1} & 1 \\ \frac{-n}{l+mm_2} & \frac{-nm_2}{1+mm_2} & 1 \end{vmatrix} = \frac{n^2(m_2-m_1)}{2(l+mm_1)(l+mm_2)}$$

$$= \frac{\frac{n^2}{2}\sqrt{[(m_1+m_2)^2-4m_1m_2]}}{l^2+lm(m_1+m_2)+m^2m_2m_2}$$

$$= \frac{n^2\sqrt{(h^2-2ab)}}{(am^2+bl^2-2hlm)} \quad \text{by (1) Ans.}$$

Ex. 12. The circle $x^2 + y^2 = a^2$ cuts off an intercept on the straight line $lx + my = 1$, which subtends an angle of 45° at the origin, show that $4[a^2(l^2 + m^2) - 1] = [a^2(l^2 + m^2) - 2]^2$

The equation of the straight lines joining the origing to the points of intersection of

$x^2 + y^2 = a^2$ ... ... (1)

and $lx + my = 1$ ... ... (2)

is $x^2 + y^2 = a^2(lx + my)^2$

or $x^2(1 - a^2l^2) + y^2(1 - a^2m^2) - 2a^2lmxy = 0$

Let the angle between them be $\phi = 45°$ (given)

$$\tan\phi = \frac{2\sqrt{\{l^2m^2a^4 - (1-a^2l^2)(1-a^2m^2)\}}}{(1-a^2l^2)+(1-a^2m^2)} \text{ by Art. 41}$$

or, $(1 - a^2l^2) + (1 - a^2m^2) = 2\sqrt{\{l^2m^2a^4 - (1-a^2l^2)(1-a^2m^2)\}}$

or, $(2 - a^2l^2 - a^2m^2)^2 = 4(a^2l^2 + a^2m^2 - 1)$

or, $4[a^2l^2 + m^2) - 1] = [a^2(l^2 + m^2) - 2]^2$ Proved.

13. Prove that the equation $m(x^3-3xy^2) + y^3 - 3x^2y = 0$ represent three straight lines equally inclined to one another. [N.U.H 1999]

Solution : Given that

$m(x^3-3xy^2) + y^3 - 3x^2y = 0$ ... ... (1)

Since the equation (1) is a homogeneous is of degree 3. So it represents straight lines through origin.

Let $y = x\tan\theta$ represented the equation (1) whene θ is the slope of the line represented by (1)

$\therefore$ (1) $\Rightarrow m(x^3 - 3x.x^2\,\text{ten}^2\theta) + x^3\tan^3\theta - 3x^2.x\tan\theta = 0$

$\Rightarrow mx^3 - 3x^3m\tan^2\theta + x^3\tan^3\theta - 3x^3\tan\theta = 0$

$\Rightarrow m - 3m\tan^2 + \tan^3\theta - 3\tan\theta = 0$

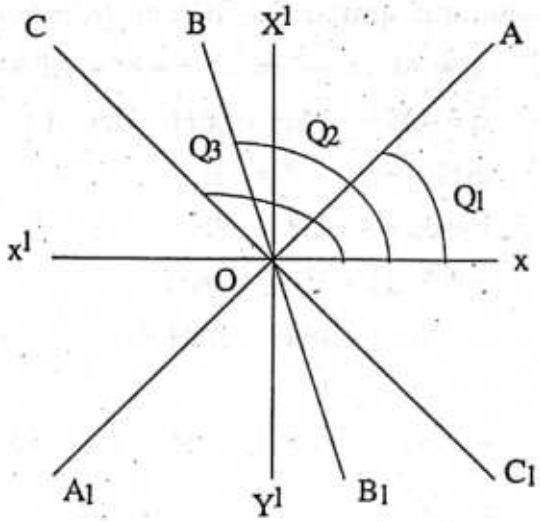
$\therefore \quad m = \dfrac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$

$\Rightarrow m = \tan 3\theta$

$\therefore \quad \tan 3\theta = \tan\alpha$ let $m = \tan\alpha$

$\therefore \quad 3\theta = n\pi + \alpha$

$\therefore \quad \theta = \dfrac{n\pi}{3} + \dfrac{\alpha}{3}$, $n = 0, 1, 2$



Let A A_1 B B_1 and CC_1 makes an angles θ_1, θ_2 and θ_3 with x – axis.

$\theta_1 = \alpha/3, n = 0$

$\theta_2 = \dfrac{\pi}{3} + \dfrac{\alpha}{3}, n = 1$

$\theta_3 = \dfrac{2\pi}{3} + \dfrac{\alpha}{3}; n = 2$

Angle between the lines AA_1 and BB_1 is $\theta_2 - \theta_1 = \dfrac{\pi}{3}$

Angle between lines BB_1 and CC_1 is $\theta_3 - \theta_2 = \dfrac{2\pi}{3} + \dfrac{\alpha}{3} - \dfrac{\pi}{3}\dfrac{d}{3} = \pi/3$

Angle between lines CC_1 and AA_1 is $\pi - (\theta_3 - \theta_1)$

$= \pi - \theta_1 + \theta_1$

$= \pi - \dfrac{2\pi}{3} - \dfrac{\alpha}{3} + \dfrac{\alpha}{3}$

$= \pi/3$

So, three lines equally inclined to one another.

14. Prove that two of the lines represented by equation $ax^4 + bx^3y + cx^2y + dxy^3 + ay^4 = 0$ will bisect angle between the other tow if $c + 6a = 0, b + d = 0$ [N.U.H . 2000,'02,'09]

Solution : Given that

$ax^4 + bx^3y + cx^2y^2 + dxy^3 + ay^4 = 0$ ... ... (1)

Since equation (1) is a homogeneous equation of degree 4. So, it represent 4 straight lines through origin.

Let $Ax^2 + 2Hxy + By^2 = 0$ ... ... (2)

be the two of the lines represented by (1).

According to the problem, other two lines represented by (1) is the bisectors of the lines represented by (2)

Equation of the bisectors of the two lines represented by (2)

$\dfrac{x^2 - y^2}{A - B} = \dfrac{xy}{H}$

$\Rightarrow \quad Hx^2 - Hy^2 - Axy + Bxy = 0$ ... ... (3)

$ax^4 + bx^3y + cx^2y^2 + dxy^3 + ay^4 = (Ax^2 + 2Hxy + By^2)(Hx^2 - Hy^2 - Axy + Bxy)$

$\Rightarrow ax^4 + bx^3y + cx^2y^2 + xy^4 = AHx^4 - AHx^2y^2 - A^2x^3y + ABx^3y$

$+ 2H^2x^3y - 2H^2xy^3 - 2AHx^2y^2 + 2BHx^2y^2 + BHx^2y^2 - B_1 + y^4 - BAxy^3 + Bxy^3$

Equating similar Co-efficient from both sides

$\therefore$ $a = AH$, $a = -BH$, $b = -A^2 + AB + 2H^2$,

$c = -AH - 2AH + 2BH + BH$, $d = -2H^2 - AB + B^2$

$\therefore$ $AH = BH$ $\therefore A = -B$

$\therefore$ $c + 6a = -3AH + 3BH$

$= -3AH + 3BH + 6AH$

$= -3AH - 3AH + 6AH = 0$

$= 0$

$b + d = -A^2 + AB + 2H^2 - 2H^2 - AB + B^2$

$= -A^2 + B^2$

$= -A^2 + A^2$ $[\because B = A]$

$= 0$ Proved.

15. If the equation $ax^2 + 2hxy + by^2 + 2gx + 2gy + c = 0$ represents two straight lines. Prove that sauare of the distance of their point of intersection from the origin is $\frac{c(a+b) - f^2 - g^2}{ab - h^2}$.

[N.U.H -2001]

Solution : Given that

$ax^2 + 2hxy + by^2 + 2gx + 2gy + c = 0$ ... ... (1)

Since equation (1) reprsents pair of st-lines

Let $l_1x + m_1y + n_1 = 0$ ... ... (2)

and $l_2x + m_2y + n_2 = 0$ ... ... (3)

by the two lines represented by (1)

$\therefore$ $ax^2 + 2hxy + by^2 + 2gx + 2fy + c$

$= (l_1x + m_1y + n_1).\quad (l_2x + m_2y + n_2)$

$\Rightarrow$ $ax^2 + 2hxy + by^2 + 2gx + 2fy + c$

$= l_1l_2x^2 + (l_1m_2 + l_2m_1)xy + m_1m_2y^2 + (l_1n_2 + l_2n_1)x + (m_1n_2 + m_2n_1)y + n_1n_2$

Equating the similar Co-efficients from both sides

$\therefore$ $l_1l_2 = a$, $m_1m_2 = b$, $l_1m_2 + l_2n_1 = 2h$, $l_1n_2 + l_2n_1 = 2g$

$m_1n_2 + m_2n_1 = 2f$, $n_1n_2 = c$.

Now, $l_1x + m_1y + n_1 = 0$

$l_2x + m_2y + n_2 = 0$

$$\therefore \frac{x}{m_1n_2 - m_2n_1} = \frac{y}{n_1l_2 - n_2l_1} = \frac{1}{l_1m_2 - l_2m_1}$$

$$\therefore x = \frac{m_1n_2 - m_2n_1}{l_1m_2 - l_1m_1},\ y = \frac{n_1l_2 - n_2l_1}{l_1m_2 - l_2m_1}$$

$\therefore$ Point of intersection of the lines (2) and (3) say A,

$$A\left(\frac{m_1n_2-m_2n_1}{l_1m_2-l_2m_1}, \frac{n_1l_2-n_2l_1}{l_1m_2-l_2m_1}\right)$$

$$\therefore OA^2=\left(\frac{m_1n_2-m_2n_1}{l_1m_2-l_2m_1}-0\right)^2+\left(\frac{n_1l_2-n_2l_1}{l_1m_2-l_2m_1}-0\right)^2$$

$$=\frac{(m_1n_2-m_2n_1)^2+(n_1l_2-n_2l_1)^2}{(l_1m_2-l_2m_1)^2}$$

$$=\frac{(m_1n_2+m_2n_1)^2-4m_1m_2\,n_1n_2+(n_1l_2+n_2l_1)^2-4l_1l_1\,n_1n_2}{(l_1m_2+l_2m_1)^2-4l_1l_2\,m_1m_2}$$

$$=\frac{4f^2-4bc+4g^2-4ca}{4h^2-4ab}$$

$$=\frac{f^2+g^2-bc-ca}{h^2-ab}$$

$$=\frac{f^2+g^2-c(a+b)}{h^2-ab}$$ (Proved)

16 If two straight lines represented by the equation $x^2(\tan^2\varphi+\cos^2\varphi)-2xytam\varphi+y^2\sin^2\varphi=0$ makes an angle α and β with x – axis respectively, then show that $\tan\alpha-\tan\beta=2$.

[N.U.H-2003, 2008]

Solution : Given equation

$x^2(\tan^2\varphi+\cos^2\varphi)-2xy\tan\varphi+y^2\sin^2\varphi=0$ ... ... (1)

since the two lines represented by the equation (1)

makes an angle α and β with x – axis.

So their equation are

$y=x\tan\alpha$ and $y=x\tan\beta$

$$\therefore \tan\alpha+\tan\beta=\frac{2\tan\varphi}{\sin^2\varphi}$$

$$=\frac{2\sin\varphi}{\cos\varphi}\times\frac{1}{\sin^2\varphi}$$

$$=2\sec\varphi\,\mathrm{cosec}\varphi$$

$$\text{and } \tan\alpha.\tan\beta=\frac{\tan^2\varphi+\cos^2\varphi}{\sin^2\varphi}$$

$$=\frac{\sin^2\varphi}{\cos^2\varphi}\cdot\frac{1}{\sin^2\varphi}+\frac{\cos^2\varphi}{\sin^2\varphi}$$

$$=\sin^2\varphi+\cot^2\varphi$$

Now, $(\tan\alpha-\tan\beta)^2=(\tan\alpha+\tan\beta)^2-4\tan\alpha.\tan\beta$

$=(2\sec\varphi\,\mathrm{cosec}\varphi)^2-4.(\sec^2\varphi+\cot^2\varphi)$

$= 4\sec^2\varphi\operatorname{cosec}^2\varphi - 4\sec^2\varphi - 4\cot^2\varphi$

$= 4\{\sec^2\varphi\,(\operatorname{cosec}^2\varphi - 1) - \cot^2\varphi\}$

$= 4\{\sec^2\varphi.\,\cot^2\varphi - \cot^2\varphi\}$

$= 4\{\cot^2\varphi\,(\sec^2 - 1)\}$

$=4\{\cot^2\varphi.\,\tan^2\varphi\}$

$\therefore (\tan\alpha - \tan\beta)^2 = 4$

$\therefore \tan\alpha - \tan\beta = 2$ (Proved)

17. If one of the straigh line given the equation $ax^2 + 2hxy + by^2 = 0$ concides with one of those given by $a_1x^2 + 2h_1xy + b_1y^2 = 0$ and the other lines represented them be perpendicular. Then show that $\frac{ha_1b_1}{b_1 - a_1} = \frac{h_1ab}{b - a} = \frac{1}{2}\sqrt{-aa_1bb_1}$ [N.U.H 1998, 2004]

Solution : Given equations

$ax^2 + 2hxy + by^2 = 0$ ... ... (1)

$a_1x^2 + 2h_1xy + b_1y^2 = 0$ ... ... (2)

Let $y = m_1x$ and $y = m_2x$ be the two lines represented by (1)

$\therefore\ m_1 + m_2 = -\frac{2h}{b}$ ... ... (3)

$m_1m_2 = \frac{a}{b}$ ... ... (4)

According to the problem, $y = m_1x$ and $y = -\frac{1}{m_2}x$ be the two lines represented by (2)

$\therefore\ m_1 - \frac{1}{m_2} = -\frac{2h_1}{b_1}$ ... ... (5)

$m_1.\ -\frac{1}{m_2} = \frac{a_1}{b_1}$ ... ... (6)

From (3) and (6) we get

$m_1 + m_2 = -\frac{2h}{b}$

and $-m_1 - \frac{a_1}{b_1}m_2 = 0$

$(+)\ m_2 - \frac{a_1}{b_1}m_2 = -\frac{2h}{b}$

$\Rightarrow\ m_2\left(1 - \frac{a_1}{b_1}\right) = -\frac{2h}{b}$

$\Rightarrow\ m_2.\frac{b_1 - a_1}{b_1} = -\frac{2h}{b}$

$\therefore\ m_2 = -\frac{2hb_1}{b(b_1 - a_1)}$

From (3); $m_1 + m_2 = -\frac{2h}{b}$

$$m_1 = -\frac{2h}{b} - m_2$$

$$= -\frac{2h}{b} + \frac{2hb_1}{b(b_1 - a_1)}$$

$$= \frac{2h}{b}\left\{\frac{b_1}{b_1 - a_1} - 1\right\}$$

$$= \frac{2h}{b} \cdot \frac{b_1 - b_1 + a_1}{b_1 - a_1}$$

$$m_1 = \frac{2ha_1}{b(b_1 - a_1)}$$

From (4) ; $m_1 m_2 = \frac{a}{b}$

$$\Rightarrow \frac{2ha_1}{b(b_1 - a_1)} \times \frac{-2hb_1}{b(b_1 - a_1)} = \frac{a}{b}$$

$$\Rightarrow \frac{4h^2 a_1 b_1}{b^2(b_1 - a_1)} = -\frac{a}{b}$$

$$\Rightarrow \frac{h^2 a_1 b_1}{b_1 - a_1} = -\frac{1}{4}ab$$

$$\Rightarrow \frac{h^2 a_1^2 b_1^2}{b_1 - a_1} = -\frac{1}{4}aa_1\, bb_1$$

$$\therefore \quad \frac{ha_1 b_1}{b_1 - a_1} = \frac{1}{2}\sqrt{-aa_1 bb_1} \qquad \ldots \quad \ldots \quad \ldots \quad (7)$$

Again From (4) ; $m_1 m_2 = \frac{a}{b} \therefore m_1 = \frac{a}{bm_2}$

From (5); $m_1 - \frac{1}{m_2} = -\frac{2h_1}{b_1}$

$$\frac{1}{m_2} - m_1 = \frac{2h_1}{b_1}$$

$$\Rightarrow \frac{1}{m_2} - \frac{a}{bm_2} = \frac{2h_1}{b_1}$$

$$\Rightarrow \frac{b - a}{bm_2} = \frac{2h_1}{b_1}$$

$$\Rightarrow m_2 = \frac{b_1(b - a)}{2h_1 b}$$

$$m_1 = \frac{a}{bm_2}$$

$$= \frac{a}{b'}\frac{2h_1b}{b_1(b-a)} = \frac{2h_1a}{b_1(b-a)}$$

From (6); $m_1.\frac{-1}{m_2} = \frac{a_1}{b_1}$

$$\Rightarrow -\frac{2h_1a}{b_1(b-a)} \times \frac{2h_1b}{b_1(b-a)} = \frac{a_1}{b_1}$$

$$\Rightarrow -\frac{4h_1^2ab}{b_1^2(b-a)^2} = \frac{a_1}{b_1}$$

$$\Rightarrow \frac{4h_1^2a^2b^2}{(b-a)^2} = -aa_1bb_1$$

$$\therefore \frac{h_1ab}{b-a} = \frac{1}{2}\sqrt{-aa_1bb_1} \quad \ldots \quad \ldots \quad \ldots \quad (8)$$

From (7) and (8) we get

$$\frac{ha_1b_1}{b_1-a_1} = \frac{h_1ab}{b-a} = \frac{1}{2}\sqrt{-aa_1bb_1}$$ (Proved)

EXERCISE V

[তারকা চিহ্নের অঙ্কগুলি অনার্স ছাত্রদের জন্য]

RECTANGULAR CO-ORDINATES 1

Art. 1. Introduction : The position of a point in space may be defined by refering it to three fixed planes. These planes have a common point, known as the **Origin** and intersect in pairs in three straight lines, known as **Co-ordinate Axes.** The planes themselves are called the **Co-ordinate planes.** When the three Co-ordinate planes and consequently the three co-ordinate axes are mutually at right angles, the axes are said to be **Rectangular.** The three planes divides the space into eight parts known as the **Octants.**

Art. 2. Co-ordinates of a point in space : Let P be any point in space. Through P draw a plane PMAN parallel to yz plane and hence perpendicular to XOX′ cutting in the point A, then OX = x is called the x co-ordinate of the point P. Again through P draw another plane PNBL parallel to zx plane and hence perpendicular to YOY′ cutting it in B then ON= y is called the y-co-ordinate of the point P. Similarly OC = z is the z-co-ordinate of the point P where C is the

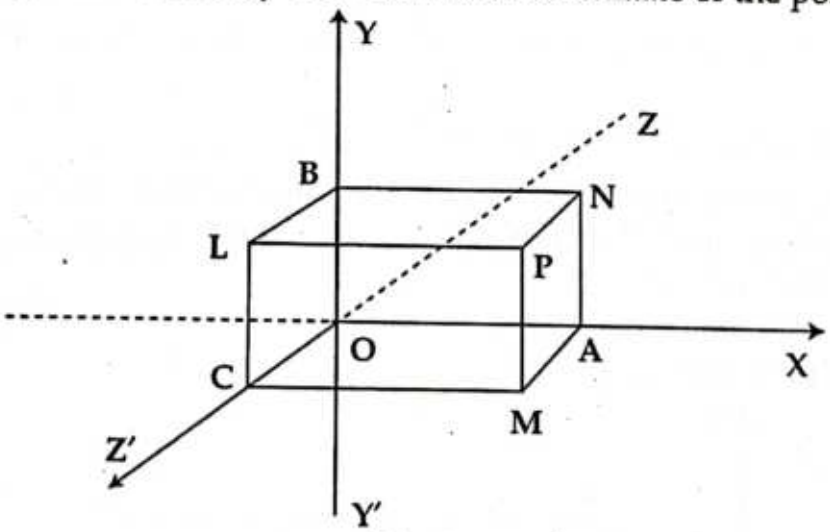


Fig : 1

point in which the plane PLCM through P parallel to xy plane cuts the z-axis ZOZ′.

The three numbers x, y, z are called the **co-ordinates of the point P and the point is represented as P(x, y, z).**

To determine the position of any point P it is not sufficient merely to know the absolute lengths of the lines LP, MP, NP we must also know the directions in which they are drawn. If lines drawn in one directions be considered as positive, those drawn in the opppsite direction must be considered negative. **We shall consider the direction as OX. OY, OZ are positive.**

The three co-ordinate planes divide the whole of the space into eight compartments which are called **octants.** The octant OXYZ in which the point P lies is called **first octant** and any point in this has each of its co-ordinates positive. Again since each of the co-ordinates of

the point P may be either positive or negative we shall have **eight such point P(x, y, z)** whose co-ordinates may be represented numerically by the three numbers x, y and z but have different signs. Below we give the eight octants and the co-ordinates of the point P in each of them situated with respect to axes just P is situated in the Octant OXYZ and has co-ordinates (x, y, z)

1. OXYZ (x, y, z)
2. OX'YZ $(-x, y, z)$
3. OXY'Z $(x, -y, z)$
4. OXYZ' $(x, y, -z)$
5. OXYZ' $(x-y, -z)$
6. OX'Y'Z' $(-x, y, -z)$
7. OX'Y'Z $(-x, -y, z)$
8. OX'Y'Z' $(-x, -y, -z)$

OCT	OXYZ	OX'YZ	OXY'Z	OX'YZ'	OXY'Z'	OX'YZ'	OX'Y'Z	OX'Y'Z''
x	+	−	+	+	+	−	−	−
y	+	+	−	+	−	+	−	−
z	+	+	+	−	−	−	+	−

Art. 3. Distance between two points: To find the distance between two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$

Through the points P, Q draw planes parallel to the co-ordinate planes to form a rectangular parallelopiped whose one diagonal is PQ. Then APCM, NBLQ, LCPB, QMAN, BPAN, LCMQ are the three pairs of parallel faces of this parallelopiped. Now $\angle AMQ = 90°$. Therefore $AQ^2 = AM^2 + MQ^2$

Also Q lies in the plane QMAN which is perpendicular to the line PA. Therefore AQ is perpendicular to PA.

Hence $PQ^2 = PA^2 + AQ^2 = PA^2 + AM^2 + MQ^2$

Now PA is the distance between the planes drawn through the points P and Q parallel to YZ plane and is therefore equal to the difference between their co-ordinates.

$$PA = x_2 - x_1$$

Similarly $MQ = AN = y_2 - y_1$

and $AM = NQ = z_2 - z_1$

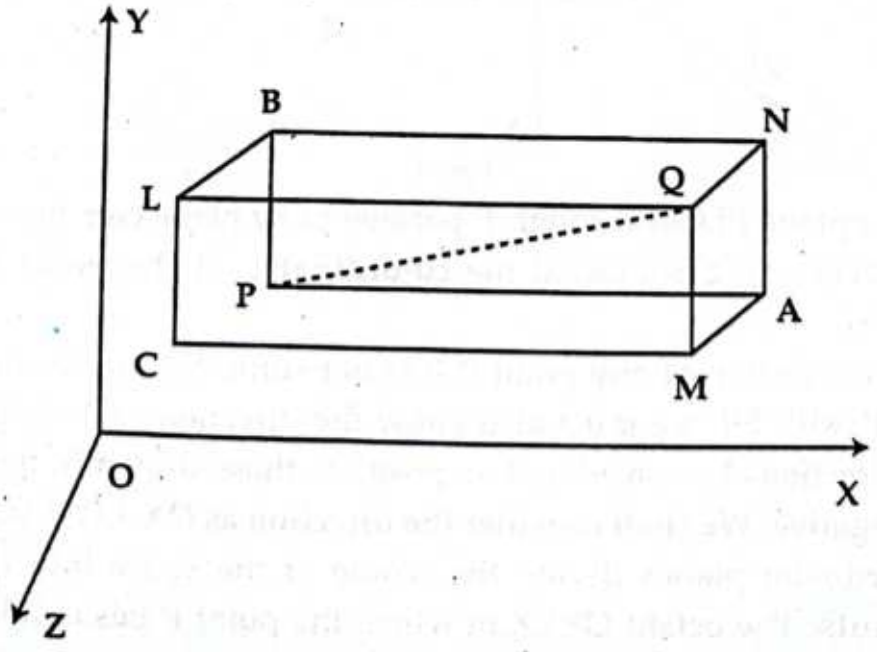


Fig : 2

Hence $BO^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2$

∴ The distance between the points (x_1, y_1, z_1) and (x_2, y_2, z_2) $PQ = \sqrt{\{(x_2 - x_2)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2\}}$ ... ... (1)

in magnitude.

Cor : Distance from the origin

when P coincides with the origin Q, we have $x_1 = y_1 = z_1 = 0$ so from (1), the distance $OQ = \sqrt{(x_2 + y_2^2 + z_2^2)}$ ... ... (2)

Art. 4. Section Ratio : To find the co-ordinates of the point which divides the straight line joining two given points in a given ratio.

Let the two given points be $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ and let $R(x, y, z)$ be a point in PQ, such that PR " RQ = m_1 " m_2

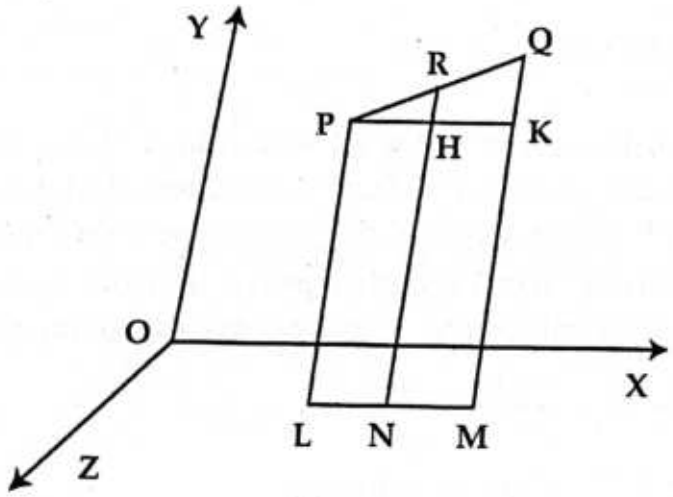


Fig : 3

Draw PL, QM, RN, paralell to OY meeting ZOX plane in L, M, N. Then the points P, Q, R, L, M, N are clearly all in one plane, and a line through P parallel to LM will be in that plane, and will therefore, meet QM, RN, in the points K,H. (suppose) Now the two triangle ΔHPR and ΔKPQ, are similar. Then

$$\frac{m_2}{m_1} = \frac{RQ}{PR} \text{ or, } \frac{m_1 + m_2}{m_1} = \frac{PQ}{PR} = \frac{KQ}{HR}$$

$$= \frac{MQ - MK}{NR - NH} = \frac{MQ - LP}{NR - LP} = \frac{y_2 - y_1}{y - y_1} \quad \therefore y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}$$

Similarly by drawing perpendiculars on XOY and YOZ and proceeding as above we can prove that

$$z = \frac{m_1 z_2 + m_2 z_1}{m_1 + m_2} \text{ and } x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}$$

Thus the co-ordinates of a point which divides the join of the point (x_1, y_1, z_1) and (x_2, y_2, z_2) in the ratio m_1 " m_2 are

$$\left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 - m_2}, \frac{m_1 z_2 + m_2 z_1}{m_1 + m_2} \right) \cdots \quad \cdots \quad \cdots \quad (3)$$

Note : If $\frac{m_1}{m_1}$ is positive, then R divides the line PQ internally.

Cor. 1. If $\frac{m_1}{m_1}$ is negative, then R divides the line PQ externaly

$$\left(\frac{m_1x_2 - m_2x_1}{m_1 - m_2}, \frac{m_1y_2 - m_2y_1}{m_1 - m_2}, \frac{m_1z_2 - m_2z_1}{m_1 - m_2}\right) \quad \ldots \quad \ldots \quad \ldots \quad (4)$$

Cor. 2. Co-ordinates of the Middle Point : If R bisects the line PQ then $m_1 = m_2$ and the co-ordinates of R are

$$\left\{\tfrac{1}{2}(x_1 + x_2), \tfrac{1}{2}(y_1 + y_2), \tfrac{1}{2}(z_1 + z_2)\right\} \quad \ldots \quad \ldots \quad \ldots \quad (5)$$

Cor. 3. General co-ordinates of a point on the line PQ.

Put $\frac{m_1}{m_2} = k$ where k is called the **position** Ratio of R with regard to P and Q. The co-ordinates of R are

$$\left(\frac{kx_2 + x_1}{k+1}, \frac{ky_2 + y_1}{k+1}, \frac{kz_2 + z_1}{k+1}\right) \quad \ldots \quad \ldots \quad \ldots \quad (6)$$

These give the co-ordinates of R for all real values of k, positive or negative. If k is positive, R lies between P and Q. Which is the case in internal division : and if k is negative, R is on the same side of both P and Q, which is the case in the external division.

Art. 5. Centre of Gravity of a Triangle : prove that the co-ordinates of the centre of gravity of the triangle whose vertices, are (z_1, y_1, z_1), (z_2, y_2, z_2), (z_3, y_3, z_3) are

$$\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3}\right)$$

Let A, B, C be the points whose co-rdinates are (x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3) respectively

Then D, the mid point of BC is

$$\left(\frac{x_2 + x_3}{2}, \frac{y_2 + y_3}{2}, \frac{z_2 + z_3}{2}\right)$$

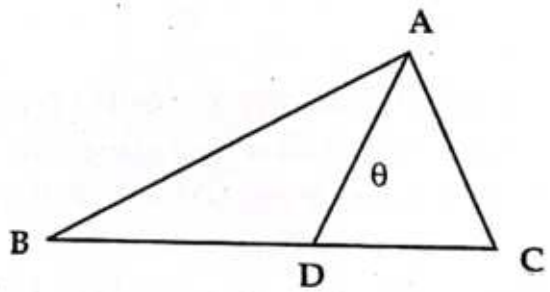


Fig : 4

Let (x, y, z) be the required co-ordinates of G, the centre of gravity of the ΔABC which divides AD in the ratio 2 " 1

Then $x = \dfrac{2.\frac{x_2 + x_3}{2} + 1.x_1}{2 + 1} = \dfrac{x_1 + x_2 + x_3}{3}$

similarly $y = \dfrac{y_1 + y_2 + a_3}{3}$ and $z = \dfrac{z_1 + z_2 + z_3}{3}$

Hence the co-ordinates of the centre of gravity of a ΔABC are

$$\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3}\right) \quad \ldots \quad \ldots \quad \ldots \quad (7)$$

Art. 6. Tetrahedron : *Definition- A tetrahedron is a solid bounded by four triangular faces. It has four vertices, each vertex arising as a point of intersection of three of the four faces. (planes). It has six edges arising as the line of intersection of two of the four faces (edges, $^4C_2 = 6$).*

Thus $ABCD$ is a tetrahedron having for triangular faces, the Δ's ABC, ACD, ADB, BCD. The points A, B, C, D, are called the VERTICES of the tetrahedron. The edges $A\,B$ and CD; BC and AD; CA and BD which do not meet are called the pairs of opposite edges of the tetrahedron.

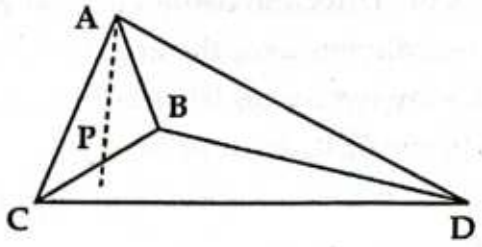


Fig : 5

Art. 7. To prove that the centre of gravity of a tetrahedron the co-ordinates of whose vertices are (x_r, y_r, z_r) where $r = 1, 2, 3, 4$ is the point.

$$\left(\frac{x_1+x_2+x_3+x_4}{4}, \frac{y_1+y_2+y_3+y_4}{4}, \frac{z_1+z_2+z_3+z_4}{4}\right)$$

Let A, B, C, D be the points (x_1, y_1, z_1); (x_2, y_2, z_2); $(x_3, y_3, z_3,)$ (x_4, y_4, z_4) If G_1 be the centre of gravity of the triangular face BCD, then the point G_1 is

$$\left(\frac{x_2+x_3+x_4}{3}, \frac{y_2+y_3+y_4}{3}, \frac{z_2+z_3+z_4}{3}\right)$$

Now we know from Statics that the C. G of a tetrahedron is on the line joining any vertex to the C. G of the opposite face and divides it in the ratio 3 " 1.

Hence if G (x, y, z) be the C. G of the tetrahedron; then

$$x = \frac{3.\frac{x_2+x_3+x_4}{3} + 1.x_1}{4} = \frac{x_1+x_2+x_3+x_4}{4}$$

Similarly $y = \frac{x_1+x_2+x_3+x_4}{4}$, $z = \frac{y_1+y_2+y_3+y_4}{4}$ ... (8)

The C. G of the tetrahedron is therefore,

$$\left(\frac{x_1+x_2+x_3+x_4}{4}, \frac{y_1+y_2+y_3+y_4}{4}, \frac{z_1+z_2+z_3+z_4}{4}\right)$$

Art. 8. Direction Cosines of a Line.

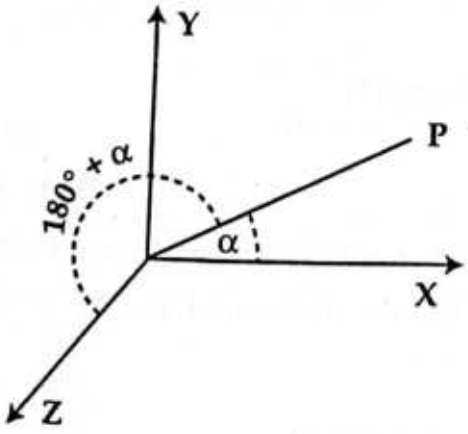


Fig : 6

If a given line OP makes angles α, β, γ, with the positive direction of axes of x, y and z respectively then cosα, cosβ, cosγ are the **Diretion Cosines** of the line OP and are generally denoted by the letters, l, m, n respectively. Again the direction cosines of PO will be cos (180 + α), cos (180 + β), cos (180 + γ) because the line through O, *i, e,* OP will now make angles 180 + α, 180 + β,

180 + γ with the axes.

The angles α, β, γ, of the line OP are called the **Direction Angle of OP.**

Note : If a line does not pass through the origin O, then its direction angles are the angles between the axes and a line drawn through O parallel to the given line (and having the same direction.)

Cor. Direction cosines (d, c, s) of the Axes of References : Now the axis of x makes with the co-ordinates axes the angles 0°, 90°, 90°, respectively and hence by definition its direction cosines are cos 0°. cos 90°, cos 90°. i, e, (1, 0, 0). Similarly the d, c, s of the axes of y and z are (0, 1, 0) and (0, 0, 1) respectively.

Art. 9. A Useful Relation : If O be the orgin and (x, y, z) the co-ordinates of a point P, then $x = lr, y = mr, z = nr$

Where l, m, n, are direction cosines of OP and r, is the length of OP.

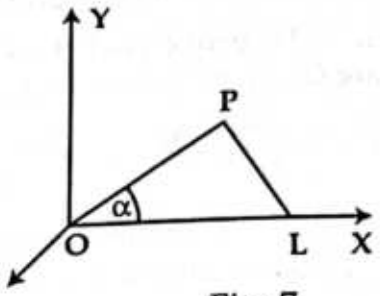


Fig: 7

Through p draw PL perpendicular to x-axis so that $OL = x$. From the right angled triangle OPL, we have

$$\frac{OL}{OP} = \cos \angle LOP = \cos \alpha \text{ or, } \frac{x}{r} = l \text{ or, } x = lr$$

Similarly we have $y = mr, z = nr$,

Therefore the co-ordinates of P are $x = lr, y = mr, z = nr$ ... ... (9)

Note : $\frac{x}{l} = \frac{yi}{m} = \frac{z}{n} = r$ ... ... 9(a)

Art. 10. *Relation between direction cosines. To prove that if l, m, n, be the direction cosines of any line then $l^2 + m^2 + n^2 = 1$, i, e. the sum of the squares of the direction cosines of every line is one.*

Let OP be drawn through the origin parallel to the given line so that l, m, n, are the cosines of angles which makes with OX, OY, OZ respectively.

Let (x, y, z) be the co-ordinates of any point P on this line

Let $OP = r$

$\therefore x = lr, y = m.\ r, z = nr$ (By eq, 9 of Art. 9)

Squaring and adding, we obtain

$x^2 + y^2 + z^2 = (l^2 + m^2 + n^2)\, r^2$

or, $r^2 = (l^2 + m^2 + n^2) r^2$ (By eq (2) of Art. (5)

or, $l^2 + m^2 + n^2 = 1$ ... ... (10)

or, $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$ [R. U. 1980]

Art. 11. Direction Ratios : *Any three numbers a, b, and c which are proportional to the direction cosines l, m, n, respectively of a given line are called the direction ratios of the given line (Rectangular axes). Now from definition of direction ratios.*

$$\frac{l}{a} = \frac{m}{b} = \frac{n}{c} = \frac{\sqrt{(l^2 + m^2 + n^2)}}{\sqrt{(a^2 + b^2 + c^2)}} = \pm \frac{l}{\sqrt{(a^2 + b^2 + c^2)}} \quad \text{(By eq. 10 of art. 10)}$$

$$l = \pm \frac{a}{\sqrt{(a^2 + b^2 + c^2)}},\ m = \pm \frac{b}{\sqrt{(a^2 + b^2 + c^2)}},\ n = \pm \frac{c}{\sqrt{(a^2 + b^2 + c^2)}}$$

Where the same sign either + ve or–ve is to be chosen throughout.

Hence of (a, b, c), be three numbers proprtional to the $d\,c$'s of OP, then the actual direction cosines of OP are

$$\frac{a}{\sqrt{(a^2+b^2+c^2)}}, \frac{b}{\sqrt{(a^2+b^2+c^2)}}, \frac{c}{\sqrt{(a^2+b^2+c^2)}} \quad \ldots \quad \ldots \quad (11)$$

and that of PO are

$$\frac{-a}{\sqrt{(a^2+b^2+c^2)}}, \frac{-b}{\sqrt{(a^2+b^2+c^2)}}, \frac{-c}{\sqrt{(a^2+b^2+c^2)}} \quad \ldots \quad \ldots \quad (12)$$

Art. 12. Projection of a point on a straight line.

The projection of a point A on a given line CD is the point B which is the foot of the perpendicular from A on CD.

This the point in which the plane through A and perpendicular to CD meets it.

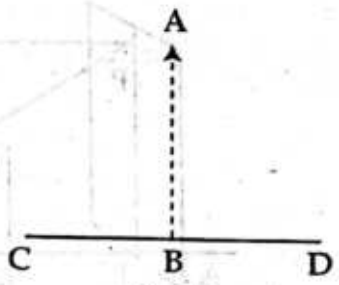


Fig : 8

Art. 13. The orthogonal projection of a given segment AB of a line on any line CD is $AB \cos\theta$ where θ is the angle between AB and CD.

Let the planes through A and B perpendicular to the line CD meet it in A', B' respectively so that A', B' is the projection of AB on CD.

Through A draw a line AP parallel to CD to meet the plane through B at P.

Now AP is parallel to CD; $\angle PAB = \theta$

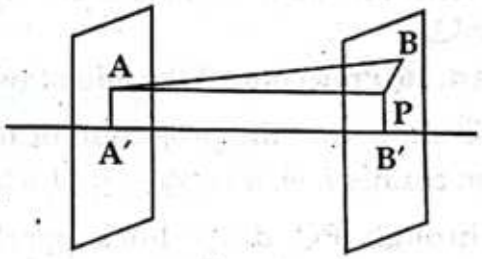


Fig : 9

Also PB lies in the plane which is perpendicular to AP.

$$\angle APB = 90^\circ$$

Hence $AP = AB \cos\theta$

Clearly $A'B'PA$ is a rectangle so that we have $AP = A'B'$

Hence $A'B' = AB \cos\theta$

Art. 14. Direction cosines of a line joining two point $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$

Let L, M be the feet of the perpendiculars drawn from P, Q. to the X-axis respectively so that $OL = x_1$, $OM = x_2$

Projection of PQ on X axis $= LM = OM - OL = x_2 - x_1$

Also if l, m, n be the direction cosines of PQ the projection of PQ on X-axis $= l\,PQ$.

Similarly pojection of PQ on Y- axis abd Z - axis, are $m\,PQ = y_2 - y_1$, $nPQ = z_2 - z_1$

$$\therefore \frac{x_2 - x_1}{l} = \frac{y_2 - y_1}{m} = \frac{z_2 - z_1}{n} = PQ. \quad \ldots \quad \ldots \quad \ldots \quad (13)$$

This the direction cosines of the line joining the two points (x_1, y_1, z_1) and (x_2, y_2, z_2) are projectional to

$$x_2 - x_1 ; y_2 - y_1 ; z_2 - z_1 \quad \ldots \quad \ldots \quad \ldots \quad (14)$$

Art. 15. Projection of a broken line conising of continuos segments on a given line.

Let $P_1, P_2, P_3 \ldots \ldots P_n$ be any number of points in space and $Q_1, Q_2, Q_3 \ldots \ldots Q_n$ be their projections on the line AB, and these points being on the same straight line, we have

Q_1Q_n = Projection of $P_1P_n = Q_1Q_2 + Q_2Q_3 + Q_3Q_4 + \ldots + Q_n^{-1} Q_n$ Where Q_1Q_2 = Projection of P_1P_2

Q_2Q_3 = Projection of P_2P_3 and so on

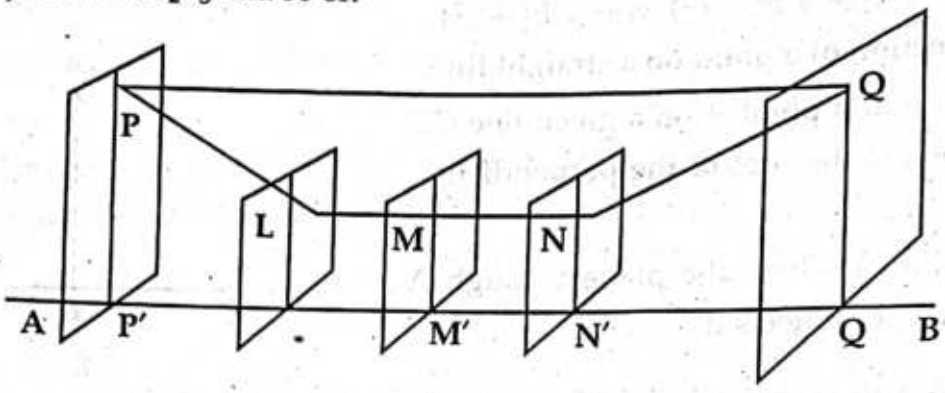


Fig : 10

Hence the sum of the projection of P_1P_2, $P_2P_3 \ldots P_n^{-1} P_n$ on AB = Projection of P_1P_n on AB = Q_1Q_n ... ... (15)

Art. 16. Projection of the join of two points on a line :

To show that the projection of the line joinig $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ on a line with direction cosines l, m, n is $(x_2 - x_1)\, l + (y_2 - y_1)\, m + (z_2 - z_1)n$.

Through PQ draw lines parallel to the co-ordinates planes to form rectangular parallelopiped whose one diagonal is PQ (Fig. 3)

Now $PA = x_2 - x_1$, $AN = y_2 - y_1$, $NQ = z_2 - z_1$

The lines PA, AN, NQ are respectively parallel to x-axis, y-axis and z-axis. Therefore, their respective projections on the line with direction cosines l, m, n are

$(x_2 - x_1)\, l$, $(y_2 - y_1)\, m$, $(z_2 - z_1)\, n_1$

As the projection of PQ on any two lines is equal to the sum of the projection of PA. AN, NQ on that line therefore the required projection is $(x_2 - x_1)\, l + (y_2 - y_1)\, m + (z_2 - z_1)\, n$ ....(16)

Art. 17. Angle between two lines : If (l_1, m_1, n_1) and (l_2, m_2, n_2) be the direction cosines of any two lines AB and CD and θ be the angle between them, then

$\cos\theta = l_1l_2 + m_1m_2 + n_1n_2$

First Method : Let OP_1, OP_2 be lines through the origin parallel to the given lines so that the cosines of the angles which OP_1 and OP_2 make with the axes are l_1, m_1, n_1 and l_2, m_2, n_2 respectively and the angle between the line is the angle between OP_1 and OP_2. Let the angle be θ.

The projection of the line OP_2 joining O (0, 0, 0) and $P_2(x_2, y_2, z_2)$ on the line OP_1 whose direction cosines are, l_1, m_1, n_1 is $(x_2 - 0)\, l_1 + (y_2 - 0)\, m_1 + (z_2 - 0)n_1$

$s = l_1x_2 + m_1y_2 + n_1z_2$ ... ... (1)

But the projection of OP_2 on OP_1 is $OP_2 \cos\theta$

$\therefore OP_2 \cos\theta = l_1x_2 + m_1y_2 + n_1z_2$ ... ... (2)

But $x_2 = l_2 OP_2$; $y_2 = m_2 OP_2$, $z_2 = n_2 OP_2$

Then from (ii), we have

$OP_2 \cos\theta = l_1l_2 OP_2 + m_1m_2 OP_2 + n_1n_2 OP_2$

or, $\cos\theta = l_1l_2 + m_1m_2 + n_1n_2$ ... ... (17)

Second method : Let $OP_1 = r_1$, $OP_2 = r_2$, $P_1P_2 = d$ and the co-ordinates of P_1, P_2 be (x_1, y_1, z_1) and (x_2, y_2, z_2) respectively.

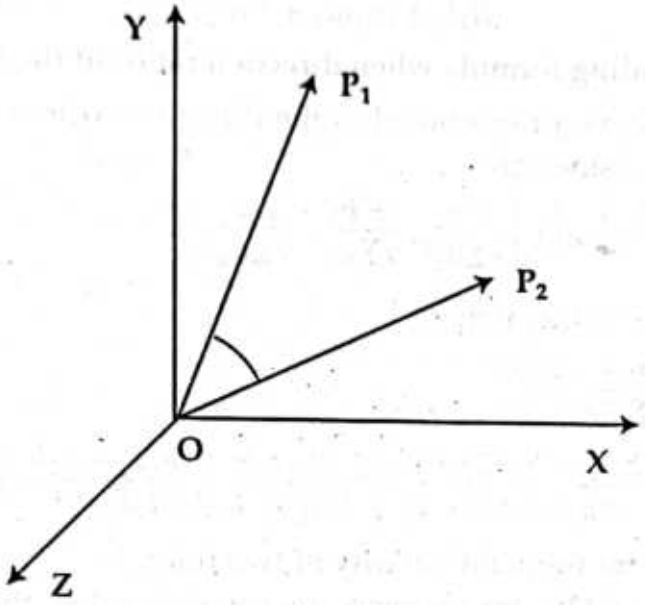


Fig : 11

Then $x_1 = r_1l_1$; $y_1 = r_1m_1$; $z_2 = r_1n_1$ ... ... (n)

and $x_2 = r_2l_2$; $y_2 = r_2m_2$; $z_2 = r_2n_2$

then by geometry, we have

$OP_1^2 + OP_2^2 - P_1P_2^2 = 2\, OP_1\, OP_2 \cos\theta$

or, $\cos\theta = \dfrac{r_1^2 + r_2^2 - d^2}{2r_1r_2}$

Now $r_1^2 = x_1^2 + y_1^2 + z_1^2$, $r_2^2 = x_2^2 + y_2^2 + z_2^2$

and $P_1P_2^2 = d_2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2$

$= r_1^2 + r_2^2 - 2r_1r_2(l_1l_2 + m_1m_2 + n_1n_2)$ by eq. (3)

$\therefore \cos\theta = \dfrac{r_1^2 + r_2^2 - r^2_1 - r_2^2 + 2r_1r_2\,(l_1l_2 + m_1m_2 + n_1n_2)}{2r_1r_2}$

or, $\cos\theta = l_1l_2 + m_1m_2 + n_1n_2$ ... ... (18)

Art.17. (a) Expression for sin θ and tan θ.

$\sin^2\theta = 1 - \cos^2\theta = 1 - (l_1l_2 + m_1m_2 + n_1n_2)^2$

$= (l^2_1 + m_1^2 + n_1^2)\,(l^2_2 + m_2^2 + n_2^2) - (l_1l_2 + m_1m_2 + n_1\,n_2)^2$

$= (l_1m_2 - l_2m_1)^2 + (m_1n_2 - m_2n_1)^2 + (n_1l_2 - n_2l_1)^2$ ... ... (19)

by Lagranges Identity

$$\text{or, } \sin^2\theta = \begin{vmatrix} l_1 m_1^2 \\ l_2 m_2^2 \end{vmatrix} + \begin{vmatrix} m_1 n_1^2 \\ m_2 n_2^2 \end{vmatrix} + \begin{vmatrix} n_1 l_1^2 \\ n_2 l_2 \end{vmatrix} \quad \ldots \quad \ldots \quad (20)$$

$$\text{or, } \sin\theta = \pm\sqrt{\{(l_1m_2 - l_2m_1)^2 + (m_1n_2 - m_2n_1)^2 + (n_1l_2 - n_2l_1)^2\}}$$

$$\text{Again } \tan\theta = \frac{\sin\theta}{\cos\theta}$$

$$= \pm\frac{\sqrt{\{(l_1m_2 - l_2m_1)^2 + (m_1n_2 - m_2n_1)^2 + (n_1l_2 - n_2l_1)^2\}}}{(l_1l_2 + m_1m_2 + n_1n_2)} \quad \ldots \quad \ldots \quad (21)$$

Art. 17. (b) Corresponding formula when direction ratios of the lines are given.

If a_1, b_1, c_1 and a_2, b_2, c_2 be proportional to the direction cosines of the lines OP_1 and OP_2, then their actual direction cosines are

$$\left\{\frac{\pm a_1}{\sqrt{\sum a_1^2}}, \frac{\pm b_1}{\sqrt{\sum a_1^2}}, \frac{\pm c_1}{\sqrt{\sum a_1^2}}\right\} \text{ and } \left\{\frac{\pm a_2}{\sqrt{\sum a_2^2}}, \frac{\pm b_2}{\sqrt{\sum a_2^2}}, \frac{\pm c_2}{\sqrt{\sum a_2^2}}\right\}$$

If θ be the angle between them, then

$$\cos\theta = \frac{\pm(a_1a_2 + b_1b_2 + c_1c_2)}{\sqrt{(\sum a_1^2)}\sqrt{(\sum a_2^2)}} \quad \ldots \quad \ldots \quad \ldots \quad (22)$$

$$\text{and} \quad \sin\theta = \pm\frac{\sqrt{\{(b_1c_2 - b_2c_1)^2 + (c_1a_2 - c_2a_1)^2 + (a_1b_2 - a_2b_1)^2}}{\sqrt{(a_1^2 + b_1^2 + c_1^2)}\sqrt{(a_2^2 + b_2^2 + c_2^2)}} \quad \ldots \quad (23)$$

Art. 17. (c) Condition for perpendicularity of two lines.

If the two lines OP_1 and OP_2 are perpendicular to each other, then $\cos\theta = 0$ as $\theta = 90°$ then the relation becomes.

$$l_1l_2 + m_1m_2 + n_1n_2 = 0 \quad \ldots \quad \ldots \quad \ldots \quad (24)$$

If we use direction ratios, then we get the required condition from eq (22)

$$a_1a_2 + b_1b_2 + c_1c_2 = 0 \quad \ldots \quad \ldots \quad \ldots \quad (25)$$

Art. 17. (d) Condition for parallelism of two lines.

If the two lines OP_1 and OP_2 are parallel to each other, then $\sin\theta = 0$, as $\theta = 0°$

Form eq (19) we have the condition

$$(l_1m_2 - l_2m_2)^2 + (m_1n_2 - m_2n_1)^2 + (n_1l_2 - n_2l_1)^2 = 0$$

Hence we have, $l_1m_2 - l_2m_1 = 0$; $m_1n_2 - m_2n_1 = 0$ and $n_1m_2 - n_2m_1 = 0$

$$\text{or, } \frac{l_1}{l_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2} = \frac{\sqrt{(l_1^2 + m_1^2 + n_1^2)}}{\sqrt{(l_2^2 + m_2^2 + n_2^2)}} = 1$$

$$l_1 = l_2\,;\ m_1 = m_2\ ;\ n_1 = n_2 \quad \ldots \quad \ldots \quad \ldots \quad (26)$$

i. e. The Direction Cosines of two lines are equal. From the eq (23). we have

$$(b_1c_2 - b_2c_1)^2 + (c_1a_2 - c_2a_1)^2 + (a_1b_2 - a_2b_1)^2 = 0$$

$$b_1c_2 - b_2c_1 = 0;\ c_1a_2 - c_2a_1 = 0\ ;\ a_1b_2 - a_2b_1 = 0$$

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \quad \ldots \quad \ldots \quad \ldots \quad (27)$$

The direction ratios of the two lines are proportional.

EXAMPLES

Ex. 1. Find the distance between the points P_1 (4, 3, −6) and P_2(−2, 1, −3)

$\therefore \quad P_1P_2^2 = (4+2)^2 + (3-1)^2 + (-6+3) = 49$

$\therefore \quad P_1P_2 = 7$ Ans.

Ex. 2. A, B, C are three points on the axes x, y, and z respectively at distance a, b, c from origin O, find the co-ordinates of the point which is equidistant form A, B, C and O.

Let P be the requred point (x, y, z) and the points A, B, C and O are $(a, 0, 0)$, $(0, b, o)$ (o, o, c) and (o, o, o) we are given that $PO = PA = PB = PC$

Taking $PO = PA$ or, $PO^2 = PA^2$, we have

$x^2 + y^2 + z^2 = (x-a)^2 + y^2 + z^2$

or, $2ax = a^2$ or $x = a/2$

Similarly taking $PO^2 = PB^2$ and $PO^2 = PC^2$ we have

$y = b/2$ and $z = c/2$

Hence the required point P is $\left(\frac{1}{2}a, \frac{1}{2}b, \frac{1}{2}c\right)$

Ex. 3. Find the ratio in which the yz plane divides the joint of the points (− 2, 4, 7) and (3, − 5, 8) and also find the co-ordiantes of the point of intersection of this line with the yz plane.

The co-ordiantes of any point on the line joining the two points are

$$\left(\frac{3\lambda-2}{\lambda+1}, \frac{-5\lambda+4}{\lambda+1}, \frac{8\lambda+7}{\lambda+1}\right)$$

If the points in the yz-plane, then its x co-ordinate should be zero.

$\frac{3\lambda-2}{\lambda-1} = 0$ or, $3\lambda - 2 = 0$ or, $\lambda = \frac{2}{3}$

Hence the required ratio is 2 " 3 and putting $\lambda = 2/3$ the required point is (0, 2/5, 37/15)

Ex 4. Find the dirction cosines of the line drawn from the origin to the point (− 6, 2, 3)

$$\cos\theta = \frac{x_2 - x_1}{\sqrt{\{(x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2\}}}$$

$$= \frac{-6-0}{\sqrt{\{(-6-0)^2 + (2-0)^2 + (3-0)^2}} = \frac{6}{7}$$

Similarly, $\cos\beta = 2/7$ and $\cos\gamma = 3/7$

The direction cosines are $-6/7, 2/7, 3/7$

Ex. 5. Find the direction cosines of the line which is equally inclined to the axes.

If the line makes angles α, β, γ with the axes then

$\cos\alpha = \cos\beta = \cos\gamma \therefore \alpha = \beta = \gamma$ (given)

$$\text{or } \frac{l}{1} = \frac{m}{1} = \frac{n}{1} = \pm\frac{\sqrt{(l^2+m^2+n^2)}}{\sqrt{(1+1+1)}} = \pm\frac{1}{\sqrt{3}}$$

$\therefore$ d, c, s are $\left(\pm\frac{1}{\sqrt{3}}, \pm\frac{1}{3}, \pm\frac{1}{\sqrt{3}}\right)$

Ex. 6. A line makes angles $\alpha, \beta, \gamma, \delta$ with the four diagonals of a cube, prove that $\cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta = \frac{4}{3}$ [N.U.H. 2000. 2007] [C.U. Hons. 1984, 1985]

Let the *d. c, s* of the line be l, m, n. The eight vertices can taken as 0 (o, o, o), P (a, a, a) A(a, o, o), B (o, a, o), C (o, o, a), L(o, a, a), M(a, o, a), N(a, a, o)

AL, BM, CN and *OP* are the diagonals of the cube Fig. (1) Art. (2). the d. c's of the four diagonals are

$(1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3}), (-1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3})$

$(1/\sqrt{3}, -1/\sqrt{3}, 1/\sqrt{3}), (1/\sqrt{3}, 1/\sqrt{3}, -1/\sqrt{3})$

The direction cosines of the line which is inclined at angles $\alpha, \beta, \gamma, \delta$ respectively to the four diagonals are

$$\cos\alpha = \frac{l}{\sqrt{3}} + \frac{m}{\sqrt{3}} + \frac{n}{\sqrt{3}} = \frac{l+m+n}{\sqrt{3}}$$

Similarly $\cos\beta = \frac{-l+m+n}{\sqrt{3}}$, $\cos\gamma = \frac{l-m+n}{\sqrt{3}}$, $\cos\delta = \frac{l+m-n}{\sqrt{3}}$

$\therefore \cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta = \frac{1}{3}.4(l^2+m^2+n^2) = \frac{4}{3}$ proved.

Ex. 7. If a line makes angles α, β, γ with the axes, show that $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$

since $l^2 + m^2 + n^2 = 1$

or, $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

or, $1 - \sin^2\alpha + 1 - \sin^2\beta + 1 - \sin^2\gamma = 1$

or, $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$ Proved.

Ex. 8 Find the condition so that the three concurrent lines with direction cosines $(l_1, m_1, n_1), (l_2, m_2, n_2), (l_2, m_3, n_3)$ are co-planar

Let (l, m, n) be the direction cosines of a line which is normal to the plane that contains the two lines whose *d c's* are (l_1, m_1, n_1) and (l_2, m_2, n_2)

$$ll_1 + mm_1 + nn_1 = 0 \quad \ldots \quad \ldots \quad \ldots \quad (1)$$

$$ll_2 + mm_2 + nn_2 = 0 \quad \ldots \quad \ldots \quad \ldots \quad (2)$$

Normal to a plane is perpendicular to every line in that plane. Again if the line whose *d c's* (l_3, m_3, n_3) also lies in plane, then the line with *d, c's* (l, m, n) is also perpendicular to it.

$$\therefore \quad ll_3 + mm_3 + nn_3 = 0 \quad \ldots \quad \ldots \quad \ldots \quad (3)$$

Eliminate l, m, n from (1), (2), (3), then we have

$$\begin{vmatrix} l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \\ l_3 & m_3 & n_3 \end{vmatrix} = 0$$

which is the required condition.

Ex. 9. If the edges of a rectangular parallelopiped are a, b, c show that the angles between the four diagonals are given by $\cos^{-1}\left(\frac{\pm a^2 \pm b^2 \pm c^2}{a^2+b^2+c^2}\right)$ [C.U. 1984]

Take one of the vertices O of the parallelopied as origin and three rectangulr faces through it as the three rectangular co-ordinate planes. See fig (1). Art 2.

Let $OA = a$, $OB = b$, $OC = c$,

The lines OP, AL, BM, CN are the four diagonals.

The co-ordinates of A, B, C are $(a, 0, 0)$, $(0, b, 0)$, $(0, 0, c)$

The co-ordinate of L, M, N are $(0, b, c)$, $(a, 0, c)$, $(a, b, 0)$

The co-ordinates of O, P are $(0; 0, 0)$; (a, b, c)

Direction cosines of OP are $\frac{a}{\sqrt{\sum a^2}}, \frac{b}{\sqrt{\sum a^2}}, \frac{c}{\sqrt{\sum a^2}}$

Direction cosines of AL are $\frac{-a}{\sqrt{\sum a^2}}, \frac{b}{\sqrt{\sum a^2}}, \frac{c}{\sqrt{\sum a^2}}$

Direction cosines of BM are $\frac{a}{\sqrt{\sum a^2}}, \frac{-b}{\sqrt{\sum a^2}}, \frac{c}{\sqrt{\sum a^2}}$

Direction cosines of CN are $\frac{a}{\sqrt{\sum a^2}}, \frac{b}{\sqrt{\sum a^2}}, \frac{-c}{\sqrt{\sum a^2}}$

The angle between OP and Al, therefore. is

$$\cos\theta = \frac{a(-a)+b(b)+c(c)}{\sqrt{\sum a^2}\sqrt{\sum a^2}} = \frac{-a^2+b^2+c^2}{a^2+b^2+c^2}$$

$$\therefore\ \theta = \cos^{-1}\left(\frac{-a^2+b^2+c^2}{a^2+b^2+c^2}\right)$$

Similarly the angle between any one of the six pairs of diagonals can be found

Hence the angles in general between the six pairs are given by

$$\cos^{-1}\left(\frac{\pm a^2 \pm b^2 \pm c^2}{a^2+b^2+c^2}\right)$$

Ex. 10. If l_1, m_1, n_1; l_2, m_2, n_2; l_3, m_3, n_3 are the $d.c.s$ of three mutually perpendicular lines whose direction cosines are proportional to $l_1 + l_2 + l_3$; $m_1 + m_2 + m_3$, $n_1 + n_2 + n_3$ make equal angles with them.

we have

$l_1l_2 + m_1m_2 + n_1n_2 = 0$, $l_2l_3 + m_2m_3 + n_2n_3 = 0$,

$l_3l_1 + m_1m_3 + n_3n_1 = 0$ ... ... (1)

Also, $l_1^2 + m_1^2 + n_1^2 = 1$, $l_2^2 + m_2^2 + n_2^2 = 1$, $l_3^2 + m_3^2 + n_3^2 = 1$ ... (2)

Therefore, the actual direction cosines of the lines are

$$\left(\frac{l_1+l_2+l_3}{\sqrt{3}}, \frac{m_1+m_2+m_3}{\sqrt{3}}, \frac{n_1+n_2+n_3}{\sqrt{3}}\right)$$

N. B. $\sqrt{[(l_1+l_2+l_3)^2 + (m_1+m_2+m_3)^2 + (n_1+n_2+n_3)^2]} = \sqrt{3}$

If the angle between the first line and the other line θ then

$\cos\theta = \{l_1(l_1 + l_2 + l_3) + m_1(m_1 + m_2 + m_3) + n_1(n_1 + n_2 + n_3)\}/\sqrt{3}$

$= 1/\sqrt{3} \therefore \theta = \cos^{-1} 1/\sqrt{3}$

Similarly, we can show that the angle between each of the other two line and the line with *d. c,s* given by (3) is also $\cos^{-1}1/\sqrt{3}$

Hence proved

Ex. 11. The direction cosines of a moving line in two adjacent positions are l, m, n and $l + \delta l$, $m + \delta m, n + \delta n$. Show that the small angle $\delta\theta$ between the positions is given by $(\delta\theta)^2 = (\delta l)^2 + (\delta m)^2 + (\delta n)^2$

Now $\cos\delta\theta = l(l + \delta l) + m(m + \delta n) + n(n + \delta n)$

$\therefore 1 - \cos\delta\theta = 1 - (l^2 + m^2 + n^2) - (l\delta l + m\delta m + \delta n)$

or, $2\sin^2\left(\frac{1}{2}\delta\theta\right) = -(l\delta l + m\delta m + n\delta n)$

Expand it and take only the first term as $\delta\theta$ is very small.

$2\left(\frac{1}{2}\delta\theta\right)^2 = -(l\delta l + m\delta m + n\delta n)$

or, $(\delta\theta)^2 = -2(l\delta l + m\delta m + n\delta n)$ ... ... (1)

Again $l^2 + m^2 + n^2 = 1$ ... ... (2)

and $(l + \delta l)^2 + (m + \delta m)^2 + (n + n\delta)^2 = 1$ ... ... (3)

Subtracting (2) from (3), we have

$(\delta l)^2 + 2l(\delta l) + (\delta m)^2 + 2m(\delta m) + (\delta n)^2 + 2n(\delta n) = 0$

or, $(\delta l)^2 + (\delta m)^2 + (\delta n)^2 = -2(l\delta l + m\delta m + n\delta n)$

Hence from (1), we have $(\delta\theta)^2 + (\delta l)^2 + (\delta m)^2 + (\delta n)^2$ proved.

Ex. 12. Find the angle between two lines whose direction cosines are given by the equations

$l + m + n = 0$ ... ... (1)

and $l^2 + m^2 - n^2 = 0$ ... ... (2)

[N. U. 1996] [D.U. Hons. 1962]

Eliminate n from (1) and (2) then

$l^2 + m^2 - (l + m)^2 = 0$ or $2lm = 0$ ... ... (3)

Either $l = 0$, i.e. $1.l + om + on = 0$, Also $l + m + n = 0$

$\frac{l}{o} = \frac{m}{+1} = \frac{n}{-1}$ ... ... (4)

If $m = 0$ in (3)

$\therefore ol + lm + on = 0$ and $l + m + n = 0$, then

Therefore, the *d.c's* are proportional to, 0, −1, 1 ; 1, 0, −1. If θ be the angle between them, then

$$\cos\theta = \pm\frac{0(1) + (-1)(0) + (1)(-1)}{\sqrt{[0^2 + (-1)^2 + 1]}\sqrt{[(1)^2 + (0)^2 + (-1)^2]}} = \pm\frac{1}{2}$$

$$\theta = \frac{1}{3}\pi, \text{ or, } \frac{2}{3}\pi$$

Ex. 13 Find the distance of A (1, −2, 3) from the line PO through P(2, −3, 5), which makes equal angles with the axes.

The direction cosines of PQ are $1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3}$

If p be the required perpendicular distance, then

$p^2 = [(-3+2)/\sqrt{3} - (5-3)/\sqrt{3}]^2 + [5-3)/\sqrt{3} - (2-1)/\sqrt{3}]^2 + [(2-1)/\sqrt{3}) - (3+2)/\sqrt{3}]^2$

$= 1/3, 14$ or $p = \sqrt{(14/3)}$ (See Art. 33).

14. Define the direction cosine of straight line. Find the acute angle between the lines whose direction cosines are given by the relations $l + m + n = 0$ and $l^2 + m^2 - n^2 = 0$ [1996, 2005]

Solution : Direction cosine : A tine makes an angles α, β, γ with the positive direction of x, y and z axes. Then $\cos\alpha, \cos\beta, \cos\gamma$ called direction cosine. It is denoted by l, m, n.

2nd part : We have

$l + m + n = 0$ ... ... (1)

$l^2 + m^2 - n^2 = 0$ ... ... (2)

From (1) $n = -(l + m)$

Putting the values of n is (2) We get

$l^2 = m^2 - (l + m)^2 = 0$

$\Rightarrow l^2 + m^2 - l^2 - 2lm - m^2 = 0$

$\Rightarrow -2lm = 0$

$\Rightarrow l = 0$ ... ... (3)

or, $m = 0$ ... ... (4)

From (1) and (3)

$l + m + n = 0$

$l + om + on = 0$

$$\therefore \frac{l}{0-0} = \frac{m}{1-0} = \frac{n}{0-1}$$

$$\therefore \frac{l}{0} = \frac{m}{1} = \frac{n}{-1} = \frac{1}{\sqrt{o^2 + 1^2 + 1^2}} = \frac{1}{\sqrt{2}}$$

$$\therefore l = 0, m = \frac{1}{\sqrt{2}}, n = -\frac{1}{\sqrt{2}}$$

Again, from (1) and (4) We get,

$l + m + n = 0$

$ol + m + on = 0$

$$\therefore \frac{l}{0-1} = \frac{m}{0-0} = \frac{n}{1-0}$$

$$\Rightarrow \frac{l}{-1} = \frac{m}{0} = \frac{n}{1} = \frac{1}{\sqrt{1^2 + 0^2 + 1^1}} = \frac{1}{\sqrt{2}}$$

$$\therefore l = \frac{-1}{\sqrt{2}}, m = 0, n = \frac{1}{\sqrt{2}}$$

Let θ be the angle betⁿ. them

$\therefore \text{Cos}\theta = \text{Cos}\theta = \pm (l_1l_2 + m_1m_1 + n_1n_2)$

$\therefore \text{Cos}\theta = \pm\left(0.\frac{-1}{\sqrt{2}} + \frac{1}{\sqrt{2}}.0 + \frac{1}{\sqrt{2}}.\frac{1}{\sqrt{2}}\right) = \pm\frac{1}{2} = \cos \pi/3 \text{ or, } \cos 2\pi/3$

$\therefore \theta = \pi/3 \text{ or, } 2\overline{\Lambda}/3$

15. The direction cosines of two st-lines are given by the relations al + bm + cn = 0. and $ul^2 + vm^2 + wn^2 = 0$. Prove that the st. lines will be Perpendicular if $a^2(v + w) + b^2(w + u) + c^2(u + v) = 0$ and parallel of $\frac{a^2}{u} + \frac{b^2}{v} + \frac{c^2}{w} = 0$. [N.U.H. 1997]

Solution : we have

$al + bm + cn = 0$ ... ... (1)

$ul^2 + vm^2 + wn^2 = 0$ ... ... (2)

From (i) $l = -\frac{bm + cn}{a}$

Putting the value of l in (2) we get $u\left\{-\frac{bm + cu}{a}\right\}^2 + vm^2 + wn^2 = 0$

$\Rightarrow u(b^2m^2 + 2bcmn + c^2n^2) + a^2vm^2 + a^2wn^2 = 0$

$\Rightarrow (b^2u + a^2v)m^2 + 2bcumn + (c^2u + a^2w)n^2 = 0$

$\Rightarrow (b^2u + a^2v)\frac{m^2}{n^2} + 2bcu\frac{m}{n} + c^2u + a^2w = 0$ ... ... (3)

Which is quadratic eqⁿ.of $\frac{m}{n}$.

Let $\frac{m_1}{n_1}$ and $\frac{m_2}{n_2}$ be the two roots of ... ... (3)

$\therefore \frac{m_1}{n_1}.\frac{m_2}{n_2} = \frac{c^2u + a^2w}{b^2u + a^2v}$

$\frac{m_1m_2}{c^2u + a^2w} = \frac{n_1n_2}{b^2u + a^2v}$

Similarly, $\frac{l_1l_2}{b^2w + c^2v} = \frac{m_1m_2}{c^2u + a^2w}$

$\therefore \frac{l_1l_2}{b^2w + c^2v} = \frac{m_1m_2}{c^2u + a^2w} = \frac{n_1n_2}{b^2u + a^2v} = k \text{ (say)}$

$\therefore l_1l_2 = k(b^2w + c^2v),\ m_1m_2 = k(c^2u + a^2w)$

$m_1n_2 = k(b^2u + a^2v)$

The two lines are perpendicular of $l_1l_2 + m_1m_2 + n_1n_2 = 0$

$\Rightarrow k(b^2w + c^2v) + k(c^2u + a^2w) + k(b^2u + a^2v) = 0$

$\Rightarrow a^2(w + v) + b^2(w + u) + c^2(u + v) = 0$

wo lines are parallel of the two roots of (3) are equal.

$\therefore (2bcu)^2 - 4(b^2u + a^2v)(c^2u + a^2w) = 0$

$\Rightarrow b^2c^2u^2 + b^2c^2u^2 + a^2b^2uw = a^2c^2uv + a^4vw$

$\Rightarrow a^2b^2uw + a^2c^2uv + a^4vw = 0$

$\Rightarrow b^2uw + c^2uv = a^2uw = 0$

$\therefore \frac{a^2}{u} + \frac{b^2}{v} = \frac{c^2}{w} = 0$ (Proved)

16. Direction cosine of two straight lines are connected by the relations $2l + 2m - n = 0$ and $mn + nl + lm = 0$ show that the lines are perpendicular each other.[N.U.H 1988]

Solution : We have

$2l + 2m - n = 0$ ... ... (1)

and $mn + nl + lm = 0$ ... ... (2)

From (1); $n = 2l + 2m$

Putting this valus in (2) we get,

$(2) \Rightarrow m(2l + 2m) + l(2l + 2m) + lm = 0$

$\Rightarrow 2lm + 2m^2 + 2l^2 + 2lm + lm = 0$

$\Rightarrow 2l^2 + 5lm + 2m^2 = 0$

$\Rightarrow (2l + m)(2m + l) = 0$

$\therefore 2l + m = 0$ ..... (3) and $l + 2m = 0$ ... ... (4)

From (1) and (3) $\therefore \frac{l}{0+1} = \frac{m}{-2-0} = \frac{n}{2-4}$

$2l + 2m - n = 0$ $\frac{l}{1} = \frac{m}{-2} = \frac{n}{-2} = 1$

$2l + m + 0n = 0$ $\frac{l}{-1} = \frac{m}{2} = \frac{n}{2} = \frac{1}{\sqrt{1^2 + 2^2 + 2^2}} = \frac{1}{3}$

$l = \frac{-1}{3}, m = \frac{2}{3} = n, \frac{2}{3}$

$\therefore$ dc's of the 1st line is $\frac{-1}{3}, \frac{2}{3}, \frac{2}{3}$

Again, (1) and (4)

$2l + 2m - n = 0$

$l + 2m + on = 0$

$\therefore \frac{l}{0+2} = \frac{m}{-1-0} = \frac{n}{4-2}$

$\frac{l}{2} = \frac{m}{-1} = \frac{n}{2} = \frac{1}{\sqrt{2^2+1^2+2^2}} = \frac{1}{3}$

$\therefore l = \frac{2}{3}, m = -\frac{1}{3}, n = \frac{2}{3}$

$\therefore$ dc's of the 2nd line is $\frac{2}{3}, \frac{-1}{3}, \frac{2}{3}$

The two lines are perpendicular of

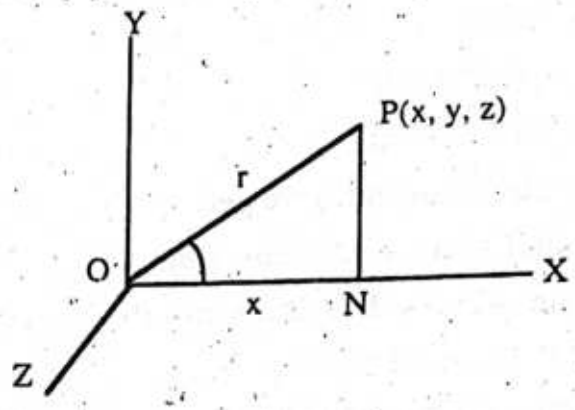
$\boxed{l_1l_2 + m_1m_2 + n_1n_1 = 0}$

Now, $-\frac{1}{3}\cdot\frac{2}{3}+\frac{2}{3}\cdot\frac{-1}{3}+\frac{2}{3}\cdot\frac{2}{3}=\frac{-2}{9}-\frac{2}{9}+\frac{4}{9}=0$

Hence The given lines are perpendicular each other.

17. Prove that $l^2 + m^2 + n^2 = 1$ where l, m, n dc's of any line [(2000) N.U.H. 2002, 2008]

Solution : Let P(x, y, z) be the co-ordinates of any point on the line



$\therefore \cos\alpha = \frac{ON}{OP} = \frac{x}{r}$

$\therefore x = r\cos\alpha = rl$

Similarly,

$y = rm$

$z = rn$

Also $Op^2 = r^2 = x^2 + y^2 + z^2$

Now, $x^2 + y^2 + z^2 = r^2l^2 + r^2m^2 + r^2n^2$

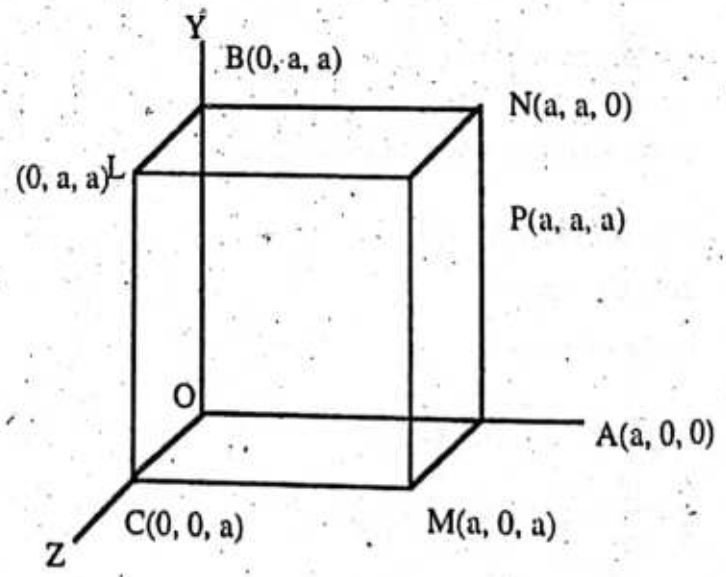
$\Rightarrow r^2 = r^2 (l^2 + m^2 + n^2)$

$\therefore l^2 + m^2 + n^2 = 1$ Proved.

\# **Find the angle between two diagonals of a cube.**

Solution.

Let OANBPBCM be a cube whose lenglth of each edges a. Let OP and CN be the two diagonals of a cube.



$\therefore$ Co-ordinate of O(0, 0, 0)

\- - - - - P (a, a, a)

\- - - - - N (a, a, 0)

\- - - - - C (0, 0, a)

Dr's of op; a – 0, a – 0, a –.0

ie, a, a, a,

Dr's of CN; a – 0, a – 0, 0 – a

ie, a, a, – a

Let θ be the angle between two diagonals Opened CN,

$$\therefore \cos\alpha = \frac{a.a + a.a + a.(-a)}{\sqrt{(a^2+a^2a^2)(a^2+a^2+a^2)}} = \frac{a^2}{\sqrt{3a^2 \times 3a^2}}$$

$\cos\theta = \frac{1}{3} \therefore \theta = \cos^{-1}\frac{1}{3}$ (Ans)

18. Prove that the straight lines whose direction cosines are given by the relations $al + lm + cn = 0$ and $fmn + gnl + hlm = 0$ are perpendicular if $f/a + g/a + h/c = 0$ and parallel if $\sqrt{af} \pm \sqrt{bg} \pm \sqrt{ch} = 0$. [N.U.H 2003]

Solution : Given that

$al + bm + cn = 0$ ... ... (1)

and $fmn + gnl + hlm = 0$ ... ... (2)

From (1) ; $n = -\dfrac{al + bm}{c}$

Putting the value of n in (2) we get $-fm\left(\dfrac{al + bm}{c}\right) - gl\left(\dfrac{al + bm}{c}\right) + hlm = 0$

$\Rightarrow -af\,lm - bfm^2 - agl^2 - bglm + chlm = 0$

$\Rightarrow agl^2 + (af + bg - ch)\,lm + bfm^2 = 0$

$\Rightarrow ag\dfrac{l^2}{m^2} + (af + bg - ch)\dfrac{l}{m} + bf = 0$ ... ... (3)

Which is quadratic equation of $\dfrac{l}{m}$.

Let $\dfrac{l_1}{m_1}$ and $\dfrac{l_2}{m_2}$ be the two roots of (3).

$\therefore$ Product of the roots

$\dfrac{l_1}{m_1} \cdot \dfrac{l_2}{m_2} = \dfrac{bf}{ag} = \dfrac{f/a}{g/b}$

$\therefore \dfrac{l_1 l_2}{f/a} = \dfrac{m_1 m_2}{g/b}$

Similarly, $\dfrac{m_1 m_2}{g/b} = \dfrac{n_1 n_2}{h/c}$

$\therefore \dfrac{l_1 l_2}{f/a} = \dfrac{m_1 m_2}{g/b} = \dfrac{n_1 n_2}{h/c} = k$ (say)

$\therefore l_1 l_2 = k.f/a,\ m_1 m_2 = kg/b,\ n_1 n_2 = k.h/c$

Two lines are perpendicular if

$l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$

$\Rightarrow k\,f/a + k.g/b + k.h/c = 0$

$\therefore f/a + g/b + h/c = 0$

Tow lines are parallel if the two roots of (3) are equal.

So $(af + bg - ch)^2 - 4.ag\,bf = 0$

$\Rightarrow (af + bg - ch)^2 = (2\sqrt{af}.\ \sqrt{bg})^2$

$\Rightarrow af + bg - ch = \pm 2\sqrt{af}.\sqrt{bg}$

$\Rightarrow (\sqrt{af})^2 \pm 2.\sqrt{af}.\sqrt{bg} + (\sqrt{bg})^2 = (\sqrt{ch})^2$

$\Rightarrow (\sqrt{af} \pm \sqrt{bg})^2 = (\sqrt{ch})^2$

$\Rightarrow \sqrt{af} \pm \sqrt{bg} = \pm\sqrt{ch}$

$\Rightarrow \sqrt{af} \pm \sqrt{bg} \pm \sqrt{ch} = 0$ (proved)

19. If l_1, m_1 n_1 and l_2, m_2 n_2 be the direction cosines of any two lines AB and CD and θ be the angle between them. Then prove that $\cos\theta = l_1l_2 + m_1m_2 + n_1n_2$. [N.U.H. 2009]

Proof :

Let l_1m_1,n_1 and l_2, m_2, n_2 be the dc's of AB and CD. Let OA = r_1 and OC = r_2, θ be the angle between AB and CD.

From the Δ OAC;

$$\text{Cos}\theta = \frac{OA^2 + OC^2 - AC^2}{2.OA.OC}$$

$\Rightarrow 2r_1r_2\cos\theta = r_1^2 + r_2^2 - \{(l_1r_1 - l_2r_2)^2 + (m_1r_1 - m_2r_2)^2 + (n_1r_1 - n_2r_2)^2\}$

$\Rightarrow 2r_1r_2\ c\text{Cos}\theta = r_1^2 + r_2^2 - l_1^2r_1^2 + 2r_1r_2l_1l_2 - l_2^2r_2^2 - m_1^2r_1^2 + 2r_1r_2m_1m_2 - m_2^2r_2^2 - n_1^2r_1^2 + 2r_1r_2n_1n_2 - n_2^2r_2^2$

$\Rightarrow 2r_1r_2\cos\theta = r_1^2 + r_2^2 - r_1^2(l_1^2 + m_1^2 + n_1^2) - r_2^2(l_2^2 + m_2^2 + n_2^2) + 2r_1r_2(l_1l_2 + m_1m_2 + n_1n_2)$

$\Rightarrow 2r_1r_2\cos\theta = r_1^2 + r_2^2 - r_1^2 - r_2^2 + 2r_1r_2(l_1l_2 + m_1m_2 + n_1n_2)$

$\therefore \cos\theta = l_1l_2 + m_1m_2 + n_1n_2$ (Proved)

20. A line makes and angles α, β, γ, δ with the four diagonals of a cube. Prove that $\cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta = 4/3$. [N.U.H. 2000, 2007, C.H.N 1984, 1985]

Solution :

From the figure, eight vertices can be taken as Co-ordinate of 0(o, o, o)

- - - - P(a, a, a)
- - - - N(a, a, 0)
- - - - C(0, 0, a)
- - - - A(a, 0, 0)
- - - - L(0, a, a)
- - - - M(a, 0, a)
- - - - B(a, 0, 0)

OP, CN, AL and BM are four diagonals of a Cube OANBLCMP.

Dr's of OP a–0, a–0, a–0

ie, a, a, a

ie, 1, 1, 1

Let $l_1, m_1 n_1$ be the de's of op

$$\therefore \frac{l_1}{1}=\frac{m_1}{1}=\frac{n_1}{1}=\frac{1}{\sqrt{1^2+1^2+1^2}}=\frac{1}{\sqrt{3}}$$

$$\therefore l_1=\frac{1}{\sqrt{3}}, m_1=\frac{1}{\sqrt{3}}, m_1=\frac{1}{\sqrt{3}}$$

li, De's of op $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}$

Similarly, dc's of CN; $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}$

de's of AL; $\frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$

de's of BM; $\frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$

Let l, m, n be The de's of the line which makes an angle $\alpha, \beta, \gamma, \delta$ with the diagonales op, CN, Al and BM.

$$\therefore \cos\alpha = l.\frac{1}{\sqrt{3}}+m.\frac{1}{\sqrt{3}}+n.\frac{1}{\sqrt{3}}$$

$$\therefore \cos\alpha=\frac{l+m+n}{\sqrt{3}}.$$

Similarly, $\cos\beta = \frac{l+m-n}{\sqrt{3}}$

$$\cos\gamma=\frac{-l+m+n}{\sqrt{3}}$$

$$\cos\delta=\frac{l-m+n}{\sqrt{3}}$$

Now, $\cos^2\alpha+\cos^2\beta\cos^2\gamma+\cos^2\delta$

$$=\left(\frac{l+m+n}{\sqrt{3}}\right)^2+\left(\frac{l+m-n}{\sqrt{3}}\right)^2+\left(\frac{-l+m+n}{\sqrt{3}}\right)^2+\left(\frac{l-m+n}{\sqrt{3}}\right)^2$$

$$=\frac{1}{3}[l^2+m^2+n^2+2lm+2mn+2ln+2lm-2mn-2nl+l^2+m^2+n^2+l^2+m^2+n^2-2lm+2mn-2nl+l^2+m^2+n^2-2lm-2mn+2nl]$$

$$=\frac{1}{3}[1+1+1+1]=4/3$$

$\therefore \cos^2\alpha+\cos^2\beta+\cos^2\gamma+\cos^2\delta = 4/3$ (Proved)

THE PLANE 2

Art. 18. (a) Definition of a palne : A plane is a surface such that if any two points are taken on it, the straight line joining them lies wholly and the surface i, e, every point of the line joining the two point will be on the plane.

Art. 18. (b) : To prove that the general equation of the first degree in x, y, z, i.e. $ax + by + cz + d = 0$ represents a plane.

Let (x_1, y_1, z_1) and $(x_2, y_2, z_2,)$ be the co-ordinates of any two points lying on the locus whose equation is

$ax + by + cz + d = 0$ ... ... (1)

$\therefore\ ax_1 + by_1 + cz_1 + d = 0$ ... ... (2)

and $ax_2 + by_2 + cz_2 + d = 0$ ... ... (3)

Multiplying (3) by k and adding to (2), we have

$a(x_1 + kx_2) + b(y_1 + ky_2) + c(z_1 + kz_2) + d(l + k) = 0$

Or, $\frac{a(x_1 + kx_2)}{l + k} + \frac{b(y_1 + ky_1)}{l + k} + \frac{c(z_1 + kz_2)}{l + k} + d = 0$ ... ... (4)

Result. (4) shows that equation (1) is satisfied by the point whose co-ordinates are $\left(\frac{x_1 + kx_2}{l + k}, \frac{y_1 + ky_2}{l + k}, \frac{z_1 + kz_2}{l + k}\right)$

But from equation (6). Art. 4, it represents any point on the line joining (x_1, y_1, z_1), (x_2, y_2, z_2); and giving different value to k. We get all the points on the line. Therefore the line completely lies on the locus; in other words, the locus is a plane.

Conclusion : Hence every equation of the first degree in x, y, z represents a plane.

Cor. 1. General equation of a given plane that passes through a given point.

The general equation of a plane is $ax + by + cz + d = 0$ ... ... (5)

Since it passes through the point (x_1, y_1, z_1), we have

$ax_1 + by_1 + cz_1 + d = 0$ ... ... (6)

Subtracting (6) from (5), we have

$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$ ... ... (28)

Cor. 2. General equation of a plane through the origin.

In case the plane passes through origin (0, 0, 0) the general equation of the plane $ax + by + cz + d = 0$ will be satisfied by (0, 0, 0) so that $d = 0$. Hence the equation reduces to the form.

$ax + by + cz = 0$ ... ... (29)

Cor. 3. The equation of a plane passing through the two points (x_1, y_1, z_1) and (x_2, y_2, z_2) is

$$L = (x-x_1)(y-y_2) - (x-x_2)(y-y_1) + R\{(x-x_1)(z-z_2) - (x-x_2)(z-z_2)\} = 0 \quad \ldots \quad \ldots \quad \ldots \quad 29(a)$$

It may easily be verified that 29 (a) is the equation of a plane passing through the points (x_1, y_1, z_1) and (x_2, y_2, z_2) because this equation is satisfied by the co-ordinates for all values of R and further in 29 (a) the terms xy cancel out, so that 29(a) is an equation of the first degree in x, y and z and as such it is of a plane.

The equation may also be written as

$$(x-x_1)(y-y_2) - (x-x_2)(y-y_1) = A\{(x-x_1)(z-z_2) - (x-x_2)(z-z_1)\} \quad \ldots \quad \ldots \quad 29(b)$$

Art. 19. Plane through three given points. To find the equation of the plane passing through the three non-collinear points.

(x_1, y_1, z_1); (x_2, y_2, z_2), (x_3, y_3, z_3)

Let the equation to the plane be

$$ax + by + cz + d = 0 \quad \ldots \quad \ldots \quad \ldots \quad (1)$$

As the given points lie on the plane (1), we have

$$ax_1 + by_1 + cz_1 + d = 0 \quad \ldots \quad \ldots \quad \ldots \quad (2)$$

$$ax_2 + by_2 + cz_2 + d = 0 \quad \ldots \quad \ldots \quad \ldots \quad (3)$$

$$ax_3 + by_3 + cz_3 + d = 0 \quad \ldots \quad \ldots \quad \ldots \quad (4)$$

Eliminating a, b, c, d from (1), (2), (3) and (4) we have

$$\begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{vmatrix} = 0 \text{ or, } \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1 \quad \ldots \quad \ldots \quad (30)$$

which is the required equation of the plane.

Art. 20. Standards forms of the equation of a plane.

(a) Intercept form : To find the equation to the plane in terms of the intercepts, which it makes on the axes.

Let the intercepts made by the plane on the axes be a, b, c. Then we can regard it as a plane passing through the points (a, o, o); (o, b, o); (o, o, c). Then by Art. 19 its equation is

$$\begin{vmatrix} x & y & z & 1 \\ a & o & o & 1 \\ o & b & o & 1 \\ o & o & c & 1 \end{vmatrix}$$

$$\text{or, } \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1 \quad \ldots \quad \ldots \quad \ldots \quad (31)$$

which is the required equation.

(b) Normal form : To find the equation of a plane in terms of p , the length of the perpendicular from orgin on the plane and the direction cosines l, m, n of this perpendicular.

Let p be the length of the perpendicualar ON from the origin on the plane. and let l, m, n be the direction cosines of perpendicualar. Let p be any point on the plane, and draw PL perpendicular on ZOX. and LM perpendicular to OX.

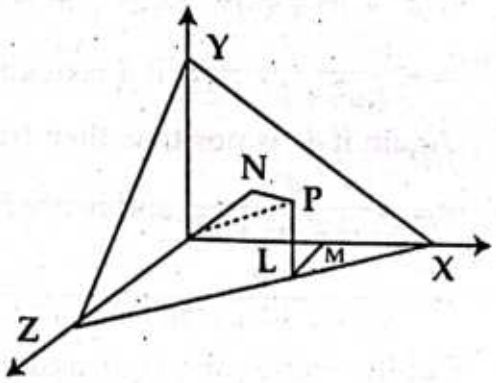


Fig. 12.

Then the projection of OP on ON is equal to the sum of the projection of OM, ML and LP on ON.

Hence if P be (x, y, z), we have

$lx + my + nz = p$ ... ... 31(a)

which is the required equation,

Note: p is always positive.

Art. 21. Reduction of the general equation of the plane

$Ax + By + Cz + D = 0$ to

(i) Intercepet form (ii) Normal form

(i) Intercept form : Transfer D to the right-hand side and divide throughout by-D so as to make R. H. S. equal to one.

$$\therefore \quad \frac{Ax}{-D} + \frac{By}{-D} + \frac{Cz}{-D} = 1$$

$$\text{or,} \quad \frac{x}{-D/A} + \frac{-y}{-D/B} + \frac{z}{-D/C} = 1$$

put $-D/A = a, -D/B = b, -D/C = c$

Then $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ which is intercept form of the equation.

(ii) Normal Form : To transform the equation

$ax + by + cz + d = 0$ ... ... (1)

to the normal form for $lx + my + nz = p$ ... ... (2)

As these two equations are of the same form on comparison we have. $\frac{-d}{p} = \frac{a}{l} = \frac{b}{m} = \frac{c}{n}$

$$\text{or } l = \frac{-ap}{d}, m\frac{-bp}{d}, n = \frac{-cp}{d} \quad \ldots \quad \ldots \quad \ldots \quad (3)$$

Since l, m, n are the direction consines of the normal to the plane we have. $l^2 + m^2 + n^2 = 1$

$$\text{or } \frac{a^2 + b^2 + c^2}{d^2} p^2 = l \quad by \quad \ldots \quad \ldots \quad \ldots \quad (4)$$

Now we always keep p positive. Hence if d be negative, then from (i) we must choose $p = \frac{d}{\sqrt{(a^2 + b^2 + c^2)}}$ and hence from (3)

$$l = +\frac{a}{\sqrt{(a^2 + b^2 + c^2)}} ; m = +\frac{b}{\sqrt{(a^2 + b^2 + c^2)}} ; n = +\frac{c}{\sqrt{(a^2 + b^2 + c^2)}}$$

Putting the values of l, m, n, and p in $lx + my + nz = p$ we have

$$\frac{a}{\sqrt{(a^2+b^2+c^2)x}} + \frac{b}{\sqrt{(a^2+b^2+c^2)y}} + \frac{c}{\sqrt{(a^2+b^2+c^2)z}}$$

$$= -\frac{d}{\sqrt{(a^2+b^2+c^2)}} \text{ if } d \text{ is negative}$$

Again if d is positive, then from (iv) we must choose

$$p = \frac{d}{\sqrt{(a^2+b^2+c^2)}} \text{ and hence from (3) } l = \frac{-a}{\sqrt{(a^2+b^2+c^2)}}$$

$$m = \frac{-b}{\sqrt{(a^2+b^2+c^2)n}}; = \frac{-c}{\sqrt{(a^2+b^2+c^2)}}$$

Put the values of l, m, n, and p in (ii), then

$$\frac{-a}{\sqrt{(a^2+b^2+c^2)^x}} - \frac{b}{\sqrt{(a^2+b^2+c)^y}} - \frac{c}{\sqrt{(a^2+b^2+c^2)^z}} = \frac{d}{\sqrt{(a^2+b^2+c^2)}} \text{ if } d \text{ is positive}$$

Thus the normal form of the equation $ax + by + cz + d = 0$ is

$$\left.\begin{aligned} -\frac{a}{\sqrt{\Sigma a^2}}x - \frac{b}{\Sigma a^2}y - \frac{c}{\sqrt{\Sigma a^2}}z = \frac{d}{\sqrt{\Sigma a^2}} &\text{ if } d \text{ is positive} \\ \frac{a}{\sqrt{\Sigma a^2}}x + \frac{b}{\sqrt{\Sigma a^2}}y + \frac{c}{\sqrt{\Sigma a^2}}z = \frac{d}{\Sigma a^2} &\text{ ; f } d \text{ is negative} \end{aligned}\right\} \quad \ldots \quad \ldots \quad \ldots \quad (32)$$

Note : Since $\frac{a}{\sqrt{\Sigma a^2}}, \frac{b}{\sqrt{\Sigma a^2}}, \frac{c}{\sqrt{\Sigma a^2}}$ are actual direction cosines of the normal to the plane $ax + by + cz + d = 0$, therefore $a, b, c,$ are proportional to the *d.c's* of the normal to the plane.

Art. 22. (a) : Angle between two planes " *Angle between two planes is equal to the angle between the normals to them from any point. Thus the angle between the two planes.*

$$ax + by + cz + d = 0 \quad \ldots \quad \ldots \quad \ldots \quad (1)$$

and $a_1x + b_1y + c_1z + d = 0 \quad \ldots \quad \ldots \quad \ldots \quad (2)$

is equal to the angle between the lines with diretion ratios a, b, c and a_1, b_1, c_1. Hence if θ be the angle between the normals, then

$$\cos\theta = \frac{aa_1 + bb_1 + cc_1}{\sqrt{(a^2+b^2+c^2)}\sqrt{(a_1{}^2+b_1{}^2+c_1{}^2)}}$$

Art. 22. (b) : Condition of perpendicularity of two planes.

If two planes are perpendicualar to each other, so are their normals whose direction cosines are porportional to a_1, b_1, c_1, and a_2, b_2, c_2, Hence the condition of perpendicualarity is

$$a_1a_2 + b_1b_2 + c_1c_2 = 0 \quad \ldots \quad \ldots \quad \ldots \quad (33)$$

Art. 22. (c) : Condition of parallelism of two planes.

Two planes $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ are parallel if and only if the co-efficients of x, y, z in their equations are proportional. i,e, if

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \quad \ldots \quad \ldots \quad \ldots \quad (34)$$

Art. 23. Some system of planes.

(i) Plane parallel to a given plane and passing through a given point.

The equation of a plane parallel to a given plane $ax + by + cz + d = 0$ is of the form

$$ax + by + cz + k = 0 \quad \ldots \ldots \ldots (1) \quad \ldots \quad \ldots \quad \ldots \quad (35)$$

where k is an arbitrary constant. If it passes through a point (x_1, y_1, z_1), then (1) is satisfied.

$ax_1 + by_1 + cz_1 + k = 0$ ... ... (2)

From (1) and (2), we have

$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$ ... ... 35(a)

Which is the required equation of the plane,

(ii) Plane perpendicular to a given plane.

Let the equation of the given plane be

$ax + by + cz + d = 0$ ... ... (1)

and that of the required plane is

$a_1x + b_1y + c_1z + d_1 = 0$ ... ... (2)

Since (1) and (2) are perpendicular to cach other, therefore,

$aa_1 + bb_1 + cc_1 = 0$ ... ... (3)

If the plane (2) passes through a point (x_1, y_1, z_1) then

$a_1x_1 + b_1y_1 + c_1z_1 + d_1 = 0$ ... ... (4)

From (2), (3) and (4) the plane cannot be determined uniquelly since one arbitrary constant remains. Infact we have infinite number of planes by rotating (2) about the normal to (1) which passes through (1)

In order to remove this constant we require another given condition. Let the plane (2) passes through another given point (x_2, y_2, z_2) then eq. (2) in satisfied.

$a_1x_2 + b_1y_2 + c_1z_2 + d_1 = 0$ ... ... (5)

Now eliminant of a_1, b_1, c_1, d_1, from (2), (3), (4) and (5), is the required equation.

$$\begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ a & b & c & 1 \end{vmatrix} = 0 \quad \ldots \quad \ldots \quad \ldots \quad (36)$$

or
$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_1 - x_2 & y_1 - y_2 & z_1 - z_2 \\ a & b & c \end{vmatrix} = 0 \quad \ldots \quad \ldots \quad (37)$$

Cor. 4. Planes perpendicular to co-ordinates planes.

Now yz plane is $x = 0$ or, $x + oy + oz = 0$ ... ... (1)

Let the equation to the plane perpendicular to yz plane be

$ax + by + cz + d = 0$ ... ... (2)

The planes (1) and (2) are perpendicular

$\therefore \quad 1.a + 0.b + 0.c = 0$ or $a = 0$

Hence the equation of the plane perpendicular to yz plane is

$by + cz + d = 0$

Similarly the equation of the planes perpendicular to zx and xy planes are $ax + cz + d = 0$ and $ax + by + d = 0$ respectively

Rule : Hence we can say that plane $ax + by + cz + d = 0$ represents planes perpendicular to yz, zx, xy planes respectively according as $a = 0$, or $b = 0$, or, $c = 0$ or according as the term of x or of y or of z is missing in its equation.

(iii) Plane Through the line of intersection of given planes.

The equation

$(ax + by + cz + d) + k(a_1x + b_1y + c_1z + d) = 0$ ... ... (1)

represents a system of planes passing through the line of intersection of the planes

$ax + by + cz + d = 0$ ... ... (2)

$a_1x + b_1y + c_1z + d_1 = 0$ ... ... (3)

k being a parameter for the equation is of the first degree in x, y, z and represents a plane : The value of k is obtained if the plane (1) is parallel or perpendicular to given plane.

Art. 24. length of the perpendicular from a point to a plane.

To find the perpendicular distance of the point $P\ (x_1, y_1, z_1)$ from the plane

$lx + my + nz = P$ ... ... (1)

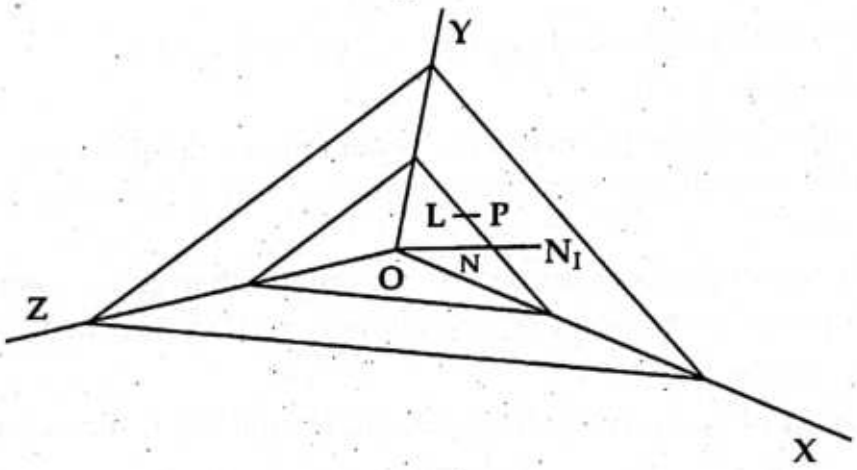


Fig : 13

When P stands for the length of the perpendicular from the origin. $P = ON$

Let the given point be (x_1, y_1, z_1)

Equation of the plane parallel to (1) is

$lx + my + nz = p_1$ ... ... (39)

and since it passes through (x_1, y_1, z_1) we have

$p_1 = lx_1 + my_1 + nz_1$ ... ... (40)

where $p_1 = ON_1$ (say) is the length of the perpendicular from origin to the plane (2) i.e the plane through p and parallel to the given plane.

Now of $PL = D$ be drawn perpendicular to the given plane from the point P, then

$D = LP = ON_1 - ON = p_1 - p = lx_1 + my_1 + nz_1 - p$ from (3)

Therefore the length of the perpendicular from a given point $P(x_1, y_1, z_1)$ is $D = lx_1 + my_1 + nz_1 - p$ ... ... (41)

Art. 25. To find the length of the perpendicular from (x_1, y_1, z_1) to the plane $ax + by + cz + d = 0$ ... ... (1)

The normal form of the equation of the plane according as d is $-ve$ or d is $+ve$ is

$$\pm \frac{ax}{\sqrt{(\Sigma a^2)}} \pm \frac{by}{\sqrt{(\Sigma a^2)}} \pm \frac{c_z}{\sqrt{(\Sigma a)^2}} \pm \frac{d}{\sqrt{(\Sigma a)^2}} = 0$$

Hence if D be the length of the perpendicular distance of (x_1, y_1, z_1) then by Art. 25 (a)

$$D = \pm \frac{ax_1 + by_1 + c^2_1 + d}{\sqrt{(a^2 + b^2 + c^2)}} \quad \ldots \quad \ldots \quad \ldots \quad (42)$$

Rule : Thus the length of the perpendicular from (x_1, y_1, z_1) to the plane is obtained by substituting (x_1, y_1, z_1) for x, y, z respectively in the expression, $ax + by + cz + d$ **and dividing the same by** $\sqrt{(a^2 + b^2 + c^2)}$

Art. 26. Projection on a plane : The projection of an area A enclosed by the curve PQR ..... on a plane is the area.

A_1 enclosed by the curve $P_1Q_1R_1$ ..... where P_1, Q_1, R_1 ..... are the feet of the perpendiculars from PQR ..... on the plane of projection. We shall assume here the result of Pure Solid Geometry that the area $A_1 = A \cos \theta$ where θ is the angle between the plane of the area A and the plane of projection.

(a) If A_x, A_y, A_z be the areas of the projections of an area A, on the three co-ordinate planes, then (pKh A_x, A_y, A_z KfjKa IãfPur Ckr A ßã©lPur u' IKnPãk ßã©lu y~ fJ yPu)

Let l, m, n be the direction cosines of the normal (IKnu') to the plane of the area A.

A_x is equal to the projection of Area on the YZ plane.

$\therefore A_x = \cos \theta = Al.$

l is the consine of the angle between the YZ plane and plane of the area A, therefore

$A_x = Al$, $A_y = Am$, $A_z = An$

$\therefore A^2_x + A^2_y + A^2_z = A^2(l^2 + m^2 + n^2) = A^2 \therefore l^2 + m^2 + n^2 = 1$

or, $A^2 = A^2_x + A^2_y + A^2_z \quad \ldots \ldots (1) \quad \ldots \quad \ldots \quad \ldots \quad \ldots \quad (43)$

(b) Area of a triangle whose vertices are (x_1, y_1, z_1), (x_2, y_2, z_2) (x_3, y_3, z_3)

$$A_x = \frac{1}{2}\begin{vmatrix} y_1 & z_1 & 1 \\ y_2 & z_2 & 1 \\ y_3 & z_3 & 1 \end{vmatrix} ; \quad A_y = \frac{1}{2}\begin{vmatrix} z_1 & x_1 & 1 \\ z_2 & x_2 & 1 \\ z_3 & x_3 & 1 \end{vmatrix}$$

$$A_z = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \quad \ldots \quad \ldots \quad \ldots \quad (2)$$

Hence A, the area of a triangle is easily obtained by (1).

$A^2 = A^2_x + A^2_y + A^2_z$.

Art. 27. Volume of a tetrahedron (Yfá~uPTr IJ~fj) -*To find the volume of a tetrahedron whose four vertices are given.*

Let A, B, C, D be the vertices whose co-ordinates respectively are (x_1, y_1, z_1), (x_2, y_2, z_3), (x_3, y_3, z_3), (x_4, y_4, z_4)

If V be the volume of the tetrahedron $ABCD$.

then $V = \frac{1}{2} p \Delta$ ... ... (1)

where p is the length of the perpendicular (uP'r ‰hWtq) from A to the opposite face (KmkrLf fu) BCD and Δ be the area of triangle BCD.

Therefore, $\Delta = \sqrt{(\Delta x^2 + \Delta y^2 + \Delta z^2)}$ ... ... (2)

Where $\Delta_x = \frac{1}{2}\begin{vmatrix} y_2 & z_2 & 1 \\ y_3 & z_3 & 1 \\ y_4 & z_4 & 1 \end{vmatrix}$; $\Delta_y = \frac{1}{2}\begin{vmatrix} x_2 & z_2 & 1 \\ x_3 & z_3 & 1 \\ x_4 & z_4 & 1 \end{vmatrix}$

$$\Delta_z = \frac{1}{2}\begin{vmatrix} x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \\ x_4 & y_4 & 1 \end{vmatrix} \quad \ldots \quad \ldots \quad \ldots \quad (3)$$

(3) The equation of the plane BCD is

$$\begin{vmatrix} x & y & z & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{vmatrix} = 0$$

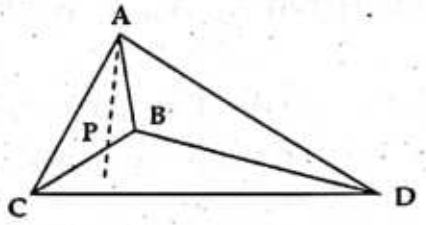


Fig : 14

or, $$\begin{vmatrix} y_2 & z_2 & 1 \\ x & y_3 & z_3 \\ y_4 & z_4 & 1 \end{vmatrix} 1 - \begin{vmatrix} x_2 & z_2 & 1 \\ y & x_3 & z_3 \\ x_4 & z_4 & 1 \end{vmatrix} 1 + \begin{vmatrix} x_2 & y_2 & 1 \\ z & x_3 & y_3 \\ x_4 & y_4 & 1 \end{vmatrix} 1 - \begin{vmatrix} x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \\ x_4 & y_4 & z_4 \end{vmatrix} = 0$$

The length of the perpendicular (uP'r ‰hWtq) from A (x_1, y_1, z_1) on the plane (4) is

$$2x_1 \Delta_x - 2y_1 \Delta_y + 2z_1 \Delta_z - \begin{vmatrix} x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \\ x_4 & y_4 & z_4 \end{vmatrix}$$

$p = \sqrt{\{4(\Delta_x^2 + \Delta_y^2 + \Delta_z^2)\}}$ ... ... (4)

$$P = \begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{vmatrix} \quad \# 2\Delta \text{ By (2)} \quad \ldots \quad \ldots \quad \ldots \quad (5)$$

$$\therefore \Delta = \frac{1}{3} P \Delta = \frac{1}{6}\begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{vmatrix} \text{ by (1)} \quad \ldots \quad \ldots \quad (6)$$

(6) Find the volume of the tetrahedron when its four faces are given.

Let the four plane faces of the tetrahedron be given by

(চতুস্তলকের চারটি তল দেওয়া থাকলে এর আয়তন নির্ণয়)

$a_r x + b_r y + c_r z + d_r = 0 \quad (r = 1, 2, 3, 4)$

Let $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \\ a_4 & b_4 & c_4 & d_4 \end{vmatrix}$

be the determinant of the above set of 4 equations.

D_1, D_2, D_3 etc. are the co-factors of d_1, d_2, d_3, etc. respectively in the determinant Δ.

মনে করি (x_1, y_1, z_1) তিনটি তলের ছেদ বিন্দু

$a_2x + b_2y + c_2z + d_2 = 0, a_3x + b_3y + c_3z + d_3 = 0$

and $a_4x + b_4y + c_4z + d_4 = 0$

অতএব (x_1, y_1, z_1) চতুস্তলকের শীর্ষ বিন্দু

Then $\frac{x_1}{A_1} = \frac{y_1}{B_1} = \frac{z_1}{C_1} = \frac{1}{D_1}$

or, $x_1 = \frac{A_1}{D_1}, y_1 = \frac{B_1}{D_1}, z_1 = \frac{C_1}{D_1}$

মনে করি, $(x_2, y_2, z_2), (x_3, y_3, z_3), (x_4, y_4, z_4)$ অন্য তিন শীর্ষ বিন্দু একইভাবে পাওয়া যায়।

চতুস্তলকের শীর্ষ বিন্দুর পরিপ্রেক্ষিতে এর আয়তন

$$V = \frac{1}{6}\begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{vmatrix} = \frac{1}{6}\begin{vmatrix} \frac{A_1}{D_1} & \frac{B_1}{D_1} & \frac{C_1}{D_1} & 1 \\ \frac{A_2}{D_2} & \frac{B_2}{D_2} & \frac{C_2}{D_2} & 1 \\ \frac{A_3}{D_3} & \frac{B_3}{D_3} & \frac{C_3}{D_3} & 1 \\ \frac{A_4}{D_4} & \frac{B_4}{D_4} & \frac{G_4}{D_4} & 1 \end{vmatrix}$$

$$= \frac{1}{6D_1D_2D_3D_4}\begin{vmatrix} A_1 & B_1 & C_1 & D_1 \\ A_2 & B_2 & C_2 & D_2 \\ A_3 & B_3 & C_3 & D_3 \\ A_4 & B_4 & C_4 & D_4 \end{vmatrix}$$

$$\therefore V = \frac{\Delta_1}{6D_1D_2D_3D_4} = \frac{\Delta_3}{6D_1D_2D_3D_4}$$

Note : **If you take the product of Δ and Δ_1 then $\Delta\Delta_1 = \Delta^4$ or $\Delta_1 = \Delta^3$ For detail consult, Author's Higher Algebra.**

EXAMPLES

Ex. 1. Find the equation of the plane through the points (2, 3, 1), (1, 1, 3) and (2, 2, 3). Find also the perpendicular distance from the point (5, 6, 7) to this plane.

First Method : The required plane is

$$\begin{vmatrix} x & y & z & 1 \\ 2 & 3 & 1 & 1 \\ 1 & 1 & 3 & 1 \\ 2 & 2 & 3 & 1 \end{vmatrix}$$

or, $2x-2y-z+3=0$

2nd Method :

The equation of any plane through (2, 3, 1) is

$a(x-2)+b(y-3)+c(z-1)=0$ ... ... (1)

Since it passes through (1, 1, 3) and (2, 2, 3) we have

$a(1-2)+b(1-3)+c(3-1)=0$ or, $a+2b-2c=0$ ... ... (2)

and $a(2-2)+b(2-3)+c(3-1)=0$ or, $b-2c=0$ ... ... (3)

Solving (2) and (3) for a, b, c by corss multiplication, we have $\frac{a}{-2}=\frac{b}{2}=\frac{c}{1}$

Putting these value in (1) we have

$-2(x-2)+2(y-3)+1(z-1)=0$ or, $2x-2y-z+3=0$

The perpendicular distance P is from (5, 6, 7) to the (4) is

$$P=\frac{2.5-2.6-7+3}{\sqrt{(4+4+1)}}=2$$

3rd Method.

See Art. 18. (a) cor. 3

The equation of plane passing through (2, 3, 1) and (1, 1, 3) is by Art. 18 (a) cor, 3, eq. 29 (b)

$(x-2)(y-1)-(x-1)(y-3)=A\{(x-2)(z-3)-(x-1)(z-1)\}$,

Since if passes through (2, 2, 3), hence

$0-(2-1)(2-3)=A\{0(2-1)-(3-1)\}$ or, $A=-1/2$

Now putting the value of A, we have

$$(x-2)(y-1)(x-1)(y-3)=\frac{1}{2}\{(x-2)(z-3)-(x-1)(z-1)\}$$

or $2x-2y-z+3=0$

Ex. 2. Show that the four points (0, −1, −1), (4, 5, 1), (3, 9, 4) and (−4, 4, 4) lie on a plane.

The four points are complanar if the determinant

$$\begin{vmatrix} 0 & -1 & -1 & 1 \\ 4 & 5 & 1 & 1 \\ 3 & 9 & 4 & 1 \\ -4 & 4 & 4 & 1 \end{vmatrix}$$ is zero

Otherwise we shall find the equation of the plane passing through any three points say $(0, -1, -1)$. $(4, 5, 1)$ and $(3, 9, 4)$ and then show that the fourth point $(-4, 4, 4)$ satisfied the plane.

Equations of any plane through $(0, -1, -1)$, $(4, 5, 1)$ and $(3, 9, 4)$ is

$5x - 7y + 11z + 4 = 0$ by example (1)

The point $(-4, 4, 4)$ clearly satisfies this plane. Hence the four points are coplanar.

N.B : For finding out the equation to the plane use, Cor, 3

Ex. 3. Find the equation of the plane passing through the intersection of the planes $x + 2y + 3z + 4 = 0$ and $4x + 3y + 2z + 1 = 0$ and the point (1, 2, 3)

Any plane through the intersection of the two planes is

$x + 2y + 3z + 4 + k(4x + 3y + 2z + 1) = 0$ ... ... (1)

Since it passes through (1, 2, 3),

$18 + 17k = 0$ or $k = -18/17$

Putting the value of k in (1) we have

$$x + 2y + 3z + 4 - \left(\frac{18}{17}\right)(4x + 3y + 2z + 1) = 0$$

or, $55x + 20y - 15z - 50 = 0$ or, $11x + 4y - 3z = 10$

which is the required plane.

Ex. 4. Find the angle between the planes $2x - y + z = 6$ and $x + y + 2z = 7$

Let θ be the angle between the planes then

$$\cos\theta = \frac{2.1 + (-1).1 + 1.2}{\sqrt{\{2^2 + (-1^2) + 1^2\}}\ \sqrt{(1^2 + 1^2 + 2^2)}} = \frac{3}{\sqrt{6}\sqrt{6}} = \frac{1}{2} \quad \therefore \theta = \frac{\pi}{3}$$

Ex. 5. Find the equation of the plane through the point (4, — 2, 1) and parallal to the plane whose direction numbers are 7, 2, —3.

The equation to the plane through $(4, -2, 1)$ is $a(x - 4) + b(y + 2) + c(z - 1) = 0$. The condition that the co-efficients a, b, c are proportional to the direction numbers 7, 2, — 3, therefore, the equation of the required plane is $7k(x - 4) + 2k(y + 2) - 3k(z - 1) = 0$ where k is constant.

or, $7(x - 4) + 2(y + 3) - 3(z - 1) = 0$ or, $7x + 2y - 3z - 21 = 0$

Ex. 6. Find the equation of the plane through the point (4, 0, 1) and parallel to the plane $4x + 3y - 12z + 6 = 0$

The equations of the parallel planes differ by a constant only. Hence the equation of the plane parallel to $2x + 3y - 12z + 6 = 0$ is $4x + 4y - 12x + k = 0$ Since it passes through (4, 0, 1).

$16 + 0 - 12 + k = 0$. or, $k = -4$

The required plane is $4x - 3y - 12z - 4 = 0$

Ex. 7. Find the equation of the plane passing through the lines of intersection of the planes $2x - y = 0$ and $3z - y = 0$ and perpendicular to the plane $4x + 5y - 3z + 1 = 0$

The plane $2x - y + k(3z - y) = 0$

i.e. $2x - (1 + k)\,y + 3kz = 0$

passes through the intersection of the two given planes whatever be the value of k.

It will be perpendicular to $4x + 5y - 3z + 1 = 0$

$\therefore\ 2.4 - (1 + k)5 + 3k(-3) = 0$ i.e. $14k = 3$ or, $k = 3/14$

Thus the required plane is $2x - y + 3(3z - y)/14 = 0$

or, $28x - 17y + 9z = 0$

Ex. 8. Find the equation of the plane through the pts. (2, 2, 1) and (9, 3, 6) and perpendicular to the plane $2x + 6y + 6z = 9$

[C.H. 1980; C.U. 1981; R.U. 1962, '78; D.H. 1976] D.U.(P) 1986]

Any plane through (2, 2, 1) is

$a(x-2) + b(y-2) + c(z-1) = 0$ ... ... (1)

Since it passes through (9, 3, 6) then (1) becomes

$a(9-2) + b(3-2) + c(6-1) = 0$ or, $7a + b + 5c = 0$ ... ... (2)

The plane (1) is perpendicular to $2x + 6x + 6z - 9 = 0$

Hence $2a + 6b + 6c = 0$ ... ... (3)

From (2) and (3) by cross multiplication.

$$\frac{a}{-24} = \frac{b}{-32} = \frac{c}{40} \text{ or, } \frac{a}{3} = \frac{b}{4} = \frac{c}{-5} \quad \ldots \quad \ldots \quad \ldots \quad (4)$$

Putting the values of a, b, c, from (4) in (1), the equation of the required plane is $3(x-2) + 4(y-2) - 5(z-1) = 0$ or, $3x + 4y - 5z = 9$

Alternate Method : The required plane is

$$\begin{vmatrix} x-2 & y-2 & z-1 \\ 9-2 & 3-2 & 6-1 \\ 2 & 6 & 6 \end{vmatrix} = 0$$

or, $3x + 4y - 5z = 9$ *by* Art. (11)

Ex. 9. Show that the equation to the plane through P (2, 3, − 1) at right angles to OP is $2x + 3y - z = 14$

The direction cosines of the line OP are $\frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}}, \frac{-1}{\sqrt{14}}$

Any plane at right angles to OP can be written as

$$\frac{2}{\sqrt{14}}x + \frac{3}{\sqrt{14}}y - \frac{1}{\sqrt{14}}z = p$$

Where $p = \sqrt{2^2 + 3^2 + (-1)^2} = \sqrt{14}$)

The required plane is $2x + 3y - z = 14$

Ex. 10. A plane meets the Co-ordinate axes in A, B, C such that the centroid of the triangle ABC is the point (p, q, r); show that the equation of the plane is [C.U. 1982]

$$\frac{x}{p} + \frac{y}{q} + \frac{z}{r} = 3$$

Let the equation of the plane be $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ ... ... (1)

$\therefore$ A is (a, o, o) B is (o, b, o) and C is (o, o, c)

Centriod of the triangle ABC is

$$G\left(\frac{a+o+o}{3}, \frac{o+b+o}{3}, \frac{o+o+c}{3}\right) \text{i,e.} \left(\frac{a}{3}, \frac{b}{3}, \frac{c}{3}\right)$$

But we are given that the centroid is the point (p, q, r)

$$\therefore p = \frac{1}{3}a,\ q = \frac{1}{3}b,\ r = \frac{1}{3}c$$

$$a = 3p,\ b = 3q,\ c = 3r$$

Therefore eq. (1) is $\frac{x}{3p} + \frac{y}{3q} + \frac{z}{3r} = 1$ or $\frac{x}{p} + \frac{y}{q} + \frac{z}{r} = 3$

Ex. 11. Find the distance of the point (2, 0, 1) and (3, — 3, 2) from the plane $x - 2y + z = 6$ and find whether the two points lie on the same side or opposite sides of the plane.

The equation of the plane is $x - 2y + z - 6 = 0$ ... ... (1)

Its distance from (2, 0, 1) is $\frac{2 - 2.0 + 1 - 6}{\sqrt{(1^2 + 2^2 + 1^2)}} = \frac{-3}{\sqrt{6}}$

Its distance from (3, — 3, 2) is $\frac{3 - (2.-3) + 2 - 6}{\sqrt{(1^2 + 2^2 + 1^2)}} = \frac{5}{\sqrt{6}}$

The distance $3/\sqrt{6}$ and $5/\sqrt{6}$

Put the point (2, 0, 1) in the L.H.S of eq. (1) this gives — 3, i.e negative value.

Put point (3, — 3, 2) in this L.H.S in eq. (1), this gives 5, i,e positive value.

The two results are of opposite signs. Therefore the two points lie on opposite sides of the plane.

Ex. 12. A variable plane is at a constant distance p from the origin and meets the axis in A, B, C. Through A, B, C, planes are drawn parallel to the co-ordinate planes. Show that the locus of their point of intersection is $x^{-2} + y^{-2} + z^{-2} = p^{-2}$

Let the equation of the variable plane be $lx + my + nz = p$.

Here l, m, n vary but p is constant as the plane is always at constant distance from the origin, l, m, n are the actual direction cosines of the plane.

It cuts the axis of x at A$(p/l.0,\ 0)$ axis of y at B$(0, p/m, 0)$ and axis of z at C$(/0, 0, p/n)$

Now the planes through A, B, C are parallel to the co-ordinate planes, then the co-ordinates of point of intersection $(p/l, p/m, p/n)$,

$$\therefore x = p/l,\ y = p/m,\ z = p/n$$

Therefore, $p^2/x^2 + p^2/y^2 + p^2/z^2 = l^2 + m^2 + n^2 = 1$

or, $x^{-2} + y^{-2} + z^{-2} = p^{-2}$

Ex. 13. A plane makes intercepts OA = a, OB = b and OC = c, on the axes of co-ordinates. Find the area of the trangle ABC made by the plane.

A_x = area of the plane triangle formed by the three points which are the projections of the vertices on the yz plane.

Clearly the points B and C are in the yz plane and the projection of the point A(a, o, o) is (o, o, o)

A_x = Area of ΔOBC which is right angled at O $A_x = \frac{1}{2}bc$

Similarly $A_y = \frac{1}{2}ca$ and $A_z = \frac{1}{2}ab$ $\therefore A = \frac{1}{2}\sqrt{(b^2c^2 + c^2a^2 + a^2b^2)}$

Ex. 14. Find the volume of the tetrahedron formed by planes whose equations are $my + nz = 0$, $nz + lx = 0$, $xl + my = 0$, $lx + my + nz = p$. [D.U. 1979]

Here $\Delta = \begin{vmatrix} o & m & n & o \\ l & o & n & o \\ l & m & o & o \\ l & m & n & -p \end{vmatrix} = -2lmnp$

Also $D_1 = lmn$, $D_2 = lmn$, $D_3 = lmn$, $D_4 = -2lmm$

Hence the volume $= \frac{2p^3}{3lmn}$

Ex. 15. Find the equation of the plane through the line $\frac{x-2}{3} = \frac{y-3}{5} = \frac{z}{7}$ **and passing through the point (1, −2, 3)**

By Cor. 3 Art. 18(a)

The general equation of the plane containing the line (given) is

$5(x-2) - 3(y-3) = k\{7(x-2) - 3z\}$ ... ... (1)

[It may be seen that this equation is obtained by spliting up. the equation into two equations $\frac{x-2}{3} = \frac{y-3}{5}$ and $\frac{x-2}{3} = \frac{z}{7}$ then multiplying crosswise and connecting the result by the variable constant k for the general case the value of k to be fixed by the additional condition]

Since this plane passes through (1, −2, 3), we have

$5(-1) - 3(-5) = k\{7(-1) - 9\}$ or $k = -5/8$

Putting this value of k in (1)

$8(5x - 3y - 1) = -5(7x - 3z - 14)$ or, $25x - 8y - 5z - 26 = 0$

16. Find the equation of the plane through the ponit (2,3,1) and (4,–5,3) and parallel to the axis of x. [N.U.H. 1996]

Solution : Let equation of the plane through the point (2,3,1) is

$a(x-2) + b(y-3) + c(z-1) = 0$ ... ... (1)

It passes through (4, – 5, 3)

$\therefore a(4-2) + b(-5-3) + c(3-1) = 0$

$\Rightarrow 2a - 8b + 2c = 0$

$\Rightarrow a - 4b + c = 0$ ... ... (2)

Dr's of x – axis

1, 0, 0

Since the plane parallel to x – axis.

So, normal to the plane and x–axis are perpendicular.

$\therefore 1.a + 0.b + 0.c = 0$

From (2); $a - 4b + c = 0$

By the rule of cross multiplication

$\therefore \frac{a}{0-0} = \frac{b}{0-1} = \frac{c}{-4-0}$

$\therefore \frac{a}{0} = \frac{b}{-1} = \frac{c}{-4} = k$ (say)

$\therefore \quad a = 0K, b = -K, C = -4K$

i,e, $a = 0, b = -1, c = -4$

Putting the values of a, b, c in (i) we get

$0(x-2) + 1(y-3) + 4(z-1) = 0$

$y + 4z - 7 = 0$

Which is the required equation.

17. Find the equation of the plane perpendicular to the planes $x - 4y + z = 0$ and $3x + 4y + z = 2$ and at a unit distance from the origin. [N.U.H 1997]

Solution : The given planes

$$x - 4y + z = 0 \quad \ldots \quad \ldots \quad (1)$$

and $$3x + 4y + z = 2 \quad \ldots \quad \ldots \quad (2)$$

Let equation of the plane which perpendicular both planes (1) and (2) is

$$ax + by + cz + d = 0 \quad \ldots \quad \ldots \quad (3)$$

Since the plane (3) perpendicular both the planes (1) and (2)

$\therefore a - 4b + c = 0$

and $3a + 4b + c = 0$

$\therefore \frac{a}{-4-4} = \frac{b}{3-1} = \frac{c}{4+12}$

$\frac{a}{-8} = \frac{b}{2} = \frac{c}{16} = k$ (say)

$\therefore a = -8k, b = 2k, c = 16k$

$a = -4, b = 1, c = 8$

Pulting the values of a, b, c in (3) we get (3) $\Rightarrow -4x + y + 8z + d = 0 \quad \ldots \quad \ldots \quad (4)$

According to the problem,

$\frac{-4.0 + 0 + 8.0 + d}{\sqrt{4^2 + 1^2 + 8^2}} = \pm 1$

$\left|\frac{d}{9}\right| = \pm 1 \therefore d = \pm 9$

(4) $\Rightarrow -3x + y + 8z \pm 9 = 0$ Which is the required plane.

18. Find the equation of the plane containing the point (1, –1, 1) and passing through the point of intersection of the planes $x - 2y + 3z + 4 = 0$ and $2x - 3y + 4z - 7 = 9$. [N.U.H-2001]

Solution : Given planes

$x - 2y + 3z + 4 = 0$ ... ... (1)

$2x - 3y + 4z - 7 = 0$ ... ... (2)

Let equation of the plane through the line of intersection of the planes (1) and (2) is $x - 2y + 3z + 4 + k(3x - 3y + 4z - 7) = 0$ ... ... (3)

Where k is an arbitrary constant.

It passes through the point (1, –1, 1).

$\therefore 1 - 2.(-1) + 3.(1) + 4 + k\{2.1 - 3(-1) + 4.1 - 7\} = 0$

$1 + 2 + 3 + 4 + k(2 + 3 + 4 - 6) = 0$

$10 + 2k = 0$

$\therefore k = -5$

Putting the value of k in (3) we get

$x - 2y + 3z + 4 - 5(2x - 3y + 4z - 7) = 0$

$\Rightarrow x - 2y + 3z + 4 - 10x + 15y - 20z + 35 = 0$

$\Rightarrow -9x + 13y - 17z + 39 = 0$

$\Rightarrow 9x - 13y + 17z - 39 = 0$

Which is the required plane.

19. Find the equatin of the plane passing through the line of intersection of the planes $2x - y = 0$ and $3z - y = 0$ and perpendicular to the plane $4x + 5y - 3z + 7 = 0$. [N.U.H – 2004]

Solution : Given planes

$2x - y = 0$ ... ... (1)

$3z - y = 0$ ... ... (2)

$4x + 5y - 3z + 7 = 0$ ... ... (3)

Let equation of the plane through the line of intersection of the planes (1) and (2) is

$2x - y + k(3z - y) = 0$ ... ... (4)

Where k is an arbitrary Constant

$\Rightarrow 2x - (1 + k)y + 3kz = 0$

Since the plane (3) and (4) are perpendicular

$a_1a_2 + b_1b_2 + c_1c_2 = 0$

$\therefore 4.2 + 5.\{-(1 + k)\} - 3.3k = 0$

$\Rightarrow 8 - 5 - 5k - 9k = 0$

$\Rightarrow 3 = 14k \therefore k = \frac{3}{14}$

Putting the values of k in (4) we get

$2x - y + \frac{3}{14}(3z - y) = 0$

$\Rightarrow 28x - 14y + 9z - 3y = 0$

$\Rightarrow 28x - 17y + 9z = 0$

Which is the required plane.

20. A point p moves on a fixed plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$. the plane through p perpendicular to op meets the axes in A, B, C respectively. If the planes are drawn through A, B, C parallel to the co-ordinate planes then show that the locus of the point inter section is

$\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{1}{ax} + \frac{1}{by} + \frac{1}{cz}$.

Solution : The given plane

$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ ... ... (1)

Let the point P(α, β, γ). passes through (1)

$\therefore \frac{\alpha}{a} + \frac{\beta}{b} + \frac{\gamma}{c} = 1$ ... ... (2)

Let equation of the plane through P(α, β, γ) perpendicular to op is

$\alpha(x - \alpha) + \beta(y - \beta) + \gamma(z - \gamma) = 0$

$\frac{\alpha x + \beta y + \gamma z}{\alpha^2 + \beta^2 + \gamma^2} = \frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha} + \frac{\alpha^2 + \beta^2 + \gamma^2}{\beta} + \frac{z}{\alpha^2 + \beta^2 + \gamma^2} = 1$ ... ... (3)

Since the plane (3) meets the axes in A, B, C.

$\therefore$ Co-ordinates of A $\left(\frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha}, 0, 0\right)$

---------- B $\left(0, \frac{\alpha^2 + \beta^2 + \gamma^2}{\beta}, 0\right)$

---------- C $\left(0, 0, \frac{\alpha^2 + \beta^2 + \gamma^2}{\gamma}\right)$

The planes through A, B, C Parallel to the Co-ordinate planes intersect at Q (say)

$\therefore$ Co-ordinate of Q $\left(\frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha}, \frac{\alpha^2 + \beta^2 + \gamma^2}{\beta}, \frac{\alpha^2 + \beta^2 + \gamma^2}{\gamma}\right)$

i.e. at the point Q

$x = \frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha}, y = \frac{\alpha^2 + \beta^2 + \gamma^2}{\beta}, z = \frac{\alpha^2 + \beta^2 + \gamma^2}{\gamma}$

Now, L.H.S $= \frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \left(\frac{\alpha}{\alpha^2 + \beta^2 + \gamma^2}\right)^2 + \left(\frac{\beta}{\alpha^2 + \beta^2 + \gamma^2}\right)^2 + \left(\frac{\gamma}{\alpha^2 + \beta^2 + \gamma^2}\right)^2$

$= \frac{\alpha^2 + \beta^2 + \gamma^2}{(\alpha^2 + \beta^2 + \gamma^2)^2}$

$$\therefore \frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=\frac{1}{\alpha^2+\beta^2+\gamma^2}$$

$$\text{R.H.S} = \frac{1}{ax}+\frac{1}{by}+\frac{1}{cz} = \frac{\alpha}{a(\alpha^2+\beta^2+\gamma^2)}+\frac{\beta}{b(\alpha^2+\beta^2+\gamma^2)}+\frac{\gamma}{c(\alpha^2+\beta^2+\gamma^2)}$$

$$= \frac{\alpha/a+\beta/b+\gamma/c}{\alpha^2+\beta^2+\gamma^2}$$

$$\therefore \frac{1}{ax}+\frac{1}{by}+\frac{1}{cz}=\frac{1}{\alpha^2+\beta^2+\gamma^2}, \text{ [By --(2)]}$$

$\therefore$ R.H.S = L.H. S (Proved)

21. Find the equation of the plane through the points (1, 0, –1) and (2, 1, 3) and perpendicular to the plane $2x + y + z = 1$. [N.U.H -2005]

Solutions : The given plane

$$2x+y+z=1 \quad \ldots \quad \ldots \quad (1)$$

Let equation of the plane through the point (1, 0, – 1) is

$$a(x-1)+b(y-0)+c(z+1)=0 \quad \ldots \quad \ldots \quad (2)$$

It passes through (2, 1, 3).

$\therefore a(2-1)+b(1-0)+c(3+1)=0$

$\Rightarrow a+b+4c=0 \quad \ldots \quad \ldots \quad (3)$

Sinec the (1) and (2) ore perpendicular.

So, $2a+b+c=0 \quad \ldots \quad \ldots \quad (4)$

From (3) and (4), we get by the rule of cross multiplication

$$\frac{a}{1-4}=\frac{b}{8-1}=\frac{c}{1-2}$$

$$\frac{a}{3}=\frac{b}{7}=\frac{c}{-1}=k \text{ (say)}$$

$\therefore a=-3k, b=7k, c=-k$

Putting the values of a, b, c in (2) we get

$(2) \Rightarrow -3k(x-1)+7k(y-0)-k(z+1)=0$

$3x-7y+z-2=0$ Which is required plane.

22. A variable plane passes through a fixed point (a, b, c) and meets the axes in A, B, C. Show that the locus of the point of intersection of the planes through A, B, C parallel to the Co-ordinate planes is $\frac{a}{x}+\frac{b}{y}+\frac{c}{z}=1$ [N.U.H-2006]

Solution : Let the variable plane be

$$\frac{x}{a_1}+\frac{y}{b_1}+\frac{z}{c_1}=1 \quad \ldots \quad \ldots \quad (1)$$

it passes through the point (a, b, c)

$$\therefore \frac{a}{a_1}+\frac{b}{b_1}+\frac{c}{c_1}=1 \quad \ldots \quad \ldots \quad (2)$$

Again, plane (1) meets the axes in A, B, C.

$\therefore$ Co-ordinates of A $(a_1, 0, 0)$

---------- B $(0, b_1, 0)$

---------- C $(0, 0, c_1)$

Planes through A, B, C parallel to the Co-ordinate planes intersect at M (say)

$\therefore$ Co-ordinate of M (a_1, b_1, c_1)

At the point M; $a_1 = x, b_1 = y, c_1 = z$.

Putting the values of a_1, b_1, c_1 in (2) we get $\frac{a}{x}+\frac{b}{y}+\frac{c}{z}=1$. which is the required locus.

23. Two system of rectangular axes have the same origin. If a plane cuts the axes at a, b, c and $a_1, b_1 c_1$ respectively from the origin. Prove that $\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}=\frac{1}{a_1^2}+\frac{1}{b_1^2}+\frac{1}{c_1^2}$

Sol : There are two rectangular system.

Eq. of the plane w. r. to the system (1) and (2) and are

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1 \quad \ldots \quad \ldots \quad (1)$$

$$\text{and } \frac{x}{a_1}+\frac{y}{b_1}+\frac{z}{c_1}=1 \quad \ldots \quad \ldots \quad (2)$$

Perpendicular distanec from the origin to planes (1) and (2) are

$$d_1=\frac{0+0+0-1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}}}=\frac{1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}}}$$

$$d_2=\frac{0+0+0-1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}}}=\frac{1}{\sqrt{\frac{1}{a_1^2}+\frac{1}{b_1^2}+\frac{1}{c_1^2}}}$$

Since Two system have same origin

$$\text{So } d_1 = d_2 \Rightarrow \frac{0+0+0-1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}}}=\frac{1}{\sqrt{\frac{1}{a_1^2}+\frac{1}{b_1^2}+\frac{1}{c_1^2}}}$$

$$=\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}=\frac{1}{a_1^2}+\frac{1}{b_1^2}+\frac{1}{c_1^2} \qquad \text{(Proved)}$$

24. Find the equation of the plane passing through the points (1, 2, –2) and (3, –2, 6) and perpendicular to the plane $2x - y - z + 7 = 0$ [N.U. H-2009]

Solution : The Given plane

$$2x-y-z+7=0 \quad \ldots \quad \ldots \quad (1)$$

Let equation of the plane through the point (1, 2, –2) is

$$a(x-1)+b(y-2)+c(z+2)=0 \quad \ldots \quad \ldots \quad (2)$$

It passes through (3, – 2, 6)

$\therefore a(3-1)\, b(-2-2) + c(6+2) = 0$

$\Rightarrow 2a - 4b + 8c = 0$

$\Rightarrow a - 2b + 4c = 0 \quad \ldots \quad \ldots \quad (3)$

According the problem, planes (1) and (2) are perpendicular.

$\therefore 2a - b - c = 0 \quad \ldots \quad \ldots \quad (4)$

From (3) and (4), by the rule of cross multiplication

$$\frac{a}{2+4} = \frac{b}{8+1} = \frac{c}{-4+1}$$

$$\frac{a}{6} = \frac{b}{9} = \frac{c}{-3} = k \text{ (say)}$$

$\therefore a = 6k, b = 9k, c = -3k$

Putting the values of a, b, c in (2) we get, $6x(x-1) + 9k(y-2) - 3k(z+2) = 0$

$$2x + 3y - z - 10 = 0$$

Which is required equation of the plane

THE STRAIGHT LINE 3

Art. 28. General equation of a straight line.

A line be determined as the intersection of any two planes through it.

No if we take two equations of the first degree in x, y and z, together, i. e. $a_1x + b_1y + c_1z + d_1 = 0 = a_2x + b_2y + c_2z + d_2$ ... ... (1)

then any point which satisfies these equations individually will satisfy both of them as well. Hence these two equations taken together represent the line of intersection of the two given planes and we know that two planes intersect in a straight line.

Then the equation (1) represents the equation of a straight line which is the line of intersection of the two planes.

Conclution : Thus a staright line in space as represented by two equations of the first degree in x, y, z.

Art. 29. Parametric equations of the line

Let the direction conines of the line L.be l, m, n and let $P_1(x_1, y_1, z_1)$ be a point on the lime. Then L is the locus of P (x, y, z) moving so that

$x - x_1 = r\cos\alpha = rl$

$y - y_1 = r\cos\beta = rm$

$z - z_1 = r\cos\gamma = rn$

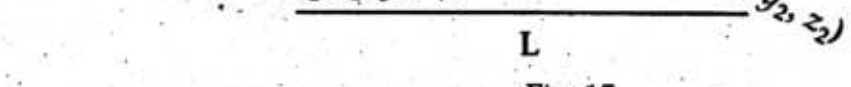


Fig. 15

$\therefore x = x_1 + rl, y = y_1 + rm$ and $z = z_1 + rn$ ... ... (43)

where the parameter represents the variable length P_1P.

If a, b, c, be direction ratios of L these equations can be written as $x = x_1 + at, y = y_1 + bt, z = z_1 + ct$ ... ... (44)

Here t does not represent the distance of the variable length P_1P.

The equation represents the co-ordinates of any point on the line L by giving different values to 't'.

Art. 30. (a) Symmetrical Form : To find the equation of a staight line through a given point (x_1, y_1, z_1) in a given direction.

Let the given point be $P_1(x_1, y_1, z_1)$ and let the given direction be defined by the direction cosines l, m, n.

Let $P(x, y, z)$ be a point on the line and let the measure of P_1P be r.

Then projecting P_1Pon the co-ordinate axes, we have

$x - x_1 = lr, y - y_1 = mr, z - z_1 = nr$

So that for all points (x, y, z) on the given line.

$$\frac{x-x_1}{l}=\frac{y-y_1}{m}=\frac{z-z_1}{n}=t \quad \ldots \quad \ldots \quad \ldots \quad (45)$$

Thus $\frac{x-x_1}{l}=\frac{y-y_1}{m}=\frac{z-z_1}{n}$ ... ... (46)

is the symmetrical form of the equation of a stright line

(b) Symmetrical form of direction ratios.

If a, b, c represent the direction ratios then the direction cosines are

$$\frac{a}{\sqrt{(\Sigma a^2)}}, \frac{b}{\sqrt{(\Sigma a^2)}}, \frac{c}{\sqrt{(\Sigma a^2)}}$$

Thus the equations of the line (1) take the form

$$\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}=r$$

$$\frac{a}{\sqrt{(\Sigma a^2)}}, \frac{b}{\sqrt{(\Sigma a^2)}}, \frac{c}{\sqrt{(\Sigma a^2)}}$$

$$\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}=t \quad \ldots \quad \ldots \quad \ldots \quad (47)$$

Here t does not represent the actual length of any point on the line from $(x_1, y_1, z_1,)$ Here is a parameter.

(c) If this line is perpendicular to one of the co-ordinate axes the equation takes one of th following :

$x = x_1, \frac{y-y_1}{b}=\frac{z-z_1}{c}$ (perpendicular to x-axis)

$y = y_1, \frac{x-x_1}{a}=\frac{z-z_1}{c}$ (perpendicular to y-axis)

$z = z_1, \frac{x-x_1}{a}=\frac{y-y_1}{b}$ (perpendicular to z-axis)

(d) If this line perpendicular to two axes two equations are sufficient to determine the line.

$x = x_1, y = y_1$, (perpendicular to x and y-axis)

$x = x_1, z = z_1$, (perpendicular to x and z-axis)

$y = y_1, z = z_1$, (perpendicular to y and z-axis)

Art. 32. Line through two given points.

If P, Q are $(x_1, y_1, z_1,)$ (x_2, y_2, z_2) the direction ratios of PQ are proportional to $x_2 - x_1$, $y_2 - y_1, z_2 - z_1$ and therefore the equations to PQ are $\frac{x-x_1}{x_1-x_2}=\frac{y-y_1}{y_1-y_2}=\frac{z-z_1}{z_1-z_2}$ ... ... (48)

Art. 31. (a) Direction ratios from equation of a straight line.

Let the eqeuations of the straight line be

$a_1x + b_1y + c_1z + d_1 = 0 = a_2x + b_2y + c_2z + d_2$ ... ... (49)

Then the equation of a line which is parallel to (1) and passes through the origin are

$a_1x + b_1y + c_1z + d_1 = 0\, a_2x + b_2y + c_2z + d_2$ ... ... (50)

Solving equations $a_1x + b_1y + c_1z = 0$

and $a_2x + b_2y + c_2z = 0$ we have

$$\frac{x}{b_1c_2 - b_2c_1} = \frac{y}{a_2c_1 - a_1c_2} = \frac{z}{a_1b_2 - a_2b_1} \quad \cdots \quad \cdots \quad \cdots \quad (51)$$

Which show the direction ratios of the line proportional to

$$b_1c_2 - b_2c_1, a_2c_1 - a_1c_2, a_1b_2 - a_2b_1 \quad \cdots \quad \cdots \quad \cdots \quad (52)$$

(b) To transform the general form of the equation (1) of 32 (a) of a straight line to symmetrical form.

In order to find the co-ordinates of any point on the line we can choose it in a number of ways. For convenience let the point be chosen to be one where the line (1) of ways. For convenince let the point be chosen to be one where the line (1) of Art. 32 (a) cuts the xy plane i.e; where $z = 0$, putting $z = 0$ in (1) we have

$$a_1x + b_1y + d_1 = 0, a_2x + b_2y + d_2 = 0$$

$$\frac{x}{b_1d_2 - b_2d_1} = \frac{y}{d_1a_2 - a_1d_2} = \frac{l}{a_1b_2 - b_2a_1}$$

$\therefore$ Hence the co-ordinates of the required point are

$$\left(\frac{b_1d_2 - b_2d_1}{a_1b_2 - a_2b_1}, \frac{d_1a_2 - d_2a_1}{a_1b_2 - a_2b_1}, 0\right) \quad \cdots \quad \cdots \quad \cdots \quad (54)$$

In case $a_1b_2 - a_2b_1 = 0$, then instead of putting $z = 0$ we may put $x = 0$ or $y = 0$

Thus, in the symmetrical form, the equations of the given line whose direction ratios are proportional to $(b_1c_2 - b_2c_1)$, $(a_2c_1 - a_1c_2)$, $(a_1b_2 - a_2b_1)$ by eq. 4 of Art. 32 (a) and which passes through a point $(x_1, y_1, 0)$ by eq (5) are

$$\frac{x - x_1}{b_1c_2 - b_2c_1} = \frac{y - y_1}{a_2c_1 - a_1c_2} = \frac{z - o}{a_1b_2 - a_2b_1} \quad \cdots \quad \cdots \quad \cdots \quad (56)$$

Art. 33. Distance of a point from a line. [C. U. 1981]

Let the length of the perpendicular distance from a point $P(x_1, y_1, z_1)$ to the line

$$\frac{x - a}{l} = \frac{y - b}{m} = \frac{z - c}{n} \text{ be } p \quad \cdots \quad \cdots \quad \cdots \quad (1)$$

If A be the point (a, b, c) on the given line and N the foot of the perpendicular from P on it, we have

$$PN^2 = AP^2 - AN^2 \quad \cdots \quad \cdots \quad \cdots \quad (2)$$

But $AP^2 = (x_1 - a)^2 + (y_1 - b)^2 + (z_1 - c)^2$

and AN = projection of AP on the given line

$= l(x_1 - a) + m(y_1 - b) + n(z_1 - c)$ provided l, m, n are the actual direction cosines of AP.

$\therefore$ $PN^2 = (x_1 - a)^2 + (y_1 - b)^2 + (z_1 - c)^2 - \{l(x_1 - a) + m(y_1 - b) + n(z_1 - c)\}$

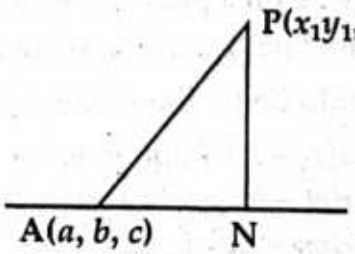


Fig. 16

$= \{x_1 - a)^2 + (y_1 - b)^2 + (z_1 - c)^2 (l^2 + m^2 + n^2)\} - \{l(x_1 - a) + m(y_1 - b) + n(z_1 - c)\}^2$

as $l^2 + m^2 + n^2 = 1$

or, $p^2 = \{l(y_1 - b) - m(x_1 - a)\}^2 + \{m(z_1 - c) - n(y_1 - b)\}^2 + \{n(x_1 - a) - l(z_1 - c)\}^2$

$$= \begin{vmatrix} x_1 - a & y_1 - b \\ l & m \end{vmatrix}^2 + \begin{vmatrix} y_1 - b & z_1 - c \\ m & n \end{vmatrix}^2 + \begin{vmatrix} z_1 - c & x_1 + a \\ n & l \end{vmatrix}^2$$

Art. 34. To find the angle between a line

$$\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n} \quad \ldots \quad \ldots \quad \ldots \quad (1)$$

and the plane $x + y + z + d = 0$ ... ... (2)

Angle between a line and a plane is the complement of the angle between the line and normal to the plane.

Since the direction cosines of the normal to the given plane and of the line are proportional to a, b, c and l, m, n respectively we have.

$$\cos(90° - \theta) = \sin\theta = \frac{al + bm + cn}{\sqrt{(a^2 + b^2 + c^2)}\sqrt{(l^2 + m^2 + n^2)}} \quad \ldots \quad (3) \ldots \ldots \quad (55)$$

where is the angle between the plane and the line and $90° - \theta$ in angle between the line and the normal

Art. 34 (a) Condition of parallelism.

The straight line (1) of Art 34 of is parallel to the plane (2), if $\theta = 0$ in eq (3) of Art. 34. **i. e** $al + bm + cn = 0$

which is also evident from the fact that a line be parallel to a plane, it is perpendicular to the normal to it.

Art. 35. Condition of perpendicularity of a line and a plane. Find the condition that a line $\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n}$ **is perpendicular to the plane** $ax + by + cz + d = 0$ ... ... (56)

If the line is perpendicular to the plane, its normal parallel to the line and hence we, have.

$$\frac{l}{m} = \frac{m}{b} = \frac{n}{c} \quad \ldots \quad \ldots \quad \ldots \quad (57)$$

which is the required condition

Art 36. To find the condition that the line

$$\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n} \quad \ldots \quad \ldots \quad \ldots \quad (1)$$

may lie on a plane $ax + by + cz + d = 0$ ... ... (2)

If the line lies in the plane, then for all values of r the point $x_1 + lr$, $y_1 + mr$. $z_1 + nr$ should lie on the plane $ax + by + cz + d = 0$

or, $a(x_1 + lr) + b(y_1 + mr) + c(z_1 + nr) + d = 0$

or, $r(al + bm + nc) + ax_1 + by_1 + cz_1 + d = 0$ ... ... (3)

$al + bm + cn = 0$

and $ax_1 + by_1 + cz_1 + d = 0$ ... ... (58)

The first condition shows that the line is parallel to the plane and the second condition shows that the point $(x_1, y_1, z_1,)$ on the line lies in the planes

The general equation of any plane containing the line (1) is $a(x-x_1)+b(y-y_1)+c(z-z_1)=0$; Where $al+bm+cn=0$.

Art. 37. Coplanar lines.

Find the condition that two lines whose equations are given in symmetrical form may intersect and determine the equation of the plane in which they lie.

Let the equation of the lines be

$$\frac{x-x_1}{l_1}=\frac{y-y_1}{m_1}=\frac{z-z_1}{n_1} \quad \ldots \quad \ldots \quad \ldots \quad (1)$$

and $$\frac{x-x_2}{l_2}=\frac{y-y_2}{m_2}=\frac{z-z_2}{n_2} \quad \ldots \quad \ldots \quad \ldots \quad (2)$$

The equation of a plane through the line (1) is

$$a(x-x_1)+b(y-y_1)+c(z-z_1)=0 \quad \ldots \quad \ldots \quad \ldots \quad (3)$$

where $$al_1+bm_1+cn_1=0 \quad \ldots \quad \ldots \quad \ldots \quad (4)$$

If the line (2) lies on the plane (3) the point $(x_2, y_2, z_2,)$ on it and the line is perpendicular to the normal to the plane (3)

This requires.

$$a(x_2-x_1)+b(y_2-y_1)+c(z_2-z_1)=0 \quad \ldots \quad \ldots \quad \ldots \quad (5)$$

where $$al_2+bm_2+cn_2=0 \quad \ldots \quad \ldots \quad \ldots \quad (6)$$

Now eliminate a, b, c from (4) and (6), then required conditon for the coplanarity of two lines is

$$\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix}=0 \quad \ldots \quad \ldots \quad \ldots \quad (59)$$

and the equation of the plane containing these lines (1) and (2) is obtained eliminting a, b, c from (3), (4) and (6) and is given by

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix}=0 \quad \ldots \quad \ldots \quad \ldots \quad (60)$$

Cor : If the equations of the lines are

$$a_1x+b_1y+c_1z+d_1=0$$

$$a_2x+b_2y+c_2z+d_2=0$$

$$a_3x+b_3y+c_3z+d_3=0$$

and

$$a_4x+b_4y+c_4z+d_4=0$$

find the condition of coplanarity.

Eliminate x, y, z from the above equations the eliminant is the required condition for the coplanarity of the two lines this is

$$\begin{vmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \\ a_4 & b_4 & c_4 & d_4 \end{vmatrix} = 0 \quad \ldots \quad \ldots \quad \ldots \quad (61)$$

Definition :- When two lines do not intersect and are parallel as well i. e, they do not lie in the same plane, then these lines are said to be non-intersecting lines. The straight line which is perpendicular to each of these two non intersecting line is called **the line of shortest distance** and length of this line intercepted between the given lines is called the **length of the shortest distance.**

(b) To find the shortest distance between two given lines whose equation are given and also obtain the equations of the shortest distance.

Let ABL and CDM be the two given lines whose equations are

$$\frac{x-x_1}{l_1} = \frac{y-y_1}{m_1} = \frac{z-z_1}{n_1} \quad \ldots \quad \ldots \quad \ldots \quad (1)$$

$$\frac{x-x_2}{l_2} = \frac{y-y_2}{m_2} = \frac{z-z_2}{n_2} \quad \ldots \quad \ldots \quad \ldots \quad (2)$$

Where the points B and D are (x_1, y_1, z_1), (x_2, y_2, z_2)

Let LM be the line which

is perndicular to both the given lines and let its direction cosines be l, m, n

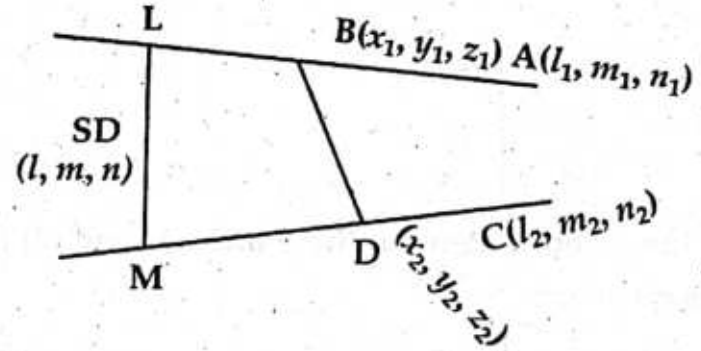


Fig : 17

Then by art 17. (c)

$ll_1 + mm_1 + nn_1 = 0$

$ll_2 + mm_2 + nn_2 = 0$

$$\therefore \frac{l}{m_1n_2 - m_2n_1} = \frac{m}{n_1l_2 - n_2l_1} = \frac{n}{l_1m_2 - l_2m_1}$$

(by cross multiplication)

$$\frac{\sqrt{(l^2 + m^2 + n^2)}}{\sqrt{\{(m_1n_2 - m_2n_1)^2 + (n_1l_2 - n_2l_1)^2 + (l_1m_2 - l_2m_1)^2\}}} = \frac{1}{\sin\theta} \quad \ldots \quad \ldots \quad (3) \text{ By Art. } 17(a)$$

$$\therefore\ l = \frac{m_1n_2 - m_2n_1}{\sin\theta},\ m = \frac{n_1l_2 - n_2l_1}{\sin\theta},\ n = \frac{l_1m_2 - l_2m_1}{\sin\theta} \quad \ldots \quad \ldots \quad (4)$$

The shortest distance (S. D) i, e. LM is the projection of BD on line LM whose direction cosines are l, m, n,

$$S.D = l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)$$

$$= \frac{(x - x_1)(m_1n_2 - m_2n_1) + (y_2 - y_1)(n_1l_2 - n_2l_1) + (z_2 - z_1)(l_1m_2 - l_2m_1)}{\sin\theta}$$

$$= \begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} \div \sin\theta \text{ by (4)}$$

$$= \begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} \div \sqrt{\Sigma(m_1n_2 - m_2n_1)^2}$$

[By (3) (6)]

Note : If the given lines are coplanar the S. D. is zero ;

Therefore, $$\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0 \quad \ldots \quad \ldots \quad \ldots \quad (62)$$

Art. 38. To find the eqution the shortest distance we see that it is complanar with both the given lines.

The eqation of the plane containing the line (1) and the line of shortest distance whose direction cosines are l, m, n,

is $$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ l_1 & m_1 & n_1 \\ l & m & n \end{vmatrix} = 0 \quad \ldots \quad \ldots \quad (5) \ldots\ldots (63)$$

Similarly the equation of the plane containing the line (2) and the shortest distance is

$$\begin{vmatrix} x - x_2 & y - y_2 & z - z_2 \\ l_2 & m_2 & n_2 \\ l & m & n \end{vmatrix} = 0 \quad \ldots \quad \ldots \quad (6) \ldots\ldots (64)$$

These two planes (2) and (2) taken together represent the equation of the line of shortest distance

N. B. There are other methods of determining the shortest distance. These methods will be explained with the help of examples.

Art. 39. Show that the general equation of a straight line contains four arbitrary constrary constants.

we can say that any equations.

$$\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n}$$

represent a straight line passing through the point (x_1, y_1, z_1) and having direction ratios proportional to l, m, n. The equations of a straight line may be written as

$$\frac{x - x_1}{l} = \frac{y - y_1}{m}, \quad \frac{y - y_1}{m} = \frac{z - z_1}{n}$$

$$\text{or, } x = \frac{l}{m}y + \frac{(mx_1 - ly_1)}{m}, y = \frac{m}{n}z + \frac{(ny_1 - mz_1)}{n}$$

$$\text{Put } \frac{l}{m} = A, + \frac{mx_1 - ly_1}{m} = B, \frac{m}{n} = C, \frac{ny_1 - mz_1}{n} = D$$

$\therefore x = Ay + B,\ y = Cz + D.$

where A, B, C, D are four arbitrary constants or parameters.

Therefore the general equations of a straight line contains four arbitrary constant.

Art. 39. (a) Intersection of three planes (তিনটি সমতলের ছেদন) কোণ শর্তে

$a_1x + b_1y + c_1z + d_1 = 0$ ... ... (1)

$a_2x + b_2y + c_2z + d_2 = 0$ ... ... (2)

$a_3x + b_3y + c_3z + d_3 = 0$ ... ... (3)

(a) সমতলগুলো একটি সাধারণ সরলরেখায় ছেদ করে,

(b) একটি ত্রিভুজাকার প্রিজমে ছেদ করে,

(c) একটি বিন্দুতে ছেদ করে,

(1), (2), (3) সমীকরণগুলোর x, y, z এর সহগগুলোর আয়তাকার সারি হতে

$\begin{vmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{vmatrix}$ পাওয়া যায়

(a) সমতলগুলো একটি সাধারণ সরলরেখায় ছেদ করার শর্ত হবে,

$\Delta_4 = 0, \Delta_3 = 0$, অথবা, $\Delta_2 = 0$, অথবা $\Delta_1 = 0$,

(b) তিনটি সমতল একটি প্রিজমে ছেদ করবে যদি,

$\Delta_4 = 0$ এবং $\Delta_1 \neq 0, \Delta_2 \neq 0, \Delta_3 \neq 0$

(c) একটি বিন্দুতে ছেদ করার শর্ত হল $\Delta_4 \neq 0$ যেহেতু $\Delta_4 = 0$ হলে সরলরেখাটি তৃতীয় সমতলের সাথে সমান্তরাল অথবা এর উপর থাকবে।

Rule (নিয়ম) : Δ_4 নির্ণয় কর। যদি $\Delta_4 \neq 0$ সমতলগুলো একটি বিন্দুতে ছেদ করবে। বিন্দুটি স্থানাঙ্কগুলো (1), (2), (3) কে সমাধান করলে x, y, z পাওয়া যাবে।

(2) Δ_4 নির্ণয় কর। যদি $\Delta_4 = 0$ আরও $\Delta_1 = 0, \Delta_2 = 0, \Delta_3 = 0$ হলে সমতল তিনটি একটি সরলরেখায় ছেদ করে

(3) $\Delta_4 = 0$ হলে যদি $\Delta_1, \Delta_2, \Delta_3$ এর মধ্যে যে কোন একটি শূন্য না হলে সমতলগুলো একটি প্রিজমে ছেদ করে।

$$\Delta_1 = \begin{vmatrix} b_1 & c_1 & d_1 \\ b_2 & c_2 & d_2 \\ b_3 & c_3 & d_3 \end{vmatrix}, \Delta_2 = \begin{vmatrix} a_1 & c_1 & d_1 \\ a_2 & c_2 & d_2 \\ a_3 & c_3 & d_3 \end{vmatrix}$$

$$\Delta_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}, \Delta_4 = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

Art. 39. (b) To find the condition that three planes may have a common line of intersection.

Let the equations of the planes be

$a_1x + b_1y + c_1z + d_1 = 0$ ... ... (1)

$a_2x + b_2y + c_2z + d_2 = 0$ ... ... (2)

and $a_3x + b_3y + c_3z + d_3 = 0$ ... ... (3)

The equation of any plane through the line of intersection of (1) and (2) is of the form

$(a_1x + b_1y + c_1z) + k(a_2x + b_2y + c_2z + d_2) = 0$

or, $(a_1 + ka_2)(x_1 + kb_2)y + (c_1 + kc_2)^2 + (d_1 + kd_2) = 0$ ... ... (4)

The plane (4) and third plane (3) will represent the same if

$$\frac{a_1 + ka_2}{a_3} = \frac{b_1 + kb_2}{b_3} = \frac{c_1 + kc_2}{c_3} = \frac{d_1 + kd_2}{d_3} = -U \text{ (say)}$$

Then $a_1 + ka_2 + Ua_3 = 0$

$b_1 + kb_2 + Ub_3 = 0$

$c_1 + kc_2 + Uc_3 = 0$

and $d_1 + kd_2 + Ud_3 = 0$

Eliminating K and U we have the required conditons given by the matrix equation

$$\begin{vmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{vmatrix} = 0 \quad \ldots \quad \ldots \quad \ldots \quad (65)$$

The notation indicates that each of the four determinants obtaines by the omitting one of the vertical columns, is zero. Thus the conditions are also expressed as

$$\begin{vmatrix} b_1 & c_1 & d_1 \\ b_2 & c_2 & d_2 \\ b_3 & c_3 & d_3 \end{vmatrix} = 0, \quad \begin{vmatrix} c_1 & d_1 & a_1 \\ c_2 & d_2 & a_2 \\ c_3 & d_3 & a_3 \end{vmatrix} = 0$$

$$\begin{vmatrix} d_1 & a_1 & b_1 \\ d_2 & a_2 & b_2 \\ d_3 & a_3 & b_3 \end{vmatrix} = 0, \quad \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

EXAMPLES

Ex. 1. Find the co-ordinates of the point where the joining the points (2, − 3, 1), (3, − 4, − 5) cuts the plane $3x + 4y + 5z = 5$.

The direction cosines of this are

$3-2, -4+3, -5-1$, or, $1, -1, -6$

Hence its equation are $\frac{x+2}{1} = \frac{y+3}{-1} = \frac{z-1}{-6} = R$ (say)

The co-ordinates of any point on the lines are

$(R+2, -R-3, -6R+1)$

If the point lines on the plane $3x + 4y + 5z = 5$. we have

$3R-6-4R-12-30R+5-5=0$ or, $R = -6/31$

Putting the value of R, we have the co-ordinates of the required point $(-\frac{56}{31}, -\frac{87}{31}, \frac{67}{31})$

Ex. 2. Find the equations of the straight lines which bisect angles between the lines

$\frac{x}{l}=\frac{y}{m}=\frac{z}{n}$ and $\frac{x}{l_1}=\frac{y}{m_1}=\frac{z}{n_1}$

Let OA, OB be the line

$\frac{x}{l}=\frac{y}{m}=\frac{z}{n}$ and $\frac{x}{l_1}=\frac{y}{m_1}=\frac{z}{n_1}$

Cut off OA = OB = 1

Then the co-ordinates of A, B, are (l, m, n), (l_1, m_1, n_1). Therefore the co-ordinates of P, the mid point of AB, are

$$\left(\frac{l+l_1}{2}, \frac{m+m_1}{2}, \frac{n+n_1}{2}\right)$$

and the co-ordinates of O are (0, 0, 0)

The equations of OP, the bisector of $\angle$A O B, are

$$\frac{x-0}{\frac{1}{2}(l+l_1)}-0=\frac{y-0}{\frac{1}{2}(m+m_1)}-0=\frac{z-0}{\frac{1}{2}(n+n_1)}-0$$

$$\text{or, } \frac{x}{l+l_1}=\frac{y}{m+m_1}=\frac{z}{n+n_1} \quad \ldots \quad \ldots \quad \ldots \quad (1)$$

Similarly the equation of the bisector of exterior angle say AOB_1 i. e. angle between OA whose direction cosines l, m, n and OB_1 whose direction cosines $-l_1, -m_1, -n_1$ are

$$\frac{x}{l-l_1}=\frac{y}{m-m_1}=\frac{z}{n-n_1} \quad \ldots \quad \ldots \quad \ldots \quad (2)$$

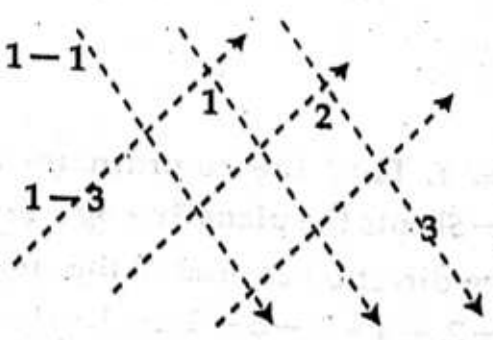
The joint equations are

$$\text{or, } \frac{x}{l \pm l_1}=\frac{y}{m \pm m_1}=\frac{z}{n \pm n_1}$$

where $l, m, n : l_1, m_1, n_1$ are the actual direction cosines

Ex. 3. Show that lines $x-y+z-5=0, x-3y+6=0$; and $y+z-5=0, 4x-2y+5z-4=0$ are parallel.

The direction ratios of the first line are from



or, 12, 4, —8

Fig. 18

The direction ratios of the second line are from

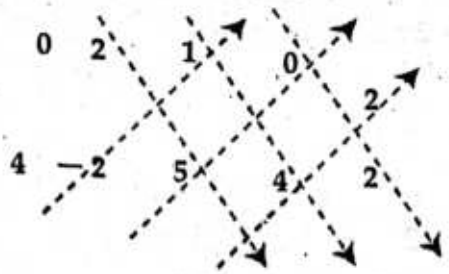


Fig : 19

or, 3, 1, −2 ... ... (2)

The direction ratios the two lines are the same. Hence they are parallel.

Note : Arrange the co-efficients of x, y, z, x, y of the planes in two parallel arrays. From the determinants with two successive columns starting from the second column i. e. the first column of the array remains uneffected. There are such three determinants which give the direction ratios of the line.

Ex. 4. Show that the lines $3x - 2y + 13 = 0, y + 3z - 26 = 0$ and $\frac{x+4}{5} = \frac{y-1}{-3} = \frac{z-3}{1}$ are perpendicular.

The direction ratios of the line are

$$\begin{matrix} 3 & -2 & 0 & 3 & -2 \\ 0 & 1 & 3 & 0 & 1 \end{matrix} \quad \text{or} -6, -9, 3 \text{ or, } 2, 3, -1$$

The direction ratios of the second line are, 5, −3, 1

Let θ be the angle between them.

then $\cos\theta = 0$

$aa_1 + bb_1 + cc_1 = 2.5 + 3(-3) + (-1)1 = 0$

Hence the lines are perpendicular

Ex. 5. Find the point where the line $5x + 8y + 4z + 4 = 0 = 6x + 5y - 2z + 15$ cuts plane $2x - y + 4z + 8 = 0$

Eliminating z from the equations, we have

$5x + 8y + 4 = 0$ and $6x + 5y + 14 = 0$ Solving for x and y

we have, $x = -4, y = 2$.

Then putting the values in any ony one of the given equations, we obtain $z = 1/2$

Hence the lines cut the plane at (–4, 2, 1/2)

Ex. 6. Find the angle between the line

$2x + 4y - 2z + 3 = 0 = 4x - 2y + 6z + 5$ and the plane

$5x - 4y + 3z - 5 = 0$

The directon ration of the line are from

$$\begin{matrix} 2 & 4 & -2 & 2 & 4 \\ 4 & -2 & 6 & 4 & -2 \end{matrix}$$

or, 20, −20, −20, or, 1, −1, −1,

The angle between the line and the plane will be the complement of the angle between the line and the normal to the plane. The direction ratios of the normal are 5, − 4, 3

$$\sin(90° - \theta) = \cos\theta = \frac{5(1) + (-4)(-1) + 3(-1)}{\sqrt{3}\sqrt{50}} = \frac{6}{\sqrt{150}} \quad \text{or, } \theta = \cos^{-1}\frac{6}{\sqrt{150}}$$

The angle between the line and the plane is $90° - \theta$

Ex. 7. find in symmetrical form of the equation of a line
$x+y+z+1=0=4x+y-2z+2$ **and find its** *d cs*

If l, m, n be the *d c's*. then

$l+m+n=0, 4l+m-2n=0,$

$$\therefore \frac{l}{-3}=\frac{m}{+6}=\frac{n}{-3} \text{ or, } \frac{l}{-1}=\frac{m}{+2}=\frac{n}{-1}=\frac{\sqrt{(l^2+m^2+n^2)}}{\sqrt{(1+4+1)}}=\frac{1}{\sqrt{6}}$$

direction cosines are $-1/\sqrt{6}, 2/\sqrt{6}, -1/\sqrt{6}$

direction ratios are $-1, 2, -1$

Putting $z=0$, we have $x+y+1=0. 4x+y+2=0$

Solving for x, and y, we have $x=-1/3$, $y=-2/3$

$$\frac{x+1/3}{-1}=\frac{y+2/3}{2}=\frac{z-0}{-1} \text{ are the required equation.}$$

Ex. 8. The axes being rectangular, find the equation to the perpendicular from the origin to the line

$x+2y+3z+4=0=2x+3y+4z+5=0$; **find the co-ordinates of the foot of the perpendicular.**

The given line can be reduced to symmetrical form by Art. 32 (b) as $\frac{x-2}{1}=\frac{y+3}{-2}=\frac{z-0}{-1}=r$ (say).

Any point on the line is $N(r+2, -2r-3, r)$ If it is foot of perpendicular from $P(0, 0, 0)$ on the given line, then PN is perpendicular to given line *d.r's* of PN are $r+2-0, -2r-3-0. r-0,$ i.e. $r+2, -2r-3, r.$

Also *dr's* of given line are $1, -2, 1$

The condition of perpendicularity gives

$1(r+2)-2(-2r-3)+1r=0$ or, $r=-4/3$

Putting the value of r, we have the foot to the perpendicular N as $(2/3 , -1/3, -4/3)$

Also, the *d.r's* of PN are $2/3, -1/3, -4/3)$

or *d. c. s* are proportional to $2, -1, -4.$

Hence the equations to the perpendicular from the origin are

$x/2=y/(-1)=z/-4$

Ex. 9. Show that the lines

$\frac{x+5}{3}=\frac{x+4}{1}=\frac{z-7}{-2}$ **and** $3x+2y+z-2=0=x-3y+2z-13$ **are coplanar and find the eqation to the plane in which they lie.**

The general equation of the plane through the second line is

$3x+2y+z-2+k(x-3y+2z-13)=0$

or, $x(3+k)+y(2-3k)+z(1+2k)-2-13k=0$

This will be parallel; to the first line if ... ... (1)

$3(3+k)+1(2-3k)-2(1+2k)=0$ or, $k=9/4$

Hence the equation of the plane containing the second line and parallel to the first is $21x - 19y + 22z - 125 = 0$ which clearly passes through the point $(-5, -4, 7)$ and so contains the first line. Thus the two lines are coplanar and lie in the plane.

$21x - 19y + 22z + 125 = 0$

Ex. 10. Find the equations of line perpendicular to both the line

$\frac{x-1}{1} = \frac{y-1}{2} = \frac{z+2}{3}$, $\frac{x+2}{2} = \frac{y-5}{-1} = \frac{z+3}{2}$ **and passing through their intersection.**

Let $\frac{x-1}{1} = \frac{y-1}{2} = \frac{z+2}{3} = r$

$\therefore x = 1 + r, y = 1 + 2r, z = -2 + 3r.$

Also let $(x + 2)/2 = (y - 5)/-1 = (z + 3)/2 = r_1$

then $x = 2r_1 - 2, y = 5 - r_1, z = 2r_1 - 3,$

If the lines meet, then

$1 + r = 2r_1 - 2, 1 + 2r = 5 - r_1, -2 + 3r = 2r_1 - 3, \therefore r = 1, r_1 = 2.$

Hence the co-ordinates of point of intersection are $(2, 3, 1)$

Let l, m, n be the direction ratios of the perpendicular which passes through (2, 3, 1) and perpendicular to the given two lines.

Hence $l + 2m + 3n = 0; 2l - m + 2n = 0$

Solve for l, m, n ; $l/7 = m/4 = n/-5$

Hence the equations of the line are

$\frac{x-2}{7} = \frac{y-3}{4} = \frac{z-1}{-5}$

Ex.11. Find the equations of the line drawn parallel to

$x/2 = y/3 = z/4$ **so as to intersect** $9x + y + z + 4 = 0 = 5x + y + 3z$; **and** $x + 2y - 3z - 3 = 0 = 2x - 5y + 3z + 3.$ [D. U. 1976]

The line is produced by the intersection of two planes

$9x + y + z + 4 + k_1 (5x + y + 3z) = 0$

or, $(9 + 5k_1)x + (1 + k_1)y + (1 + 3k_1)z + 4 = 0$ ... ... (1)

and $x + 2y - 3z - 3 + k_2 (2x - 5y + 3z + 3) = 0$

or, $(1 + 2k_2)x + (2 - 5k_2)y + (-3 + 3k_2)z + (-3 + 3k_2) = 0$ ... ... (2)

Since the line is parallel to $x/2 = y/3 = z/4$ hence from (1) and (2) we have

$2(9 + 5k_1) + 3(1 + k_1) + 4(1 + 3k_1) = 0$ or, $k_1 = -1$

and $2(1 + 2k_2) + 3(2 - 5k_2) + 4(-3 + 3k_2) = 0$ or, $k_2 = +4$

Hence the equations of the line are

$2x - z + 2 = 0, x - 2y + z + 1 = 0$ or, $(x + 1)/2 = y/3 = z/4$

Ex. 12. Find the distance of the point $(-1, -5, -10)$ from the point intersection of the line

$\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{+12}$ [R. U. 1985]

the plane $x - y + z = 5$

Let $(x - 2)/3 = (y + 1)/4 = (z - 2)/12 = r$ ... ... (1)

or, $x = 3r + 2, y = 4r - 1, z = 12r + 2$

The line (1) meets the plane $x-y+z=5$

Putting the values of x, y, z in (2)

$\therefore 3r+2-4r+1+12r+2=5$ or, $r=0$

The co-ordinates of the point are $(2,-1,2)$

The required distannce $=\sqrt{\{(2+1)^2+(-1+5)^2+(2+10)^2\}}=13$

12. (a) Find the distance of the pt. $(-1, 2, 5)$ from the st. line through the pt. $(3, 4, 5)$ whose direction cosincs are propotional to $(2, -3, 6)$ [D. U. 1982, 1983]

Actual directions are $\frac{2}{\sqrt{(9+16+36)}}=\frac{2}{7}, \frac{-3}{7}, \frac{6}{7}$, The distance,

$$\delta^2=[\{3-(-1)\}^2+(4-2)^2+(5-5)^2]-[\{3-(-1)\}\times 2/7+(4-2)\times\frac{-3}{7}+(5-5)\times 6/7]^2$$

$=16+4-\{(8-6)/7\}^2=20-4/49=976/49$

$\therefore \quad \delta=4\sqrt{61}/7$. By Art. 15.

Ex. 13. Find the shortest distance between the lines

$$\frac{x-3}{3}=\frac{y-8}{-1}=\frac{z-3}{1}, \frac{x+3}{-3}=\frac{y+7}{2}=\frac{z-6}{4}$$

Find also equation and the points in which it meets the given lines

[C. U. 1986; C. U. (Hons.) 1987; D. U. 1979]

Let $\frac{x-3}{3}=\frac{y-8}{-1}=\frac{z-3}{1}=r_1$ or, $x=3r_1+3, y=8-r_1, z=3+r_1$

and $\frac{x+3}{-3}=\frac{y+7}{2}=\frac{z-6}{4}=r_2$ or, $x=-3r_2-3, y=2r_2-7, \quad z=4r_2+6$

Let L $(3r_1+3, 8-r_1, 3+r_1)$; M $(-3r_2-3, 2r_2-7, 4r_2+6)$ by any two points L, M on the given lines

The direction ratios of the LM are

$3r_1+3r_2+6, -r_1-2r_2+15, +r_1-4r_2-3$

If the line is perpendicular to both the given lines, then we have

$3(3r_1+3r_2+6)-1(-r_1-2r_2+15)+(r_1-4r_2-3)=0$

and $-3(3r_1+3r_2+6)+2(-r_1-2r_2+15)+4(r_1-4r_2-3)=0$

Simplifying, we have $11r_1+7r_2=0$ and $7r_1+29r_2=0$

These two given $r_1=r_2=0$ and hence the point, L and M are $(3, 8, 3)$ and $(-3, -7, 6)$. The equation of the the line LM which is the S. D. is

$$\frac{x-3}{3+3}=\frac{y-8}{8+7}=\frac{z-3}{3-6} \text{ or, } \frac{x-3}{-2}=\frac{y-8}{-5}=\frac{z-3}{1}$$

The length of S. D. between L and M

$=\sqrt{[(3+3)^2+(8+7)^2+(3-6)^2]}=\sqrt{(270)}=3\sqrt{(30)}$

Ex. 14. Find the magnitude and the equations of the line of shortest distance between the lines

$$\frac{x-8}{3}=\frac{y+9}{-16}=\frac{z-10}{+7} \quad \ldots \quad \ldots (1)\ ; \quad \frac{x-15}{3}=\frac{y-29}{8}=\frac{z-5}{-5}$$

Let. $l, m,$ be the direction cosines of the lines L M of shortest distance. As it is perpendicular two lines we have

$3l-16m+7n=0$. and $3l+8m-5n=0$

$\therefore\ l/24=m/36=n/72$ or, $l/2=m/3=n/6=1/7$

Hence $l=2/7, m=3/7, n=6/7$

The magnitude of the shortest distance is the projcetion of the joint of the pts. $(8,-9,10)$, and $(15,29,5)$ on

ML and is $=(15-8)\,2/7+(29+9)3/7+(5-10)\,6/7=10$

$\therefore$ S. D. $=10$

Again the equation of the plane containing the first of the two given lines and the line of the shortest distance is

$$\begin{vmatrix} x-8 & y+9 & z-10 \\ 3 & -16 & 7 \\ 2 & 3 & 6 \end{vmatrix}=0 \quad \text{Or, } 117x+4y-41z-490=0$$

Also the equation of the plane containing the second line and the S. D. is

$$\begin{vmatrix} x-15 & y-29 & z-5 \\ 3 & 8 & -5 \\ 2 & 3 & 6 \end{vmatrix}=0 \quad \text{Or, } 9x-4y-z-14=0$$

The equation of the shortest distance is

$117x+4y-41z-490=0=9x-4y-z-14$

Ex. 15. Through the pt. A $(-1, 1, 2)$ a line is drawn parallel to the line of intersection of the planes $x-2y+z=3$ and $x+6y-5z=0$ This line cuts the plane $x-3y+2z=2$ in B. Find the eqation of the line AB and the co-ordinates. [R. U. 1962]

The derection ratios of the line

$x-2y+z-3=0=x+6y-5z$ ... ... (2)

are from

$$\therefore \quad \begin{matrix} 1 & -2 & 1 \\ 1 & 6 & -5 \end{matrix} \quad \begin{matrix} 1 & -2 \\ 1 & 6 \end{matrix} \quad \text{or, } 4, 6, 8, \text{or } 2, 3, 4$$

The direction ratios of the line parallel to (l) are 2, 3, 4

The equation of the line through A $(-1, 1, 2)$

are $\frac{x+1}{2}=\frac{y-1}{3}=\frac{z-2}{4}=r$ (say) ... ... (2)

If (2) meets the plane $x-3y+2z-2=0$ ... ... (3)

then $(2r-1)-3(3r+1)+2(2+4r)-2=0$ or, $r=2$.

Point of intersection of (2) and (3) is

B $(2r-1, 3r+1, 2+4r)$ or, B $(3, 7, 10)$

Eq. to AB is $\frac{1}{2}(x+1) = \frac{1}{3}(y-1) = \frac{1}{4}(z-2)$

Ex. 16. $ax + hy + gz = 0, hx + by + fz = 0, gx + fy + cz = 0$

সমতলগুলো একটি সরলরেখায় অতিক্রম করবে যদি

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

এবং সরলরেখাটির দিকানুপাতগুলো $\frac{l^2}{\frac{\delta\Delta}{\delta a}} = \frac{m^2}{\frac{\delta\Delta}{\delta b}} = \frac{n^2}{\frac{\delta\Delta}{\delta c}}$ সমীকরণকে সিদ্ধ করে।

Δ হতে $\Delta_3 = 0, \Delta_4 = 0$ এ দুটি প্রদত্ত শর্ত যদি l, m, n সমীকরণ সরলরেখাটির দিকানুপাত হয়, তখন দু সরলরেখা হতে

$$l/(hf - bg) = m/(gh - af) = n/(ab - h^2) \quad \ldots \quad \ldots \quad \ldots \quad (1)$$

$$l/(bc - f^2) = m/(fg - ch) = n/(hf - bg) \quad \ldots \quad \ldots \quad (2)$$

$$l/(hc - fg) = m/(g^2 - ac) = n/(af - hg) \quad \ldots \quad \ldots \quad \ldots \quad (3)$$

(1) এবং (2) হতে

$l/(hf - bg) = n/(ab - h^2); l/(bc - f^2) = n/(hf - bg)$

এ দুটি সমীকরণকে গুণ করলে

$$l^2/(bc - f^2) = n^2/(ab - h^2) \quad \ldots \quad \ldots \quad \ldots \quad (4)$$

আবার (2) এবং (3) হতে একই প্রকারে,

$$l^2(bc - f^2) = m^2(ac - g^2) \quad \ldots \quad \ldots \quad \ldots \quad (5)$$

(4) এবং (5) একত্রে $l^2/(be - f^2) = m^2/(ac - g^2) = n^2/(ab - h^2)$

কিন্তু $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

এবং $\delta\Delta/\delta a = bc - f^2, \delta\Delta/\delta b = ca - g^2, \delta\Delta/\delta c = ab - h^2$

Ex. 17. Find the locus of lines which intersect the three lines $y = b, z = -c;\ z = c$ $x = -a; x = a, y = -b.$

$(y = b, z = -c;\ z = c; x = -a, x = a, y = -b.$ সমতল তিনটি দ্বারা ছেদিত সরলরেখাগুলে সঞ্চারপথ নির্ণয় কর)

যদি $y - b + \lambda_1(z + c) = 0, z - c + \lambda_2(x + a) = 0, x - a + \lambda_3(y + b) = 0$

সমতলগুলো একটি সরলরেখায় ছেদ করে। তা হলে সরলরেখাটি তিনটি সরলরেখার সাথে মিলিত হবে। অতএব

$$\begin{vmatrix} 0 & 1 & \lambda_1 & -b + \lambda_1 c \\ \lambda_2 & 0 & 1 & -c + \lambda_2 a \\ 1 & \lambda_3 & 0 & -a + \lambda_3 b \end{vmatrix} = 0$$

দুটি সরলরেখার

$y-b=0=z+c$ ... ... (1)

$z-c=0=x+a$ ... ... (2)

মধ্যে দিয়ে অতিক্রান্ত সরলরেখার সমীকরণ

$y-b+\lambda(z+c)=0=(z-c)+\mu(x+a)$ ... ... (3)

সরলরেখাটি $x-a=0=y+b$ সরলরেখাকে ছেদ করে, (3) নং সমীকরণে একে স্থাপন করলে

$\lambda(z+c)=2b, z-c+2a\mu=0$, এই সমীকরণ হতে z-কে অপসারণ করলে

$2b/\lambda-c=c-2a\mu$ or, $2b/\lambda+2a\mu=2c$ or, $b+a\lambda\mu=c\lambda$

λ এবং μ এর মান (4) এ বসালে

$$b+a-\left(-\frac{z-c}{x+a}\right)\left(-\frac{y-b}{z+c}\right)=c\left(-\frac{y-b}{z+c}\right) \quad \ldots \quad \ldots \quad \ldots \quad (4)$$

or, $ayz+bzx+cxy+abc=0$ or, $\lambda_1\lambda_2\lambda_3+1=0$ ... ... (5)

$\lambda_1\lambda_2\lambda_3 a-2c\lambda_1\lambda_2+2b\lambda_3-a=0$

সরলরেখা (3)-এর যে কোন বিন্দু প্রদত্ত তিনটি সরলরেখার সাথে মিলিত হলে, ইহা

$y-b+\lambda_1(z+c)=0, z-c+\lambda_2(x+a)=0,$

$x-a+\lambda_3(y+b)=0,$

সমতলগুলোকে সিদ্ধ করবে। সমতলগুলো হতে $\lambda_1, \lambda_2, \lambda_3$ অপনয়ন করলে

$$\lambda_1\lambda_2\lambda_3=\frac{x-a}{y+b}\cdot\frac{z-c}{x+a}\cdot\frac{y-b}{z+c}=1 \text{ or, } ayz+bzx+cxy+abc=0$$

এটি একটি অধিবৃত্তকের সমীকরণ।

Ex. 18. Show that the eq. to the plane containing the line $y/b+z/c=1, x=0$; and parallel to the line $x/a-z/c=1, y=0$ is $x/a-y/b-z/c+1=0$ and if $2d$ is the S. D. Prove that $\frac{1}{d^2}=\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}$ (দেখাও $y/b+z/c=1, x=0$ সরলরেখাটি যে তলে এবং তলটি $x/a-z/c=1, y=0$ সরলরেখার সাথে সমান্তরাল এর সমীকরণ $x/a-y/b-z/c+1=0$ হবে, যদি সরলরেখাদ্বয়ের মধ্যে ক্ষুদ্রতম দূরত্ব (S. D). $2d$ হলে প্রমাণ কর যে $\frac{1}{d^2}=\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}$) [C. H. 1984,'89]

$1/d^2=1/a^2+1/b^2+1/c^2$ [C. H. 1984]

যে কোন তলে $y/b+z/c=1, x=0$ সরলরেখাটি থাকলে এর সমীকরণ হবে

$(y/b+z/c-1)+\lambda x=0$ ... ... (1)

যেহেতু এটি $x/a-z/c=1, y=0$; অর্থাৎ

$(x-a)/a=y/0=z/c$ ... ... (2)

সরলরেখার সাথে সমান্তরাল

সুতরাং $\lambda a+c.\lambda/c=0$, or, $\lambda=-1/a$

অতএব সমতলের সমীকরণটি হবে

$x/a - y/b - z/c = -1$

সরলরেখা (2) এর উপর $(a, 0, 0)$ বিন্দু হতে (3) সমতলের (S. D) ক্ষুদ্রতম দূরত্ব

$2d = \frac{2}{\sqrt{(1/a^2 + 1/b^2 + 1/c^2)}}$ or, $1/d^2 = 1/a^2 + 1/b^2 + 1/c^2$

Ex. 19. Find the eq. to the planes through the point $(1, 0, -1)$ and the lines $4x - y - 13 = 0 = 3y - 4z - 1$ and $y - 2z + 2 = 0 = x - 5$ and show that the equations to the line through the given point which intersects the two given lines can be written as $x = y + 1 = z + 2$.

$(1, 0, -1)$ বিন্দু এবং $4x - y - 13 = 0 = 3y - 4z - 1$ এবং $y - 2z + 2 = 0 = x - 5$ সরলরেখাগামী সমতলের সমীকরণ নির্ণয় কর, এবং দেখাও যে প্রদত্ত বিন্দু ও সরলরেখাগামী সমতলের দ্বারা ছেদক সরলরেখার সমীকরণকে লিখা যায় $x = y + 1 = z + 2$)

প্রদত্ত সরলরেখাগুলোর সমীকরণ যথাক্রমে $4x - y - 13 + \lambda_1 (3y - 4z - 1) = 0$ এবং $y - 2z + 2 + \lambda_2 (x - 5) = 0$ এরা $(1, 0, -1)$ বিন্দুগামী, সুতরাং

$4.1 - 0 - 13 + \lambda_1 (3.0 - 4. - 1 - 1) = 0$ or, $\lambda_1 = 3$

$0 - 2(-1) + 2 + \lambda_2 (1 - 5) = 0$ or, $\lambda_2 = 1$

অতএব নির্ণেয় সমতলগুলোর সমীকরণ

$4x - y - 13 + 3(3y - 4z - 1) = 0$ or, $4x + 8y - 12z - 16 = 0$

or, $x + 2y - 3z - 4 = 0$ ... ... (1)

আবার,

$y - 2z + 2 + 1(x - 5) = 0$ or, $x + y - 2z - 3 = 0$ ... ... (2)

(1) এবং (2) হতে বিয়োগ করলে $y - z - 1 = 0$

or, $y - 1 = z$ or, $y + 1 = z + 2$ ... ... (3)

একে (1)-এ বসালে

$x + 2y - 3(y - 1) - 4 = 0$

or, $x - y - 1 = 0$ or, $x = 1 + y$ ... ... (4)

সুতরাং (3) এবং (4) হতে $x = 1 + y = z + 2$

Ex. 20. Find the eqs to the planes through the point $(-1, 0, 1)$ and the lines $4x - 3y + 1 = 0 = y - 4z + 13$; $2x - y - 2 = 0 = z - 5$ and show that the eq. to the line through the given point which intersects the two given lines can be written as $x = y - 1 = z - 2$.

$(-1, 0, 1)$ বিন্দু এবং $4x - 3y + 1 = 0 = y - 4z + 13$; $2x - y - 2 = 0 = z - 5$ সরলরেখাগামী সমতলের সমীকরণ নির্ণয় কর এবং দেখাও যে সমতল দুটি ছেদককে $x = y - 1 = z - 2$ আকারে লিখা যায়।)

Sol. Any plane through $4x - 3y + 1 = 0 = y - 4z + 13$ is $4x - 3y + 1 + \lambda(y - 4z + 13) = 0$ ... ... (1)

It passes through $(-1, 0, 1)$, then $-4 + 1 - \lambda(4 - 13) = 0$ or $\lambda = 1/3$.

Putting the value of λ in (1), we have $4x - 3y + 1 + (y - 4z + 13)/3 = 0$

The required plane is $3x - 2y - z + 4 = 0$ ... ... (2)

Again any plane through $2x-y-2=0=z-5$ is $2x-y-2+\lambda_1(z-5)=0$

... ... (3)

This passes through $(-1, 0, 1)$ then $-2-0-2+\lambda_1(1-5)=0$ or, $\lambda_1=-1$

Hence the equation of the other prance is from (3),

$2x-y-z+3=0$ ... ... (4)

Any line through $(-1, 0, 1)$ is $\frac{x+1}{l}=\frac{y}{m}=\frac{z-1}{n}=r$ ... ... (5)

Any point on it is $(lr-1, mr, nr+1)$ If this is the point of intersection with the line $4x-3y+1=0=y-4z+13$, this $4(lr-1)-3(mr+1)=0$ and $mr-4(lmr+1)+13=0$

or, $4lr-3mr-3=0$ ... ... (6)

$0-mr+4nr+9=0$ ... ... (7)

$\therefore$ (6) $\times$ 3; $12lr-9ms-9=0$

(7) $\times$ 1; $0-mr+4nr-9=0$

Subtracting ; $12lr-8mr-4nr=0$

or, $3l-12m-n=0$ ... ... (8)

Similarly from (5) if we take $\frac{x+1}{l}=\frac{y}{m}=\frac{z-1}{n}=r_1$, then $(x, y, z)=(lr_1-1, mr, nr_1+1)$ is the point on the line, then from $2x-y-2=0=z-5$, we have

$2(lr_1-1)-mr_1-2=0$ and $nr_1+1-5=0$

or, $2lr_1-mr_1-4=0$, $nr_1=4$ or, $r_1=4/n$

$\therefore 2l\,4/n-m4/n-4=0$ or, $8l-4m-4n=0$

or, $2l-m-n=0$ ... ... (9)

solving (8) and (9)

$3l-2m-n=0$

$2l-m-n=0$

$\therefore \frac{l}{2-1}=\frac{m}{-2+3}=\frac{n}{-3+4}$ or, $\frac{l}{1}=\frac{m}{1}=\frac{n}{1}$

Hence the line (5) becomes

$\frac{x+1}{1}=\frac{y}{1}=\frac{z-1}{1}$ or, $x+1=y=z-1$

or, $x+1=y, y=z-1$

or, $x=y-1, y-1=z-2$

$\therefore x=y-1=z-2$ is a line.